قناة تانية تمريضيانو على التليجرام

Arab Republic of Egypt
Ministry of Health and Population
Central Administration for Human Resource Development
General Administration for Technical Health Education
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Technical Secondary Schools for Nursing

General Surgery
Second Grade

introduction

To our sons and daughters, students of nursing schools, the subject of Surgery is one of the most important branches of medicine to study and practice. Surgery requires knowledge, skill, and experience. At the same time, the success of surgery depends on complete vigilance, accurate diagnosis, and speed and good management, whether in critical cases (in reception), in the operating room, or in the various surgical departments.

Nursing plays a major role in the success of surgical treatment, by assisting the surgeon whether in operations, in reception, or in the surgery department by following up on the patient after operations.

Surgical diseases are not isolated from other specialties. A patient may suffer from more than one disease at the same time. Likewise, the doctor, along with the nursing staff and assistants, works to treat the patient. This is the goal of studying surgical diseases. Therefore, the study of surgery must focus on the patient's person, not just on studying the disease.

We hope this book helps explain surgery with the required clarity. And we note that the questions in this book are merely examples of questions that may appear in the exam, and we wish you all the best.

Chapter One

Inflammations

Definitions:-

Inflammation is the body's reaction against any external stimulus and includes microbes, viruses, and physical factors like electricity and burns, and chemical factors like exposure to acids and alkalis. The most common in surgery is bacterial inflammation.

Types of Bacterial Inflammation include:

- Acute inflammation
- Chronic inflammation

1. Specific like (Tuberculosis):
2. Non-specific, which is caused by any type of microbe that was not treated completely.

Acute inflammation

- Non-specific microbe like boils, abscesses, carbuncles, cellulitis, and inflammation of wounds after operations.
- Specific microbe like Tetanus, Tuberculous lymphadenitis, and Gas Gangrene. The inflammation may be localized only or generalized when the microbe passes into the blood causing microbial blood poisoning (septicemia).

Non-specific inflammations :

1. Abscess : is a collection of pus under the skin or in any internal organ like the liver.
2. Carbuncle : is a group of boils that occur in diabetic patients in the skin tissues, they are productive and occur on the patient's back or behind the neck.
3. Cellulitis : Inflammation of the tissues under the skin and does not lead to the formation of an abscess. Its danger lies in the spread of the microbe to the blood and causing bacterial poisoning (Bacteremia).

4. Inflammation of wounds after operations: Occurs on the third or fifth day in the form of redness in the wound, pain, with pus discharge from the wound.

Symptoms:

A- Local symptoms:

- Pain at the site of inflammation.
- Swelling at the site of inflammation.
- Redness of the skin and warmth.
- Inability to move the inflamed part.
- Impairment of the function of the inflamed organ (e.g., inability to move in limb inflammations).

B- General symptoms:
1- Elevated body temperature, feeling chills and shivering, and sweating.
2- Patient lethargy and reduced movement.
3- Increased pulse.

Treatment:
1- Complete rest for the patient and the inflamed organ (e.g., limbs).
2- Elevating the inflamed limb to reduce pain.
3- Sufficient amount of fluids.
4- Administering appropriate antibiotics and analgesics.
5- Applying compresses to the inflamed organ.
6- Draining pus in cases of abscess, inflammation of wounds after operations, and carbuncles.

Note:
The skin should not be opened in cases of cellulitis to prevent blood poisoning (septicemia).

B- Specific microbial inflammations: (Specific Inflammation)

1- Erysipelas:

It is an inflammation of the skin caused by streptococcal microbe. The skin becomes bright red, slightly raised above the surrounding skin, with serous blisters on the edge. This disease is contagious and requires isolating the patient and not touching the wound except with gloves, and also not opening the inflammation.

2- Inflammation of lymph glands: (Lymphadenitis)

Characterized by enlarged lymph glands, whether in the neck (Figure 1), under the armpit, or in the upper thigh (groin). Lymph gland inflammation can be specific, e.g., due to Tuberculous lymphadenitis (TB), or non-specific, in cases of lymph gland inflammation resulting from skin inflammation.

3- Tetanus: -

It is a local inflammation by anaerobic microbes which lead to poisoning of nerve cells. It affects wounds that occur on the street, farms, animal dung, and also the umbilical cord in children.

Symptoms appear from the second day of infection up to the third week and are:

- Headache , Sweating and elevated temperature.
- Inability to open the jaw (Trismus), jaw stiffness.
- Inability to breathe, swallow, or move the neck.
- Occurrence of general convulsions before death.

Prevention is better than cure in these cases:

1- Vaccination against the disease for children and stable workers.
2- Administering Tetanus serum to the patient (1500) units when the patient is wounded.
3- Good cleaning of the wound.

Gas Gangrene:-
Occurs as a result of wound infection with anaerobic microbes which produce gases and toxins. It appears as swelling at the wound site, and the skin color becomes dark red with foul-smelling discharge.
And the presence of gas bubbles under the skin with muscle blackening around the wounds. The patient must be isolated.
And treated with Gas Gangrene serum and antibiotics, and excision of the affected tissues of the skin and muscles, or amputation of the affected limb in case of loss of viability.

Note:
The Gas Gangrene microbe may infect wounds after operations in rare cases, and the patient must be isolated to prevent the spread of infection.

Secondly Wounds and Burns

Wounds

Definition:
A wound is a cut in the skin and the lining tissues of the human body. Wounds occur either due to injury, burn, or surgical procedure.
Wounds are either clean, dirty, or susceptible to contamination.

Types of wounds:
1- Closed wounds: like abrasions, contusions, and blood collection under the skin (hematoma).
2- Open wounds: punctures, bites, and scratches.
3- Compound wounds: which involve loss of a part of the skin with muscle injury and bone fracture in the limbs or rib fractures with organ injury (e.g., car and train accidents and gunshot wounds).
4- Surgical wounds: which are clean wounds.

The symptoms of wounds are due to:

- The effect of the injury.
- The body's reaction to the injury.
- Complications, if complications occur in the wound.

Symptoms resulting from the injury:

Local symptoms:
- Pain.
- Bleeding.

- Disorder in the structure and function of the affected organ.

General symptoms:
Poisoning in case the body absorbs components of the injured tissues, especially in cases of severe muscle injury.

The body's reaction to the injury:

- Immediate general reaction: neurogenic shock from pain in case of a large wound.
- Immediate local reaction: constriction of blood vessels and production of inflammatory secretions.
- Delayed general reaction: general signs of inflammation and systemic response to inflammation.
- Delayed local reaction: the process of inflammation and healing.

The healing of the wound passes through three stages: the secretion stage, where inflammatory fluid is secreted and cells gather in the area of inflammation, then the formation stage, where the body forms tissues to complete healing, then the maturation stage, where a balance occurs between tissue breakdown and formation.

Factors that delay wound healing:

- Presence of necrotic or devitalized tissue in the wound.
- Nutritional factors: deficiency of proteins and vitamin C.
- Hormonal factors: like cortisol hormone.

Local factors including:
1- Local insufficiency in blood circulation.
2- Microbial inflammation.
3- Exposure to radiation.
4- Effusion resulting from blockage in veins or lymphatic vessels.
5- Repeated injuries to the wound.

Factors that help wound healing:
1- Occurrence of previous wounds that have healed, which speeds up the healing process.
2- Cleaning the wound and removing any foreign bodies and devitalized tissues.
3- Stopping bleeding by surgical ligation if necessary.
4- Proper nutrition.
5- Prevention and treatment of any infections in the wound.

In the case of a clean wound (incised or surgical), it is preferable to leave it closed, while in the case of a contaminated wound or one exposed to contamination, it is preferable to perform regular dressing changes with removal of any foreign bodies.

Complications of wounds:

1- Extension to adjacent organs.

2- Contamination: Contamination occurs due to the attack of microbes on the body's immune system. And delayed surgical intervention to clean the wound.

3- Improper healing: due to local insufficiency in blood circulation or deep inflammation or adhesion of the body to deep tissues (muscles or bones). Improper healing leads to the formation of:
Atrophic scar, or hypertrophic scar, or multiple scars, or keloid scar.

Treatment of Wounds
First: General treatment

1- Good ventilation.
2- Stopping bleeding from any of the body's organs.
3- Treatment of shock.
4- Administering analgesics in case of pain.
5- Administering prophylactic antibiotics.

Secondly: Local treatment

1- Prevention of contamination and infection: removal of foreign bodies, dressing the wound, and giving antibiotics.
2- Cleaning the surrounding skin well, with gauze moistened with saline solution or Betadine.
3- Cleaning the wound well with sterile saline solution to remove foreign bodies.
4- Ensuring the integrity of the wound edges and excision of unhealthy skin.
5- Suturing the wound within 6-8 hours of injury.
6- Draining the wound in case of expected blood effusion, pus formation, or rupture of an internal organ.

7- Dressing the wound:
The goal of dressing is to protect the wound from injury and infection, keep the wound dry, and apply pressure on the wound to limit effusion. The dressing consists of a layer of gauze that allows secretions and effusions to exit, and cotton between the gauze to absorb them to keep the wound dry. And a bandage or plaster for pressure on the wound.

In the case of a clean, sutured wound, Betadine or alcohol can be used.
In the case of an open wound, alcohol is not used, rather Betadine is used.

8- Prophylactic antibiotics if needed.

9-Immobilization of movable organs to allow for healing.
10-Administering Tetanus serum in cases of injury on streets and farms.

Burns (Burns)

Causes:

A - Physical factors like heat (dry and steam), radiation, and electricity.
B - Chemical factors like acids and caustic substances.

Degrees of Burns:

(1) First degree: Redness of the skin (Erythema).
(2) Second degree: Serous blisters (vesicles).
(3) Third degree: Burning of the skin while preserving the sense receptors and sweat glands, and the patient feels acute pain.
(4) Fourth degree: Complete burning of the skin.
(5) Fifth degree: Burning of muscles.
(6) Sixth degree: Burning of bones. The burn can be minor when it affects less than 15% of human skin, or severe when it affects more than 15%.

General symptoms:

1- Respiratory suffocation cases like fires.
2- Shock: (a) neurogenic from severity of pain.
(b)- Hemorrhagic from deficiency of fluids, plasma, and blood.
4- Septic shock occurring on the third day due to burn contamination.
5- Renal failure in severe cases.

Treatment of Burns:

- Minor:
  - Does not require patient hospitalization.
  - Washing the wound with saline solution.
  - Applying Dermazin cream.
  - Covering the wound well.
- Treatment of severe burns:
  A-First aid:
  - Covering the burn with sterile dressings.
  - Administering an anesthetic or analgesic.
  - Administering Tetanus vaccine.

B -Treatment in the burn unit:

- Administering anesthetic regularly.
- Warming the room.
- Ensuring the patient's breathing is safe.
- Administering antibiotics.
- Administering fluids according to the doctor's orders, as well as plasma and blood.
- Making a fluid chart and inserting a urinary catheter.
- Cleaning the wound and dressing it by the doctor in operations.

Surgical Infections

Surgical infections are those that require surgical intervention for treatment or that occur after a surgical operation. Infection occurs as a result of an infectious microbe transmitted to a closed cavity or surface or tissue where blood circulation is weak, in a patient susceptible to infection due to immunodeficiency.

The most common microbes causing surgical infections are the streptococcal microbe, the staphylococcal microbe, and the coliform microbe.

The infection can spread either directly to adjacent tissues, or through the secretion of toxins into the circulatory system.

Complications of Surgical Infections

1- Tissue destruction if the microbe spreads through the muscles or the subcutaneous layer.
2- Abscess if the infection is contained and pus forms inside a cavity.
3- Inflammation of lymphatic vessels and lymph glands. If the infection spreads through the lymphatic vessels.
4- Pleural inflammation and pericarditis.
5- Fistula formation as in the case of a colon abscess, or a sinus in the case of chronic subcutaneous infection.
6- Weakening of immunity and occurrence of secondary infections.

Symptoms of Infection

General symptoms: elevated temperature, feeling of fatigue, sweating, loss of appetite.
Local symptoms: redness, swelling, pain, and inability to move the affected part.

Treatment of Infection

Incision and cleaning: In the case of an abscess, incision and thorough cleaning are necessary to remove bacteria, necrotic tissues, and toxins.
Excision: In the case of appendix, gallbladder, and gangrene.
Antibiotics: Appropriate antibiotics for the microbe causing the infection must be chosen.

The best method for selection is based on the result of a culture and sensitivity test from wound secretions:

Penicillin derivatives like Ampicillin, Amoxicillin, and unasyn are used in case of superficial infections (Gram-positive microbe Gram +vc i.e., streptococcal or staphylococcal).

And Gentamicin, nebcin, (Claforan), are used in the case of Gram-negative microbe.

And Flagyl or tienam or Meronem (Atovaquone?) are used in the case of anaerobic microbes. And stimulating blood circulation and increasing blood flow. In case of insufficient blood circulation, good nutrition is necessary, as the body needs all nutritional elements.

Cellulitis

It is a non-suppurative inflammation that affects the connective tissues in the limbs, caused by the streptococcal microbe.
It appears as dark redness raised above the skin surface. It is accompanied by inflammation of the lymphatic vessels.
It is treated with rest, warm compresses, elevating the affected part, and using penicillin. If no improvement appears within 24 hours, one must suspect the presence of an abscess or a highly resistant microbe.

Chapter Two

Fluids and Electrolytes

Water constitutes about 60% of body weight and is distributed as follows:

- Plasma 5% of body weight.
- Interstitial fluids 15% of body weight. (Note: 25% mentioned in text seems high, 15% is more standard for interstitial)
- Intracellular fluids 40% of body weight.

Electrolytes

Electrolytes are distributed inside and outside cells. The basic salts inside cells are: Potassium, Phosphate, and Bicarbonate. The basic salts outside cells are: Sodium and Chloride.

Physiological factors that maintain fluids and Electrolytes

1 - The cell membrane separates the inside from the outside of the cell and regulates the transfer of sodium and potassium via the sodium pump, and regulates the movement of water to and from the cell.
2 - The wall of the blood capillaries. Allows the passage of water and salts but does not allow proteins to exit from inside the blood vessels.
3 - The process of internal exchange of fluids in the body. There is secretion and absorption of fluids inside the body cavities. As happens in the stomach, intestines, and gallbladder.

Water balance in the body

The volume of water in the body depends on the balance between gain and loss of water. Water increases in the body through:

- Drinking water, about one liter, and a person drinks water according to the sensation of thirst.
- Oxidation processes in the body add half a liter of water daily.

And the body loses water through:

- The kidneys, which excrete about one liter of water daily in urine.
- Through the skin (sweat) and lungs (with exhalation) about half a liter of water daily.

Electrolytes

The most important Electrolytes in the body are Sodium, Potassium, Chloride, Calcium, Magnesium, and Phosphate.

Sodium and Potassium: enter the body through food and exit through the kidneys.
Magnesium: enters the body in food and exits through the kidneys and feces.

Water deficiency in the body (Dehydration)

Dehydration occurs in cases of:

1- Loss of water in large quantities without compensation, as in cases of:
Severe diarrhea, repeated vomiting, and intestinal obstruction.

2- Not compensating for lost water during surgical operations via solutions.
Symptoms: Thirst, dry mouth, headache, and sometimes elevated temperature. Upon examining the patient, dry tongue is noted, decreased urine output, and this appears clearly if there is a urinary catheter after surgical operations.

Treatment: Replacement with Glucose 5% intravenously or glucose saline solution (Glucose/Saline).

Potassium K +

The normal level of potassium in the blood is 3.5 - 5.0 mEq/L.

And potassium deficiency is diagnosed if the level falls below 3.5 mEq/L.

Causes of potassium deficiency:

1- Transfer of potassium into the cells, and its most important causes are administering drugs like Insulin, Dopamine, and Adrenaline.
2- Loss of potassium from the stomach, and its causes include continuous vomiting and gastric suction. And loss of potassium from the colon in cases of continuous diarrhea.
3- Loss of potassium in urine, and its causes include diuretic drugs, Cortisone, and Magnesium deficiency.

Symptoms of potassium deficiency:

Weakness of muscle movement, and abdominal distension due to weak intestinal movement.

Treatment:
Administering potassium orally in the form of potassium syrup, or intravenously. Potassium is given intravenously in ampules added to solutions, and slowly. Care must be taken that injecting potassium intravenously quickly may lead to cardiac arrest.
Increased potassium is diagnosed if the potassium level in the blood rises above 5.0 mEq/L.

Causes of increased potassium:
1- Renal failure and impaired kidney function.
2- Some drugs prescribed for high blood pressure patients, and some diuretics, and some analgesics.

3- Blood transfusion especially if stored for a long time.
4- Hemoconcentration as in cases of shock.
5- Neglecting to take insulin doses in diabetic patients, and high blood glucose levels at high rates.

Symptoms of increased potassium:
Weakness of body muscle movement, and impaired heart function, and changes appear on the ECG when potassium exceeds7.0 mEq/L.

Treatment:
1- Administering 25% Glucose intravenously with added Insulin.
2- Administering Lasix intravenously to the patient.
3- Administering Calcium intravenously slowly.

Calcium Ca++

Calcium deficiency

Causes of calcium deficiency:

1- Renal failure
2- Deficiency of parathyroid hormone (Parathyroid Gland) especially after thyroidectomy operations
3- Some diuretic drugs

Symptoms of calcium deficiency:

Drop in blood pressure, feeling of tingling around the mouth, and hand muscle spasms, and sometimes general muscle spasms.

Treatment:

Injecting Calcium Gluconate in solution intravenously slowly, and calcium should not be given quickly, otherwise it causes cardiac arrest.

Increased calcium

Causes of increased calcium:

1- Increased secretion of parathyroid hormone.
2- Some drugs like Theophylline.
3- Taking vitamin D in high doses.
4- Some chronic diseases like Tuberculosis.
5- Some tumors like Lymphoma.

Symptoms:

Fatigue, muscle weakness, loss of appetite, nausea, vomiting, abdominal pain, excessive thirst, increased urination rate, formation of stones in the kidney and urinary tract, bone pain and bone deformities.

Treatment :

Administering saline solution intravenously with added Lasix
Administering Hydrocortisone intravenously or Prednisone tablets according to the severity of the condition.

Sodium Na+

Increased sodium:

Causes of increased sodium in the blood:

1- Administering saline solution in quantities and concentration greater than the body's ability to get rid of it, especially after surgical operations.
2- Loss of water in quantities greater than the loss of sodium as in cases of continuous vomiting and diarrhea.

Symptoms:

In cases of acute increase in blood sodium levels, convulsions occur, and the level of concentration and consciousness decreases, and the patient may go into a coma.

Treatment :

In case of dehydration, saline solution must be administered intravenously quickly to avoid hypovolemic shock, and water replaced slowly. In case of increased sodium without dehydration, the patient should be given Lasix intravenously.

Sodium deficiency:

Causes :

1- Drugs like diuretics
2- Diarrhea
3- Heart failure
4- Renal failure
5- Impaired liver function

Symptoms:

Convulsions, fatigue, and coma in acute cases.

Treatment:

Sodium should not be replaced quickly, because rapid replacement leads to brainstem damage. Treatment depends on administering concentrated saline solution to the patient slowly.

Chapter Three

Shock

Shock is a drop in blood circulation, resulting from an imbalance between blood volume and circulatory volume, either due to blood loss, or due to expansion of the circulatory volume. This acute disturbance in circulation leads to severe insufficiency in blood flow and oxygen in the tissues.

Types of Shock

Surgical shock, neurogenic shock, and septic shock.

Stages of Shock

Shock passes through two consecutive stages:

- The first stage called Reversible Shock: which responds to treatment, and improvement appears in the vital signs, which are pulse, pressure, and urine output.
- The second stage called Irreversible Shock: which does not respond to treatment, and the patient shows no improvement in vital signs.

Surgical Shock

Occurs as a result of massive loss of blood or fluids from the body, as happens in cases of bleeding, burns, vomiting, and severe diarrhea.

Symptoms:

1- Facial pallor and profuse sweating.

2- Rapid and weak pulse.

3- Low blood pressure.

4- Rapid and shallow breathing.

Treatment:

1- Place the patient lying on their back with head lowered.

2- Administer blood intravenously.

3-Administer solutions like glucose saline or Ringer's or Lactated Ringer's.

4-Administer vasopressors for circulation and Hydrocortisone which helps the circulation retain fluids.

5-Treat the cause.

Neurogenic Shock

Occurs as a result of severe pain.

Symptoms:

1 - Slow and weak pulse.

2 - Low blood pressure.

3 - Pallor and profuse sweating.

Treatment:

Place the patient in a lying position with head lowered, stimulate the patient, and administer solutions.

Septic Shock

Occurs as a result of severe inflammation in the body. The inflammation may be in the urethra, pneumonia, purulent or gangrenous infection of the wound, internal abscess, or advanced peritonitis.

Symptoms:

1 - Elevated temperature.

2 - Rapid pulse.

3 - Low blood pressure.

4 - Local symptoms.

Treatment:

1 - Treat the cause, and usually surgical treatment is the most important type of treatment.

2 - Administer solutions intravenously and transfuse plasma and blood.

3-Strong antibiotics appropriate for the expected type of bacteria.

4-Administer Zantac intravenously to prevent stomach ulcers.

5-Improve breathing to increase oxygen in the blood.

Chapter Four

Postoperative Complications

Surgical complications result from a disease the patient suffered from before the operation, or from the disease for which the patient underwent the operation, or from the effect of the surgery itself in addition to complications of anesthesia.

Complications of Surgical Operations

1- Complications of anesthesia
2- Complications of fluid transfusion
3- Complications of the wound
4- Respiratory complications
5- Complications of the heart and veins
6- Digestive system complications
7- Liver complications
8- Kidney and urinary system complications
9- Elevated temperature after surgery

Complications of Anesthesia

First: Complications of General Anesthesia

1- Malignant hyperthermia: May appear during surgery or within 24 hours of the operation, and its complications include increased pulse and pressure, elevated temperature, and muscle rigidity.
2- Nausea and vomiting
3- Vocal cord spasm.
4- Urinary retention.
5- Decreased temperature.
6- Injury of some nerves due to wrong positioning of the patient on the operating table, which is very rare.

Secondly: Complications of Spinal Anesthesia

1- Drop in blood pressure.

2- Respiratory disturbance: due to high level of spinal anesthesia.

3- Headache: especially in young people, women, and men.

4- Urinary retention.

5- Inflammation of the spinal cord membranes (meningitis): which is a very rare condition.

6- Nerve injury: very rare as in general anesthesia.

Complications of Solutions Transfusion

Local complications:

Leakage of fluids under the skin and around the cannula: Symptoms include swelling around the cannula, pain, and stopping of the drip. Treated by moving the cannula to another site and using an ointment to help the body absorb the fluids.

Phlebitis: Appears as swelling, bluish discoloration, and pain with warmth in the vein. Treated by stopping the drip and applying an anti-inflammatory ointment, and warm compresses.

Microbial inflammation at the cannula or injection site, and its transmission rates with using sterilization before setting up the drip.

General complications

1**- Pyrogenic reaction**: A sudden rise in temperature with injecting solutions, or after it, and happens due to the presence of proteinaceous materials in the drip. Resulting from impurity of the drip.

symptoms: Sudden rise in temperature, chills, and shivering.

treatment: Stopping the drip and giving the patient anti-allergy drugs like Apil (likely antihistamine) or Decadron (Dexamethasone).

Air embolism: It is a rare condition that may occur with the insertion of a central venous catheter .
And it must be prevented by not introducing air into the central venous catheter, or into the solution bottles, and placing a clamp on the catheter when changing solutions or setting up a three-way stopcock on the catheter. And if it happens, symptoms of shock appear on the patient with bluish discoloration (cyanosis), and the head of the bed must be lowered. And placing the patient lying on their left side to keep the air embolism in the right side of the heart until it is absorbed by the tissues, while giving oxygen.

3- Pulmonary embolism: Occurs as a result of a clot forming in a peripheral vein, and part of it travels to the lung vessels.

symptoms: Rapid pulse, elevated pressure, rapid breathing, and cyanosis.
treatment: Transfer the patient to intensive care to give oxygen and artificial respiration sometimes, with blood thinning drugs like Heparin and Streptokinase.

4- Increased amount of fluids in the body: Occurs as a result of injecting cardiac, renal, young children, or elderly patients with large amounts of solutions too quickly, greater than the body's ability to get rid of fluids in urine, sweat, and respiration.

symptoms: Difficulty breathing in the lying position, cyanosis, distention of neck veins, high blood pressure, persistent cough, and edema.

Treatment: Stop the solutions, place the patient in a sitting position with head elevated 45 degrees, with injection of a diuretic like Lasix.

Complications of the Wound

1- Hematoma: Resulting from bleeding due to a shortcoming in surgery or the patient's predisposition to bleeding.

2- Seroma: Occurs after operations like mastectomy or hernia repair due to dissection and cutting of lymphatic vessels under the skin. It appears as swelling at the site. If the swelling is mild, applying a pressure dressing on the swollen wound is sufficient, with giving a drug that helps the body absorb the collected fluids under the skin like Alpha Chymotrypsin. And if the swelling increases and becomes large in size, it can be

Aspiration of the serous fluid using a syringe after sterilizing the wound, with the application of a pressure dressing. In the case of a seroma appearing after vascular surgery, it is left without intervention. If it increases, it is explored in the operating room under complete aseptic conditions.

Wound Dehiscence: Results from improper suturing of the muscles or after operations for peritonitis or acute pancreatitis. The wound first secretes serous fluid, then the wound dehisces. In this case, urgent exploration is needed to ensure there is no internal cause, and then the wound is resutured.

Respiratory Complications

These occur most frequently after chest, heart, and upper abdominal surgeries. They also occur in elderly patients and patients with chest diseases.

Atelectasis: Results from bronchial obstruction and weak breathing. It appears as a rise in temperature, rapid breathing, and rapid pulse. Prevention involves early mobilization after surgery and breathing exercises. The condition is treated with bronchodilators and expectorants.

Bronchitis: Results from the aspiration of fluids over a period of time and their entry into the lungs. It occurs more often after emergency surgeries.

Symptoms: Elevated temperature, difficulty breathing, and decreased blood oxygen levels. Prevention involves fasting before surgery and inserting a nasogastric (NG) tube before emergency surgeries to empty the stomach, along with administering Zantac. Treatment begins with tracheal suction, hydrocortisone, and antibiotics.

Pneumonia: Results from a lung infection by a microbe. Diagnosis is confirmed by plain X-ray or CT scan. Prevention is similar to preventing lung aspiration, and treatment involves antibiotics, rest, antipyretics, and expectorants. A ventilator is used in case of lung function failure and low blood oxygen levels.

Cardiac and Venous Complications After Surgical Operations

1- **Cardiac arrhythmias**, including impaired heart function, coronary insufficiency, and myocardial infarction. These complications may occur in cardiac patients who undergo surgery following heart attacks, or within 6 months of heart attacks, or due to an undiagnosed congenital heart defect before surgery.

2- **Deep Vein Thrombosis (DVT)**

Deep vein thrombosis occurs in the leg veins after surgical operations, due to circulatory stasis from prolonged immobility during or after surgery. It also occurs at a higher rate in the elderly and in cases of obesity.

Prevention:

Prevention is the most important element in confronting venous thrombosis. The preventive steps are:

1-Use of anticoagulant medications before and after surgeries, such as low molecular weight heparin (LMWH) like Clexane (Enoxaparin), administered subcutaneously.

2-Avoid dehydration by giving the patient appropriate doses of fluids.

3-Encourage and assist the patient to move after surgery, as much as possible.

Symptoms of Deep Vein Thrombosis

1-Swelling of the leg compared to the other leg
2-Leg pain, especially upon movement
3-Slight rise in temperature

4-Diagnosis relies on performing an ultrasound with Doppler on the legs (Duplex scan).

Treatment of Deep Vein Thrombosis

1. Administer heparin in regular doses, intravenously using control pumps, to ensure continuous heparin infusion. The heparin dose must be adjusted by performing a daily PTT test. Until the required dose is reached, which is the dose that doubles the PTT time.

2. Bed rest until the acute symptoms of the clot subside, on average for one week.
3. High oral fluid intake, in addition to the IV fluids containing heparin, to stimulate circulation, reduce blood viscosity, and avoid dehydration.

3-Pulmonary Embolism

The most dangerous complication of leg clots. It occurs when a part of the leg clot dislodges and travels to the pulmonary artery, causing obstruction of the main artery or one of its branches. Obstruction of the main pulmonary artery leads to death:

Symptoms

1. Sudden difficulty breathing
2. Cyanosis (bluish discoloration)
3. Rapid and weak pulse
4. Drop in blood pressure

Diagnosis is confirmed when:

1. Changes appear on the ECG.
2. A decrease in oxygen percentage and an increase in carbon dioxide percentage are shown in blood gases.

Treatment of Pulmonary Embolism

1. Administer oxygen to raise the blood oxygen level.
2. Administer fluids to raise blood pressure and stimulate circulation.
3. Continuously inject heparin intravenously using a control pump.
4. Placing a filter in the inferior vena cava in case of a large clot in the deep leg veins, or in case of recurrent pulmonary embolism.

Complications of the Digestive System

1-Paralytic Ileus: Resulting from potassium deficiency and after operations for peritonitis and pancreatitis.

2-Gastric Dilatation: Occurs due to stomach operations without using a nasogastric (NG) tube or without suction during and after the operation.

Symptoms: Severe abdominal distension and difficulty breathing. Treatment is done by inserting an NG tube and applying suction from the stomach.

3-Intestinal Obstruction: Resulting from adhesions within the peritoneal tissue or an internal hernia. It can be distinguished from paralytic ileus by the presence of noticeable distension, colicky pain, or increased gastric reflux in the NG tube, as well as by a plain upright abdominal X-ray.

Treatment is initially conservative by stopping oral food intake, inserting an NG tube, and administering fluids.

4-Fecal Impaction: Occurs more in elderly patients and may lead to distension and paralytic ileus. The most important diagnostic step is the digital rectal exam. Treatment is by evacuation. It is best done in the operating room.

Liver Complications

1-Hepatitis due to anesthesia allergy.

2-Viral hepatitis resulting from blood transfusion. Transmission of Hepatitis B or C virus is possible due to contaminated blood. Therefore, blood tests must be meticulously performed before transfusion.

3-Obstructive jaundice due to ligation or injury of the bile duct, particularly during gallbladder and stomach operations.

Kidney and Urinary Tract Complications

1-Urinary Retention: After hernia operations, urinary retention may occur due to pain, or after prostate operations, or in elderly men due to pelvic congestion in case of prostate enlargement. Prevention involves giving treatment for the prostate and pain relief after hernia operations. Treatment involves giving analgesics, helping the patient relax, and attempting to urinate. If conservative measures fail, a urinary catheter is used under complete aseptic conditions.

2-Urethritis: Resulting from acquiring a microbe from the hospital after bladder catheterization. Prevention involves proper sterilization when inserting the catheter. Treatment involves increased fluids and appropriate antibiotics.

Fever After Surgical Operations

Fever occurs in 40% of patients after surgery.

Causes:

1-Within 48 hours of surgery: Anesthesia - IV solutions - Blood transfusion - Atelectasis - The original disease.

2-After 48 hours of surgery: Wound contamination - Intra-operative inflammation - Cannula site inflammation - Pneumonia - Urinary tract infection.

3-From the fifth to the seventh day: Wound infection - Internal contamination after surgery - Abscess at the operation site - Deep leg vein thrombosis.

Blood Transfusion

Complications of Blood Transfusion:

1. Complications of fluid transfusion.
2. Blood reaction due to incompatibility.
3. Disease transmission.
4. Jaundice.

Incompatibility Reaction: Is the most dangerous complication of blood transfusion.

Causes:

١-Error in performing the crossmatch between the patient's blood and the donor's blood.

2-Error in recording the blood type on the blood bag.

3-Error in recording the donor's or patient's name, leading to mixing of blood bags.

4-Error in using or recording the type of serum used in the compatibility confirmation test.

5-Presence of incompatibility between minor blood groups, for which testing is not routinely done.

Symptoms of Incompatibility

1. Elevated temperature and chills
2. Rapid pulse
3. Drop in blood pressure
4. Pain in the kidney region
5. Difficulty breathing
6. Cyanosis of the lips and extremities

Prevention of Incompatibility

1. Ensure the sera used to determine the patient's and donor's blood types are correct.
2. New tools must be used for each compatibility test.
3. The names of both the patient and donor must be accurately recorded on each serum sample.
4. Before injecting blood into the vein, the patient's and donor's names and the accuracy of the blood type must be verified.
5. Blood should be given slowly at the beginning, especially the first 100 ml, and a nurse must be present next to the patient throughout this period to ensure no incompatibility reaction occurs.
6. Symptoms of incompatibility must be quickly recognized and treatment started immediately.

Treatment of Incompatibility

1. Stop the blood transfusion immediately.
2. Inject corticosteroids intravenously.
3. inject Mannitol intravenously (200 ml).
4. Inject calcium intravenously very slowly.

Disease Transmission via Blood

There are many diseases that can be transmitted through blood transfusion. These include AIDS (HIV), Acquired Immunodeficiency Syndrome, Syphilis, and Malaria. Therefore, all necessary tests must be performed before blood transfusion to avoid infection with these diseases, and with the availability of tests, the cost of blood transfusion increases.

Jaundice

Jaundice is transmitted via blood, resulting from the patient being infected with Hepatitis B and C if the donor is infected with one of these viruses, and the donor is in the incubation period where the virus does not appear in the test, or if the test was not accurate.

Other Causes of Jaundice

1. Transmission via contamination of the needle used for blood testing or transfusion, or contamination of the IV set. Therefore, single-use needles and devices must be used.
2. Hemolysis if the blood transfused to the patient was stored for a long time.
3. Hemolysis due to incompatibility, appearing several days after treating the incompatibility and the patient recovering from it.

Determining the Patient's Blood Group

After taking a sample of the patient's blood, the following test is performed:

1. Two drops of the patient's blood are placed on a glass slide with a clear space between them.
2. A drop of Anti-A serum is placed on one drop, and a drop of Anti-B serum is placed on the second drop. Each drop is mixed well with the serum, keeping a distance between them.
3. Clotting (agglutination) is observed at either of the two drops.

If no clotting occurs at all, the blood group is O.

If clotting occurs in both drops, the blood group is AB.

If clotting occurs only with Anti-A serum, the blood group is A.

If clotting occurs only with Anti-B serum, the blood group is B.

There is another type of blood group called the Rh factor, and people are divided into:

Rh Positive (Rh +ve), representing 85% of people.

Rh Negative (Rh -ve), representing 15% of people.

The Rh factor is important for women. If a pregnant woman is Rh -ve and was previously exposed to Rh+ve blood, and the fetus is Rh+ve, a reaction between the mother's and fetus's blood may occur, leading to jaundice in the fetus, which is sometimes serious and may lead to the death of the fetus.

Chapter five

Common Surgical Diseases

Hernia

Definition:

A hernia is the protrusion of viscera from the cavity that contains them, through an opening in the wall of the cavity. The viscera are contained in a sac called the hernial sac.

Types of Hernia:

Hernias can be either internal or external.

Internal Hernia: Occurs inside the body and does not appear on the body surface, such as a diaphragmatic hernia.

External Hernia: Appears on the body surface, such as inguinal or femoral hernia.

There are common types of external hernias, as well as rare types. The common types of hernias are inguinal, femoral, and umbilical. A hernia can be either congenital or acquired.

In congenital hernia, the hernial sac is present at birth, and the hernia may appear at birth or later at puberty. Acquired hernia appears in adults.

Causes of Acquired Hernia:

1. Weakness of abdominal muscles
2. Increased intra-abdominal pressure due to persistent cough, chronic constipation, prostate enlargement, or urinary tract obstruction.

The hernial sac contains intestine or omentum.

Incisional Hernia:

A hernia that occurs after a surgical operation, away from the natural hernia openings. It occurs due to improper suturing of the wound, or due to severe inflammation in the wound or inside the abdomen before or after the operation. Inflammation leads to failure of the abdominal muscles to heal.

Symptoms of Hernia:

1. Appearance of a swelling that increases in size with walking, exertion, and coughing.
2. Pain with exertion due to tension on the hernia contents.
3. The hernia returns inside the abdomen at rest and when lying on the back.

Complications of Hernia

1. Irreducibility: Inability to return the hernia into the abdomen, due to adhesions between the hernia contents or between the contents and the hernial sac.
2. Obstructive Ileus: Due to obstruction of the intestine within the hernia.
3. Strangulation: Due to obstruction of the blood vessels supplying the intestine within the hernia. In this case, the patient needs urgent surgery, otherwise, gangrene occurs in this intestine.

Treatment of Hernia

Surgical operation to remove the hernial sac and reinforce the posterior wall of the inguinal canal, either by repair or by placing a mesh. Before surgery, any cause of increased intra-abdominal pressure such as cough, constipation, and urinary obstruction must be treated. It must also be confirmed that there is no focus of inflammation in the body or inflammation in the hernia. The patient needs an urgent operation in case of obstruction or strangulation.

Thyroid Gland

The thyroid gland is one of the endocrine glands. The thyroid gland secretes thyroid hormones T3 and T4 and plays a fundamental role in cell function.

Causes of Thyroid Gland Enlargement (Goitre) (Image 2):

1. Simple goiter
2. Toxic goiter (Hyperthyroidism)
3. Goiter with thyroid hormone deficiency (Hypothyroidism)
4. Thyroid tumor
5. Thyroiditis (Inflammation of the thyroid)

Simple Goiter:

Occurs due to a relative deficiency of thyroid hormones, either due to iodine deficiency or due to increased demand for hormones in cases of stress and increased physical or psychological burden.

It can be a physiological goiter like that which occurs in girls during puberty. In this case, the thyroid enlargement is symmetrical and non-nodular. Treatment is medical: thyroid hormone (Thyroxine - Eltroxine). In case of iodine deficiency, treatment is with iodine.

Or it can be a nodular goiter (Nodular Goitre) when nodules or granules form in the thyroid gland due to the relative deficiency of thyroid hormones. This is a surgical condition treated with surgery.

Symptoms: Presence of a swelling in the neck. A feeling of suffocation may occur when lying on the back, and a change in voice is possible. Upon examination, nodules can be felt in the absence of symptoms of increased hormone levels. Treatment is partial thyroidectomy in the case of nodular or granular goiter.

2-Toxic Goiter (Hyperthyroidism):

In this case, the enlarged thyroid gland secretes a high percentage of T3, T4 hormones. It can be primary hyperthyroidism, i.e., resulting from increased secretion from the thyroid gland itself, or secondary hyperthyroidism resulting from increased secretions of the pituitary gland. Increased hormones lead to increased pulse and blood pressure, nervous tension leading to insomnia, and protruding eyes with exophthalmos. In primary toxic goiter, the gland enlarges symmetrically and without nodular formation, and there is an increase in pituitary thyroid-stimulating hormones.

Treatment is medical. In secondary toxic goiter, nodules or granules form in the thyroid gland and there is a deficiency in pituitary secretions, and its treatment is surgical. Before surgical operation, the normal level of thyroid hormone must be stabilized. In the surgical operation, partial or subtotal thyroidectomy is performed.

3-Goiter with Hormone Deficiency (Hypothyroidism):

Manifests as slowness in thinking, movement, and pulse, weight gain, and low blood pressure. Treatment is medical by giving thyroid hormone (Elthroxine).

4- Tumors of the Thyroid Gland:

There are benign and malignant tumors of the thyroid gland. Benign tumors appear as simple enlargement of the thyroid gland. Malignant tumors cause additional symptoms due to local pressure on the trachea and recurrent laryngeal nerve. Treatment is surgical with partial excision for benign tumors and total excision for malignant tumors.

Salivary Glands:

Salivary glands are exocrine glands responsible for secreting saliva. They include the parotid glands (two glands), the submandibular glands (two glands), and the sublingual glands, which are numerous. All salivary glands connect via a duct to the oral cavity.

Diseases of the Salivary Glands:

1-Acute Inflammation of the Salivary Gland:

The inflammation may be viral, in which case it affects both glands simultaneously. Its symptoms include enlargement of the glands, pain in the glands, fever, pus, pain when eating, and feeling tired and fatigued. The patient needs treatment with rest, antipyretics, and antibiotics. The patient should consume warm fluids and avoid cold and acidic fluids.

2-Chronic inflammation

Usually results from the presence of stones in the salivary gland or its duct. Symptoms include slight enlargement of the gland size, with pain during swallowing. Stones are more common in the submandibular salivary gland due to food residues in the mouth after eating and the viscosity of this gland's secretions, and because the secretion flow is from the bottom upwards. Stones appear on a plain X-ray. The definitive treatment in this case is surgical excision of the gland. If the stone is in the gland's duct, it may be possible to remove the stone without excising the gland.

Tumors of the Salivary Glands:

Salivary gland tumors appear in the parotid gland more than any other salivary gland. They are mostly
benign tumors. A benign tumor grows slowly without causing any pain. After a period of its appearance, the patient resorts to the doctor complaining of a noticeable swelling above the angle of the jaw. A malignant tumor sometimes leads to paralysis of the facial nerve (CN VII). This paralysis leads to facial asymmetry and inability to close the eyelids. A CT scan can differentiate between benign and malignant tumors. A benign tumor is treated by surgical enucleation with preservation of the facial nerve. A malignant tumor is treated by complete excision of the parotid gland, preserving the facial nerve. If the tumor cells extend into the sheath of the facial nerve, the nerve is excised along with the gland.

Breast Diseases

Mammals are characterized by the presence of breasts. The breast is a modified sebaceous gland designed for the function of lactation. The breast is affected by several diseases, including mastitis, fibrocystic disease, benign tumors, and malignant tumors.

Mastitis (Inflammation of the Breast):

Occurs during lactation in the form of blockage and pain. The inflammation results from blockage in one of the milk ducts with the presence of staphylococcal bacteria. In the first stage, inflammation can be treated with antibiotics and warm compresses for a week. Breastfeeding from the affected breast continues while taking antibiotics that do not affect the infant.

Breast Abscess:

Sometimes the inflammation progresses to abscess formation. In this case, the abscess must be opened under local anesthesia and cleaned well. The patient needs antibiotics with continuous dressings.

Fibrocystic Disease of the Breast (Fibrocystic Changes):

Affects many patients at a young age. Sometimes in the pre-menopausal period or after it. It occurs due to abnormal sensitivity of the breast to hormonal changes. It appears in the form of granules in the breast. The disease usually affects both breasts, but appears in one before the other. The patient suffers from breast pain that increases in the last week before menstruation. A swelling appears

in the breast which is painful on pressure, and secretions may discharge from the nipple. Cysts form in the breast. Diagnosis is confirmed by imaging (mammography/ultrasound). In case of a swelling suspicious in nature, the tumor is completely excised and sent for pathological analysis.

Benign Tumors

Usually occur in young women before the age of forty. The tumor may be multiple, meaning more than one tumor exists at the same time. The tumor is painless. Its size is easily determined within the breast and it moves easily within the breast tissue, due to its existence inside a capsule separating it from the adjacent breast tissue. The tumor does not cause pain except when pressed. If left, this tumor enlarges. Therefore, this tumor must be excised.

Breast Cancer

This tumor affects 1 in 16 women. Its incidence increases if the disease is present in the mother or sister, or in case of not breastfeeding children naturally, or if the tumor appears in one breast, which increases the probability of its appearance in the other breast. These are considered risk factors. The tumor initially appears as a small, ill-defined mass and moves easily within the breast tissue, then it infiltrates the adjacent tissues and thus becomes difficult to define its edges or move it within the breast. Then the tumor metastasizes to the axillary lymph nodes via the lymphatic vessels. Then it metastasizes to other organs in the body upon the arrival of cancer cells into the bloodstream. Diagnosis depends on taking a biopsy and its pathological analysis. The most accurate biopsies are the surgical biopsy (where the entire tumor is excised for analysis) or the core needle biopsy. It is necessary to perform imaging on both breasts, an ultrasound on the abdomen, and a bone scan.

Treatment: Treatment is determined according to the stage of the tumor.

First: Surgical Treatment:

1. Conservative excision of the tumor with surrounding tissues and partial excision of the axillary lymph nodes.
2. Complete excision of the breast and axillary glands (Mastectomy with axillary clearance).

Complementary Treatment:

Surgical treatment is complemented by the following treatments:

1. Chemotherapy if there is enlargement of the axillary lymph nodes.
2. Local radiotherapy on the tumor site or the excised breast and on the bones in case of tumor metastasis to the bones.

Esophagus:

The esophagus is 25 cm long and connects the pharynx to the stomach. It passes through the neck, then the chest, then through the diaphragm to the abdomen. The length of the esophagus inside the abdomen is1- 2cm. It transports food and drink to the stomach. It consists of a mucous membrane lining and muscles that work to move the cavity contents towards the stomach.

Cardiac Achalasia (Image 3)

Cardiac Achalasia occurs due to the failure of the lower esophageal sphincter muscle to relax upon eating food. This is due to atrophy of the nerves of the lower esophageal muscle. The patient suffers from three symptoms: difficulty swallowing, regurgitation of food into the pharynx, and weight loss. The disease appears on a barium swallow X-ray as severe narrowing at the lower end of the esophagus with dilation of the esophagus above the narrowing ("bird's beak" appearance) and it also appears on endoscopy.

Treatment: Medical (dilatation) first in early cases, then surgical treatment in advanced cases.

Surgical Treatment:

Surgical splitting of the lower esophageal muscle (Heller's cardiomyotomy) or the same procedure using surgical endoscopy.

Reflux Esophagitis (GERD):

Occurs due to relaxation of the lower esophageal sphincter, which causes reflux of stomach acid into the esophagus. The reflux causes inflammation of the lower esophagus.

Symptoms: Heartburn and pain behind the sternum. The pain increases in the evening, after eating, and after eating fatty, heavy foods. Regurgitation of acid and stomach contents into the mouth. If the patient is not treated, stricture may occur in the lower esophagus, or fibrosis. In cases of increased fibrosis, shortening of the esophagus results. Malignant transformation (Barrett's esophagus leading to adenocarcinoma) may occur in the lower esophagus.

First: Medical Treatment:

1. Abstinence from smoking, coffee, citrus fruits, chocolate, and fatty foods.
2. Eating light meals throughout the day, avoiding hunger, and avoiding fullness and eating fatty and spicy foods.
3. Eating at least two hours before sleep, with dinner being light.
4. Antacid medications, prokinetics (e.g., Metoclopramide), and medications to reduce stomach acid secretion (Zantac or Losec/Omeprazole).

Second: Surgical Treatment:

To restore the normal length of the esophagus in the abdomen with a "Nissen" fundoplication operation, which can be performed by open surgery or laparoscopy.

Esophageal Tumors

Benign Tumors

1. Muscular tumors (Leiomyoma): Mostly found in the middle and lower third. They cause difficulty swallowing. Surgical treatment is excision if they are larger than 5 cm.
2. Esophageal cysts: Diagnosed in children during the first year. In adults, they do not cause symptoms unless inflammation or bleeding occurs inside the cyst.

Malignant Tumors (Image 4)

Occur more in men than women. Symptoms appear as difficulty swallowing, pain on swallowing, severe weight loss, and bloody vomiting may occur. Diagnosis depends on performing an esophageal endoscopy, taking a biopsy, and pathological analysis of the biopsy.
Treatment: Excision of the esophagus (esophagectomy) through an opening in the diaphragm or through the chest (transthoracic).

Stomach

The stomach consists of the fundus, body, and antrum. It begins at the cardia and ends at the pylorus. The stomach digests proteins by secreting hydrochloric acid and the enzyme pepsin. The stomach also plays an important role in iron digestion.

Gastric and Duodenal Ulcers:

Gastric and duodenal ulcers occur due to erosion of the layer that protects the stomach wall from stomach acid, and therefore some cases of gastric ulcer occur with increased stomach acid secretion and some cases with decreased stomach acid secretion. The erosion of this barrier occurs due to a number of factors such as smoking, excessive coffee consumption, irregular eating, nervous stress, and due to bacterial infection with H. pylori bacteria, which is present in most cases of gastric ulcer and must be treated along with the ulcer treatment to avoid recurrence of the gastric ulcer (Images 5 and 6).

Symptoms:

1. Pain in the upper abdomen that appears with eating.
2. Pain upon pressure on the upper abdomen.

Examinations:

1. Endoscopy of the esophagus, stomach, and duodenum with biopsy if an ulcer is present.
2. Barium meal X-ray.

Treatment:

Antibiotics

Antihistamines: Zantac (Ranitidine) and Famotec.

Proton pump inhibitors: Losec.

Triple therapy for H. pylori: Losec (or Zantac) with Flagyl with Erythromycin or (Amoxicillin).

Surgery: In cases of:

1. Failure of drug therapy for one year.
2. Patient non-compliance with treatment.
3. Occurrence of complications: Gastric outlet obstruction, ulcer bleeding, or ulcer perforation.

Stomach Tumors

1. Benign tumor: From the mucous membrane, can be excised by endoscopy or surgery.
2. Stomach Cancer (Gastric Carcinoma)

Symptoms:

Persistent, ill-defined pain, weight loss, nausea, and vomiting.

Examinations:

1. Blood picture: Anemia.
2. Barium meal X-ray.
3. Gastroscopy with biopsy.

Treatment:

1. Surgery to excise the tumor, part of the stomach (partial gastrectomy) or the entire stomach (total gastrectomy) and connect the remaining part to the intestine, or connect the esophagus to the intestine in case of total gastrectomy.
2. Chemotherapy: After surgery, especially in the presence of secondary tumors (metastases).

Colon

The colon extends from the small inte

stine to the rectum and consists of the ascending, transverse, descending, and sigmoid colon. The colon is the organ responsible for absorbing water and electrolytes, especially sodium, and absorbing bile salts. Urea is broken down in the colon by the action of colon bacteria and is absorbed in the colon.

Inflammatory Bowel Disease (IBD - Ulcerative Colitis)

The disease affects patients around the age of fifteen in 15% of cases and may occur later at an older age.

Symptoms:

Appears as recurrent bloody diarrhea, abdominal pain with weight loss. It is sometimes accompanied by manifestations in the skin, joints, and eyes. In 7% of cases (likely typo, perhaps 15-25% for severe flare), the disease appears in an acute form and leads to death if urgent surgical intervention is not performed. In the acute state, obstruction of the colon, severe abdominal distension, and high fever occur.

Treatment:

1. Avoid dairy products.
2. Salazopyrin, its dose determined by the degree of disease progression.
3. Prednisone (a corticosteroid derivative), especially in acute cases.

Surgery in cases of:

1. Severe bleeding.
2. Acute inflammation with colon perforation.
3. Colon obstruction.
4. Appearance or suspicion of cancer development.
5. Lack of response to treatment.

Surgery: Complete removal of the colon (colectomy) with connection of the small intestine to the rectum, or complete removal of the colon and rectum (proctocolectomy) with permanent diversion of stool flow (ileostomy).

Colon Cancer

More than 70% of colon tumors affect the descending colon. Symptoms depend on the size and location of the tumor. Tumors of the left colon cause colon obstruction and bleeding. They may cause acute intestinal obstruction. Tumors of the right colon cause weight loss, loss of appetite, and anemia due to microscopic bleeding, and appear as a palpable abdominal mass. Diagnosis is confirmed by the result of colonoscopy and biopsy.

Treatment:

1. Surgical by excising the part of the colon carrying the tumor. The patient may need a temporary colon diversion - colostomy.
2. Chemotherapy.
3. Radiotherapy.

Colostomy

It is a diversion of the stool flow and is performed in cases:

1. Temporary diversion in case of acute colonic obstruction, where the colon cannot be reconnected in the presence of stool. Therefore, diversion is done temporarily until the colon is prepared for reconnection in a second stage.

2. Permanent diversion in case of removal of the colon and rectum due to a malignant tumor in the colon. A colostomy can be performed in the right or left colon.

Complications of Colostomy:

1. Obstruction.
2. Prolapse of the colostomy into the abdomen.
3. Hernia around the colostomy.

Hemorrhoids

They are varicose veins and congestion of the anal veins. They appear in the form of bleeding, a feeling of heaviness, or prolapse through the anus. Prolapse appears first with straining and disappears after defecation. Then it becomes permanent prolapse afterwards (Image 7). There are four grades of hemorrhoids. The first and second grades can often be treated with medical therapy. The third and fourth grades need surgical treatment.

Complications:

1. Acute bleeding.
2. Strangulation.
3. Ulceration.

Medical Treatment:

For Grade I and II in the absence of bleeding:

1. Proctoglyvenol or Anusol (ointment or suppository).
2. Daflon tablets.
3. Avoid constipation.

Surgical Treatment for cases of Grade III and IV and for bleeding hemorrhoids.

Anal Fissure

Occurs due to severe constipation with straining during defecation.

Symptoms: Sharp pain during defecation, and severe constipation.

Treatment:

1. Analgesic to overcome the pain.
2. Xylocaine ointment topically to relieve pain.
3. Lactulose or Duphalac to overcome constipation.

Surgical treatment in case of no improvement with medical treatment for three weeks, or upon the occurrence of anal stenosis, or the transformation of the fissure into a chronic fissure.

Gallbladder

The gallbladder is a store for the bile secreted by the liver. It is located under the right lobe of the liver and is ٧،٥-12cm long. The gallbladder empties its bile into the bile duct via the cystic duct. The bile duct empties its bile into the duodenum after joining with the pancreatic duct. The size of the gallbladder is about 5 cm.

Gallstones:

Gallstones are formed from the deposition of cholesterol, calcium, or carbonates around a small nucleus in the center formed by mucus debris surrounding bacteria. Gallstones form due to an increased percentage of cholesterol in the bile, increased breakdown of red blood cells, or the occurrence of inflammation.

Gallbladder Inflammation (Cholecystitis):

Stones may exist without causing symptoms, or they may cause inflammation of the gallbladder. In the absence of symptoms, observation and follow-up of the patient are sufficient. If inflammation occurs, it can be chronic or acute.

Symptoms of Calculous Inflammation:

1. Pain in the right upper quadrant under the ribs, and the pain increases with eating high-fat foods. It may be accompanied by a feeling of nausea and may cause vomiting.
2. Symptoms of Acute Inflammation: Severe pain with high fever, and a feeling of nausea and vomiting. Complications may occur in case of acute inflammation.

Complications of Calculous Gallbladder Inflammation:

1. Stone obstruction of the bile duct (Choledocholithiasis).
2. Acute peritonitis.
3. Acute cholecystitis (Empyema/Gangrene).
4. Acute inflammation of the bile ducts (Cholangitis).

Complications of Acute Inflammation:

1. Gallbladder abscess if obstruction of the cystic duct occurs.
2. Acute peritonitis if the inflammation spreads to the peritoneal membrane.
3. If a rapid surgical operation is not performed, gallbladder rupture may occur, leading in this case to acute peritonitis.

Examinations: Ultrasound on the abdomen in addition to routine tests.

Treatment: Removal of the gallbladder (cholecystectomy) is the only way to treat symptoms and complications, and can be done by open surgery or laparoscopy.

Bile Duct Obstruction:

The main causes of bile duct obstruction are the presence of stones in the bile ducts and the occurrence of strictures in the bile ducts.

Symptoms:

1. Yellowing of the eyes and skin (Jaundice).
2. Dark red (tea-colored) urine and whitish (clay-colored) stool.
3. Loss of appetite, and the patient complains of nausea and fatigue.

Examinations:

Analysis of bilirubin level in the blood shows an increase in the direct bilirubin percentage and an ultrasound on the abdomen. Accurate diagnosis requires endoscopy of the biliary and pancreatic ducts (ERCP).

Treatment:

Surgical operation to remove the gallbladder (cholecystectomy) and explore the common bile duct (CBD exploration). After exploration, the bile duct is connected to the duodenum (choledochoduodenostomy) or a T-tube is placed in the bile duct. This tube is withdrawn after confirming (by X-ray) that the bile duct is free of stones.

Pancreas

The pancreas is a mixed gland (endocrine and exocrine) located inside the curve of the duodenum behind the stomach. The endocrine part secretes the hormones insulin, glucagon, and somatostatin. The exocrine part secretes enzymes that pour through the pancreatic duct into the duodenum and help digest proteins.

Acute Pancreatitis:

One of the causes of acute abdominal pain. The main causes of acute pancreatitis are the presence of stones in the bile ducts (gallstones), alcohol consumption, obstruction of the pancreatic duct, high blood calcium levels, viral inflammation, and external injury.

Symptoms:

1. Severe abdominal pain, nausea, and vomiting.
2. High temperature, pain, and abdominal distension.
3. Low blood pressure and fluid loss in the blood.

Tests

1. CT scan of the abdomen.
2. Elevated blood amylase level

Treatment:

1. Conservative treatment: intravenous fluids, subcutaneous Sandostatin, antipyretics, Zantac and intravenous calcium.
2. Surgical treatment in the event of a pancreatic abscess or acute pancreatic erosion.
3. Delayed treatment if the cause is gallstones. The cholecystectomy is performed after recovery from the acute condition, i.e., after three or four weeks.

Pancreatic Tumors

Malignant tumors of the pancreas are more common than benign tumors. If the tumor arises in the head of the gland, it appears early due to obstruction of the cystic duct and the appearance of obstructive bile. If it arise in the neck of the gland or in the tail of the pancreas, it appears late due to the absence of bile.

Symptoms:

1. Obstructive jaundice, if the tumor is in the head of the gland near the pancreatic duct or bile duct.
2- Abdominal pain
3-Loss of appetite and weight loss

Tests

1. CT scan of the abdomen.
2. Endoscopy of the bile ducts and pancreas
3. Blood analysis to determine bile levels

Treatment:

1. Tumor removal
2. radiation or chemotherapy
3. If the tumor is difficult to remove, the goal of treatment is to treat pain and jaundice and obtain a sample of the tumor for analysis.

Liver

Liver Abscess

1- Purulent abscess

causes:

1. Inflammation of the bile ducts
2. Spread of the microbe through the blood
3. General blood poisoning
4. Direct spread of peritoneal inflammation
5. There are other causes such as liver injury.

2- Amoebic abscess:

It occurs due to the amoeba reaching the liver from the colon through the pyloric circulation.

Symptoms:

1- High temperature.
2- Nausea and vomiting.
3- pain .
4- Hepatomegaly.

Tests

1- Elevated white blood cell count
2-Ultrasound
3- CT scan

Treatment:

1- Intravenous antibiotics for two weeks
2-Draining the abscess with a percutaneous catheter under ultrasound guidance
3- Surgical drainage

Liver Tumors:

1. Benign tumors: Hemangiomas or adenomas, treated by tumor excision.
2. Malignant tumors: Causes include exposure to food toxins, pesticides, and cirrhosis.

Symptoms:

1. Weight loss
2. Pain
3. Hepatomegaly (Liver enlargement)
4. Jaundice
5. Ascites

Investigations:

1. Liver function tests
2. CT scan

Treatment:

1. Surgical resection
2. Liver transplant

Spleen

Causes of Splenomegaly (Enlarged Spleen):

1. Infection: viral or parasitic.
2. Blood disease: malignant and immune anemias, and leukemia.
3. Portal vein obstruction
4. Spleen tumors and cysts

Reasons for Splenectomy (Spleen Removal):

1. Spleen injury in an accident or surgically.
2. In cases of sickle cell anemia or pernicious anemia.
3. Platelet destruction .
4. Along with stomach resection in cases of stomach cancer.
5. Staging of lymph node tumors.

Splenectomy before the age of four is sometimes followed by severe infections, particularly by the pneumococcus bacteria. Therefore, if it is necessary to perform a splenectomy in children under four years old, pneumococcal vaccine must be administered.

Bowel Obstruction (Ileus)

It is the cessation of the passage of gastrointestinal contents in the normal direction. It can occur due to a mechanical obstruction or due to intestinal paralysis (paralytic ileus). Obstruction has four main symptoms: pain, vomiting, distension, and constipation.

Causes of Mechanical Obstruction:

1. Congenital defect in intestinal formation: congenital atresia or absence of part of the intestine.
2. External obstruction: strangulated hernia - volvulus - intussusception - adhesions.
3. Obstruction of the intestinal lumen: tumor.

Causes of Intestinal Paralysis (Paralytic Ileus):

1. After abdominal surgeries
2. Acute peritonitis
3. Hypokalemia (low blood potassium)
4. Intestinal circulatory insufficiency (ischemia)
5. Deterioration of the patient's general condition: renal or liver failure - chemotherapy.

Large Bowel (Colon) Obstruction

Occurs due to a colon tumor or due to ulcerative colitis.

Diagnosis of Intestinal Obstruction:

1. Symptoms: pain, vomiting, bloating, constipation.
2. Investigations: Plain X-ray in upright position, barium enema X-ray in case of colon obstruction.
3. Blood test: sodium and potassium, blood gases.

Treatment:

1. Correction of fluid and electrolyte imbalances in the blood.
2. In case of mechanical obstruction, treatment is surgical according to the case: hernia repair, volvulus reduction, or tumor resection and bowel resection in case of gangrene.
3. In case of paralytic ileus: IV fluids + NPO (nil by mouth) + nasogastric tube insertion.

Peritonitis

It is an inflammation of the peritoneal membrane lining the abdominal viscera. Inflammation occurs due to bacteria reaching the peritoneal membrane, usually gram-negative bacteria, aerobic and anaerobic.

Causes:

1. Leakage of stomach and intestinal contents into the peritoneal cavity after gastrectomy, intestinal, and colon resections.
2. Without a clear cause, such as cases of primary peritonitis in hepatic or renal ascites.
3. Perforation in the stomach or intestine: perforated gastric ulcer or intestinal gangrene or diverticulum.
4. Bacterial spread without perforation: acute appendicitis, or intestinal circulatory insufficiency (ischemia).
5. Bacteria reaching from the outside: surgical contamination.
6. Bacteria reaching via the fallopian tubes: pelvic inflammatory disease (PID).
7. Bacteria reaching via the blood: septicemia.

Symptoms

1. High fever with chills and rigors.
2. Low blood pressure with increased pulse rate.
3. Dry tongue and frequent vomiting.
4. Severe abdominal pain.
5. Vomiting, constipation, and abdominal distension.

Investigations:

Tests to identify the cause, and a blood count shows a significant increase in white blood cells.

Plain X-ray of the abdomen, to show paralytic ileus.

Treatment:

1. Correction of the patient's general condition: with IV fluids, NG tube, antibiotics, analgesics, blood and plasma transfusion.
2. Treating the cause.

Chapter Six

Injuries

Injuries are a major cause of death. The treatment of accident victims begins at the accident site and ends with discharge from the hospital with rehabilitation and physical therapy. The success of treatment depends on correct diagnosis and on prioritizing the management of the patient and the treatment.

Causes and timing of death after injuries:

1. Immediate death: (50%) due to severe injury to the brain, spinal cord, or heart.
2. Early death: (30%) in the few hours following the injury. These are considered the "golden hours" during which the victim can be saved and death prevented. Death occurs due to airway obstruction, respiratory system injury, or shock.
3. Late death: (20%) after days or weeks. Due to infection or organ failure.

Diagnosis is based on a patient assessment process:

Primary Assessment: To assess the severity of the condition and threat to the patient's life
Secondary Assessment: For a comprehensive assessment of all injuries

Primary Assessment

Known in English by the letters ABCDE which stand for:
Airway
and Breathing
and Circulation
and Disability (Neurological status)
and Exposure & environment (Exposing the patient and controlling the environment)

Method for primary patient assessment:

١-Airway: Open the airway using a chin lift and observe chest movement and air entry/exit from the nose.

2 - Pulse: A rapid weak pulse indicates shock due to bleeding. A slow weak pulse indicates neurogenic shock.

3 - Observe injury sites before moving the patient, so that incorrect movement does not cause a serious injury.

If a neck vertebra fracture is suspected, it should be immobilized using a cervical collar, and the patient should not be moved until it is applied.

4 - Remove any remaining clothes from the patient and maintain their temperature using blankets and warming the surrounding environment (heaters or air conditioning).

Secondary Assessment

A comprehensive assessment of the patient's condition to avoid neglecting a serious injury. It is done as follows:

1. Palpate the scalp to observe bruises, contusions, and skull fractures.
2. Examine the pupil for dilation, response to light, and equality on both sides.
3. Examine the ears, nose, and mouth: bleeding or clear fluid discharge (cerebrospinal fluid) means a base of skull fracture.
4. Examine facial bones then the neck to detect fractures. The patient should not be moved if a neck fracture is suspected.
5. Examine the chest.
6. Examine the abdomen and pelvis.

7. Examine the vertebrae and spine, taking care with the patient rolling technique.
8. Perform necessary tests and scans according to the patient's condition. Generally, in case of multiple injuries, the examination includes basic scans: neck X-ray, chest X-ray, pelvic X-ray, and ultrasound on the abdomen and pelvis, in addition to a head CT scan. Then the patient is transferred for X-rays on the limbs and back if fractures are suspected there.

Triage

If more than one patient is injured in an accident, cases are divided according to priorities into highest priority, medium priority, and low priority. They are transferred to the hospital and treated according to the priority order.

Highest Priority : Patient suffering from:-

1. Difficulty breathing
2. Severe shock
3. Acute bleeding
4. Open wounds to the abdomen or chest
5. Shock due to acute heart failure, heart attacks, or diabetic complications
6. Cardiac arrest
7. Severe head injuries

Medium Priority : Patient suffering from

1. Burns
2. Multiple major fractures
3. Fractures affecting the spinal cord

Third Priority (Low):

1. In case of victim's death.
2. Minor and superficial wounds, injuries, and burns that do not require hospital admission, intensive care, or surgical or thoracic intervention. Assessment tables (see table)

Head, Brain, and Nervous System Injuries

If a patient is involved in an accident, they must be asked about their state of consciousness after the injury. If the patient lost consciousness, even for moments after the accident, they must be considered suspected of having a brain concussion. In this case, they must be placed under observation for 24 hours. Observation is done by monitoring the Glasgow Coma Scale.

The Glasgow Coma Scale (GCS) is calculated by observing eye opening, motor response, and verbal response. If the Glasgow Coma Scale result is 8 or less, the patient has a severe nervous system injury and needs endotracheal intubation and placement on a ventilator.

Glasgow Coma Score

Component	Points
Eye Opening	
Spontaneous	4
To Voice	3
To stimulation	2
None	1
Motor Response	
To command	6
Localize	5
Withdrawal	4
Abnormal flexion	3
Extension	2
None	1
Verbal Response	
Oriented	5
Confused but comprehensible	4
Inappropriate or incoherent	3
incomprehensible (no words)	2
None	1

Types of Head and Skull Injuries
First: Head Injuries
1-Laceration:
Lacerations on the scalp bleed heavily, so the wound must be sutured to stop the bleeding. Suturing is done under local or general anesthesia depending on the number and length of wounds. Until suturing is done, a sterile gauze dressing should be placed on the wound and secured with a pressure bandage to stop the bleeding.
2-Scalp Contusion:
Must be differentiated from a skull fracture by performing plain or CT scans of the head. Treatment of the contusion is conservative with observation to ensure it decreases in size and does not increase.
Secondly: Skull Injuries
Can be classified into three types:
1-Fracture of facial and jaw bones
Appears as swelling, bleeding, or deformity of the nose, mouth, and upper and lower jaw. The presence and type of fracture are confirmed by plain X-ray or CT scan. Some fractures are treated conservatively, while others are fixed surgically.
2-Skull fracture
There are multiple types of skull fractures. It is divided into simple fracture without a scalp wound and compound fracture where there is a scalp wound with skull bone fracture. It is classified according to the degree of depression of the fractured skull bone into linear fracture without depression, and depressed fracture accompanied by depression of the skull bones.
3-Basilar skull fracture

Often results from severe injuries affecting the hard bones of the skull base. It leads to leakage of cerebrospinal fluid from the nose or ear. It may lead to injury of the seventh nerve (facial nerve) or the eighth nerve (auditory and balance nerve). If the leakage persists, surgical intervention is necessary, otherwise the patient is at risk of meningitis.

Third: Brain Injuries

1. Concussion stage: A short stage lasting seconds or minutes. The patient temporarily loses consciousness. Usually, the patient complains of not remembering the injury, not knowing what happened to them, or how they came to the hospital.
2. Post-concussion stage. This is a transitional period, after which the patient either returns to their normal state, or shows signs and symptoms of increased intracranial pressure. Therefore, the patient must be observed, and necessary scans performed during this stage.

Brain Hemorrhage

First: Bleeding outside the brain tissue:

1-Epidural Hematoma (Bleeding outside the outer brain membrane - Dura mater)

Occurs due to a tear in the middle meningeal artery, causing immediate and severe bleeding (Image 8 in appendix). The patient usually enters a temporary period of improvement (lucid interval), and then lapses into coma, and signs of increased intracranial pressure appear. Diagnosis is clinical and via CT scan, and the patient needs rapid surgical intervention, which often achieves good results (Image 9 in appendix).

2-Subdural Hematoma (Bleeding inside the outer brain membrane - Dura mater)

Usually occurs in the elderly or due to moderate or minor trauma, and the bleeding may be immediate and associated with epidural bleeding. However, the bleeding is often chronic and appears days or weeks after the trauma. If symptoms are severe and accompanied by specific deficits in brain function, with a large volume of bleeding, the patient needs surgery to evacuate the hematoma. If the symptoms are mild, general, and non-localized, and the bleed volume is small and not compressing the brain, and does not cause brain shift, the patient is treated conservatively, with good results (Image 10 in appendix).

Secondly: Intracerebral Hemorrhage (Bleeding inside the brain tissue)

Diagnosis of intracerebral hemorrhage depends on the trauma patient remaining in a coma from one hour after the trauma, or for 6 hours after the trauma, with seizures, and the presence of a skull fracture. The patient shows symptoms of increased intracranial pressure. Diagnosis relies on CT scan. The patient needs conservative or surgical treatment depending on the size, location of the hemorrhage, and the degree of its effect on intracranial pressure.

Neck Injuries

The neck is anatomically divided into a number of zones.

Zone I: The thoracic inlet zone, located between the thoracic inlet and the laryngeal cartilage. The most dangerous injuries occur here, to the great vessels. Exploration is necessary if there are signs of hemorrhage.

Zone II: Located between the laryngeal cartilage and the angle of the jaw. It is the most exposed area of the neck, and the least dangerous.

Zone III: Above the angle of the jaw. Bleeding in this area may lead to hemorrhage at the base of the skull.

Diagnosis of Neck Injuries

If the patient is conscious, the neck is examined for injuries, wounds, or contusions. If none are found, the patient is asked to move their neck under medical observation. Moving the neck without pain means no neck fracture, and a neck X-ray can be avoided. If the patient is unconscious or not fully conscious or feels neck pain upon movement, the neck remains immobilized and an X-ray of the neck is performed. The neck should be palpated to rule out pain, hematoma, or air under the skin. If vascular injury is suspected, a Doppler ultrasound of the neck vessels is advised.

If there is a suspicion of an esophageal injury, it is recommended to perform a CT scan of the neck.

Chest Injuries

1-Rib Fractures and Cracks

Rib fractures can be simple and involve one or more ribs, or they can be severe and dangerous. The patient complains of pain with breathing. Diagnosis depends on performing plain X-rays of the chest: PA and lateral views. Treatment involves prescribing analgesics and advising rest. The rib is not bound or strapped, so as not to impede natural rib movement and not to affect breathing.

2-Pleural Cavity Injuries

The pleura is a thin membrane surrounding the lung, consisting of two layers, between which there is a serous fluid in a thin layer. Thanks to this layer, the lung can move, contracting and expanding with exhalation and inhalation. The pleura separates the lung from the chest cavity, allowing it to move easily. If the injury penetrates the pleura, air enters the pleural cavity from the outside, resulting in a collection of air in the pleural cavity (Pneumothorax), or bleeding occurs due to lung injury or injury to a blood vessel. This results in a collection of blood in the pleural cavity (Hemothorax).

Pneumothorax (Air collection in the pleural cavity)

Occurs due to a penetrating chest wound. There are two types:

1. Open Pneumothorax: Allows air to enter and exit.
2. Tension Pneumothorax: Allows air to enter but not exit, thus the air collection increases progressively, and pressure increases in the pleural cavity, leading to displacement of the heart and great vessels to the opposite side, threatening the patient's life.

Symptoms:

In case of pneumothorax, the lung collapses and air entry decreases. Decreased air flow on the affected side is noted using a stethoscope. Plain X-ray of the chest shows the air collection and lung collapse. In case of bleeding, the hemothorax, or hemopneumothorax (air and blood) appears, along with lung collapse.

Treatment:

Insertion of a chest tube (Image 11 in appendix), done under local anesthesia, urgently, to save the patient's life. In case of tension pneumothorax, the chest wound must be covered and closed.

The chest tube is inserted in the area between the 6th and 7th ribs to drain an air or blood collection. If urgent drainage of a pneumothorax is needed, the tube is inserted between the 2nd and 3rd ribs. The tube is fixed with a suture to the skin and covered with gauze and plaster. The tube is connected via tubing to a sealed bottle, from which a tube exits to vent air. The tube is removed when:
a) Air and blood stop draining from it
b) The patient's symptoms disappear
c) Plain chest X-ray shows the lung expanded and clear, and the air and fluids have disappeared from the pleural cavity.

3-Lung Injuries:
Sometimes a lung injury occurs in the form of laceration or contusion. The injury leads to decreased oxygen levels in the blood (hypoxemia). Diagnosis depends on blood gas analysis, and CT scan of the lung if available. The patient is treated by reducing crystalloid IV fluids (like saline) and giving diuretics; colloidal fluids (like Hespan/Hetastarch) are given. If blood oxygen drops below92%, it is preferable to intubate the patient and place them on a mechanical ventilator.

Chest Tubes

Indications for chest tube insertion:
1. Pneumothorax (Air in the pleural cavity)
2. Hemothorax (Blood in the pleural cavity)
3. Empyema (Pus in the pleural cavity)
4. After chest surgeries

Complications of chest tube insertion:
1. Bleeding during insertion.
2. Infections in the chest cavity.
3. Persistence of hemothorax or pneumothorax.

The chest tube is removed when:
1. Drainage of fluids stops for 24 hours
2. Air leak stops for 24 hours
3. Air flow returns to normal during breathing in the lung
4. The lung returns to its normal position as seen on plain X-ray
5. The tube is clamped for 24 hours and an X-ray is repeated before tube removal

After tube removal, breathing must be monitored, and a repeat plain chest X-ray is done after 24 hours.

Abdominal Injuries
If a patient suffers a blunt abdominal injury, a careful abdominal examination is necessary to diagnose internal bleeding (hemorrhage) and intestinal injuries. An ultrasound of the abdomen and pelvis must be performed in case of any direct abdominal injury. Diagnostic Peritoneal Lavage (DPL) is required in case of:
1. Low blood pressure (hypotension)
2. Patient being unconscious

If the patient's condition is stable, a CT scan of the abdomen is performed.
All cases of penetrating abdominal wounds, whether by a sharp object or gunshot, require surgical exploration, as does the presence of air under the diaphragm on a plain abdominal X-ray in an upright patient. If a patient arrives at the hospital with a penetrating abdominal wound from which viscera are protruding outside the peritoneal cavity, the viscera must be covered with gauze soaked in warm saline solution. The viscera should not be pushed back into the peritoneal cavity, and the patient must be prepared for surgery as soon as possible.
In all cases of abdominal injury requiring surgical intervention, a Ryle's Tube (NG tube) and a Foley's catheter must be inserted. In case of abdominal injury among multiple injuries, the known sequence ABCD must be followed before diagnosing and treating abdominal injuries.

Spleen Injury:
Diagnosis is based on the presence of injury to the 9th and 10th ribs on the left side with pain upon palpation, and low blood pressure. Diagnosis is confirmed by ultrasound and CT scan. In case of spleen injury with persistent signs of bleeding, splenectomy is performed.

Liver Injury:
Diagnosis is based on the presence of injury to ribs 7 to 10 on the right side with pain upon palpation and low blood pressure.

Bowel Injury:
Diagnosis is based on the presence of a penetrating wound or abdominal pain that increases upon palpation, and the presence of air under the diaphragm visible on a plain abdominal X-ray with the patient upright.

Pelvic Injuries:
A pelvic fracture may be stable or unstable. A pelvic fracture can lead to bleeding or injury to the urethra, rectum, reproductive system, or placenta in pregnant women.
A pelvic X-ray must be performed in all cases of multiple injuries, even in the absence of signs of injury. An orthopedic specialist must be consulted to determine the patient's need for fixation or urgent surgery. The possibility of a urethral injury must be considered before inserting a urinary catheter.

Signs of urethral injury:
1. Bleeding from the urethra
2. Presence of an anterior pelvic fracture
3. Large bruise on the pubis or scrotum
4. Diagnosis is confirmed by performing a retrograde urethrogram.

In case of a urethral injury, a suprapubic cystostomy is performed to divert urine.

Bladder Rupture
Manifests by the presence of blood in the urethra or the patient's inability to urinate, with the appearance of a painful swelling in the lower abdomen. Diagnosis is confirmed by a cystogram. A urinary catheter can be inserted in case of bladder injury with an intact urethra. Severe injuries require surgery.

Chapter Seven
Laser Surgery

Definition of Laser:
Laser is a beam of concentrated light resulting from the use of radiation. Laser is used in various applications in medicine and different fields.
In surgery, laser is used for:

- Cutting tissues
- Destroying tumors and pathological tissues
- Cauterizing blood vessels to stop bleeding

Laser is described as the "light scalpel" as it is used to cut tissues remotely, precisely, and without bleeding. It is used in the following medical specialties:
Ophthalmic surgery: to treat retinal diseases and glaucoma
Plastic and dermatological surgery: to remove tumors, skin growths, excess hair, and treat pigmented skin lesions
Cardiac catheterization: to treat coronary artery blockages
ENT surgery: to remove benign laryngeal tumors, growths, and nasal bleeding
Dentistry: to treat cavities and gum diseases
Brain and nerve surgery: to remove tumors
Orthopedic surgery : to treat inflammation of the uterine lining (endometriosis)
Urological surgery: to treat prostate enlargement
Chest surgery: to treat bronchial tumors
Gynecological surgery: to treat some cases of ectopic pregnancy and breast tumors
Advantages of Laser Surgery:

1. Reduced bleeding
2. Reduced inflammation
3. Performing surgery without large wounds or through minimal intervention via small incisions.

Disadvantages of Laser Surgery:

1. High price and cost of the laser device
2. Necessity of trained medical staff to protect themselves and patients from laser complications
3. Misuse or misdirection of the laser beam can cause damage to the skin or tissues around the surgical site
4. Bleeding due to mistakenly directing the laser beam to blood vessels

Laparoscopic Surgery

Laparoscopic surgery is the use of a scope and special surgical instruments inserted through small openings to perform the same traditional surgical operations, with the same steps except for the wound opening steps. It differs from gastrointestinal endoscopy.
Main fields of laparoscopic surgery:

1. Laparoscopy (Abdominal scope surgery)
2. Thoracoscopy (Chest scope surgery)
3. Arthroscopy (Knee joint scope surgery).

Principles of Laparoscopic Surgery:

1. Laparoscopic surgery relies on creating a surgical cavity (space).
2. Carbon dioxide gas is used to insufflate the abdomen with a quantity of gas around 4-6liters to create this cavity, ensuring the pressure inside the abdominal cavity does not exceed 15 mmHg.
3. The scope is inserted into the abdomen, connected to a camera, and the image is displayed on aregular television screen or monitor.

Advantages of Laparoscopic Surgery:

1. Performing surgery through small incisions is better cosmetically
2. Reduced wound complications
3. Reduced post-operative pain
4. Shorter hospital stay
5. Faster return to normal activity and work
6. Better for overweight patients
7. Image magnification provides better vision for the surgeon

Disadvantages of Laparoscopic Surgery:

1. Requires special equipment and instruments, as well as requires skill, training, and special experience
2. Cost is higher than traditional surgery, although it may equalize in total cost when including early return to work.

There are surgeries where it is confirmed that performing them laparoscopically is better than traditional surgery:

1. Laparoscopic cholecystectomy
2. Repair of diaphragmatic hernia
3. Repair of GERD (Gastroesophageal reflux disease)
4. Thoracic sympathectomy
5. Diagnostic laparoscopy
6. Bariatric (weight loss) surgery by laparoscopy
7. Laparoscopic repair of inguinal and femoral hernia

And there are surgeries that can be performed laparoscopically, although the advantages of laparoscopy do not significantly surpass traditional surgery:

1. Laparoscopic colectomy
2. Laparoscopic gastrectomy
3. Laparoscopic nephrectomy (kidney removal)
4. Laparoscopic adrenalectomy (adrenal gland removal)

And there are surgeries still in the development stage, including:
Laparoscopic esophageal resection, pancreatic resection, bladder resection, and spinal fusion by laparoscopy.

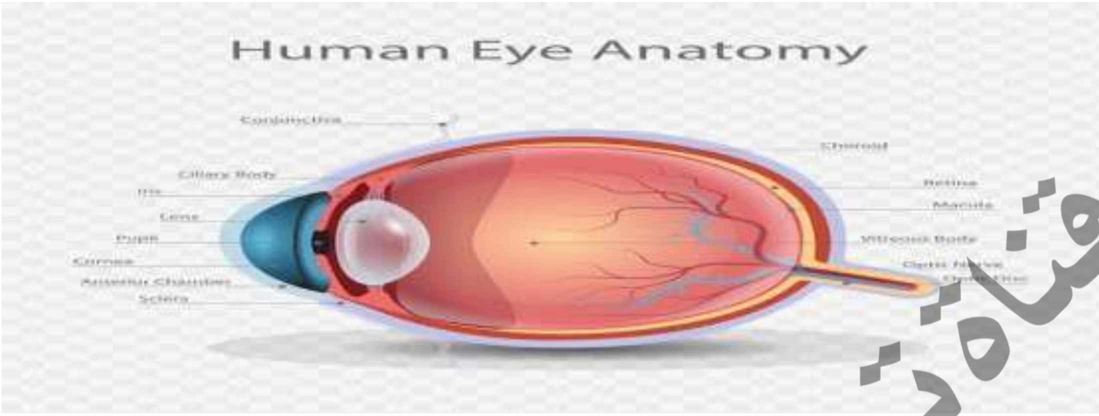
Organ Transplantation

With the beginning of the twentieth century, organ transplant operations began by transferring a healthy organ from a living or recently deceased donor to the patient's body, in case of failure of that organ's function in the patient:

Cases of organ transplantation:

- Kidney transplant in cases of chronic renal failure.
- Whole or partial liver transplant from a living donor in cases of advanced liver failure, and in some cases of liver tumors.
- Pancreas and islet cell transplantation for diabetic patients and radical intestinal resection, still experimental in a limited number of global centers.
- Lung and heart transplantation, experimental in a limited number of advanced global centers for patients with lung and heart failure.
- Corneal transplantation, where the cornea is transferred from a recently deceased person to patients with corneal opacity, giving them the possibility of vision. If the cornea is left in the [globe?], it deteriorates within hours. Kidney and liver transplant operations are performed successfully in Egypt for several years through donation by relatives of the patients (kidney or part of the liver) but they are costly operations and require post-operative immunosuppressive drugs so the body does not reject the new transplanted organ.

Chapter Eight
Common Eye Diseases



- The eye is a sphere with a diameter of 24 mm.

The eye wall consists of three layers:-

(1) The Outer Layer:

It consists of the cornea (Cornea) and the sclera (Sclera). The function of this layer is to protect the eye, and the cornea refracts light rays falling on the eye.

(2) The Middle Layer :-

It consists of the iris (Iris) + the ciliary body (Ciliary body) + the choroid (Choroid). The function of this layer is to nourish the eye tissues and the ciliary body helps in accommodating the eye's lens.

(3) The Inner Layer :-

It is the retina, which forms the optic cup, and the retina receives light rays falling on the eye.

The interior of the eye is divided into three chambers:-

1. **Anterior chamber**: Between the iris and the cornea.
2. **Posterior chamber:** Between the iris and the lens.
3. **Vitreous chamber:** Behind the eye's lens.

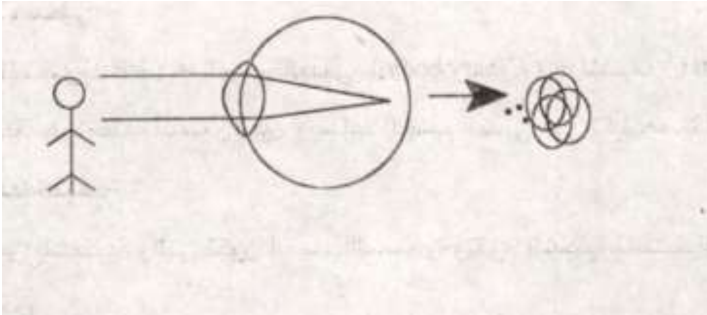
Accessory structures of the eye :-

For the eye to perform its function efficiently, it needs:-

1. Eyelids
2. Conjunctiva.
3. Lacrimal apparatus
4. Eye muscles so it can move in all directions.

The main function of the eye is vision. How does this happen?

1. The cornea and lens refract light rays falling on the eye and focus them on the center of vision in the retina (macula).
2. The image is converted into nerve signals that are transmitted to the visual center in the posterior part of the brain via the optic nerve.
3. The nerve signals are translated into an image that can be seen, and this happens in the visual centers of the brain.



Common Eyelid Diseases

1-Blepharitis
Its most common type is ulcerative blepharitis where there are scales on the eyelashes and eyelid margin. It usually affects patients with oily skin. The patient complains of burning in the eyelid and is treated with drops and ointments.

2-Chalazion

- Arises from inflammation of one of the eyelid glands (meibomian gland).
- The patient complains of eyelid inflammation. It may be accompanied by severe pain if there is a microbial infection.
- Treated with drops and may need surgical intervention to evacuate the contents of the chalazion through an incision from the conjunctival side.

3-Rubbing lash
This is the inward direction of eyelashes, scratching the surface of the cornea and conjunctiva, and may cause corneal ulcers. Trichiasis needs surgical treatment.

4-Entropion
This is the inward turning of the eyelid margin, often arising from an old trachoma infection (Trachoma) and needs surgical correction.

5-Ectropion This is the outward turning of the eyelid margin, often affecting the elderly due to weakness of the eyelid muscles and tissues. It leads to continuous tearing and is treated surgically.

6-Ptosis Drooping of the eyelid is its most common type and is treated surgically by strengthening the levator palpebrae muscle.

Symptoms complained of by eye patients -:

1. Gradual weakening of vision (blurring), e.g., refractive errors and cataracts.
2. Rapid loss of vision: e.g., retinal detachment.
3. Burning and itching of the eye: e.g., allergic diseases.
4. Eye pain: as in cases of glaucoma.
5. Discharge and redness of the eye: conjunctivitis.
6. Deviation and squint (strabismus) of the eye.
7. Continuous tearing: as in obstruction of the lacrimal duct.

Common Conjunctival Diseases

- The conjunctiva is the membrane lining the eyelids and covering the sclera.
- Inflammation of the conjunctiva is called conjunctivitis

1-Muco - purulent Conjunctivitis

- Most common type.
- The most important bacterium is Haemophilus or Koch(-Weeks?).
- Transmitted to the eye by flies, hands, or towels.
- The patient complains of redness in the eyes and the presence of mucopurulent discharge, especially in the morning.
- Treated with antibiotic drops (e.g.: Tobrex - Tobramycin) with frequent eye washing with warm water.

2-Ophthalmia neonatorum

- It is inflammation of the eye conjunctiva during the first month of life.
- Often caused by a bacterium (most famously Chlamydia) which reaches the conjunctiva during birth.
- Tobrex drops (Tobramycin) are instilled three times daily, along with treatment of the mother if vaginal infections are present.

3-Trachoma

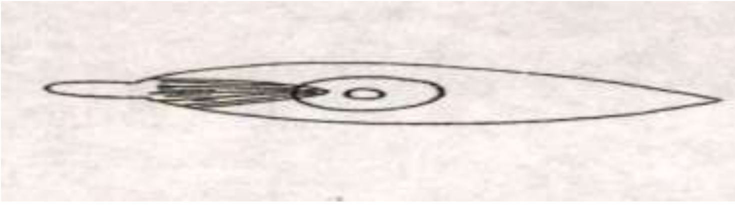
- It is a chronic inflammation affecting the conjunctiva and the upper part of the cornea.
- Caused by the microorganism Chlamydia trachomatis.
- The disease is characterized by the presence of follicles and papillae on the upper eyelid and superficial inflammation of the cornea.
- Leads to the formation of subconjunctival fibrosis and "PTD" follicles (likely refers to Herbert's pits).
- Potential complications include: Trichiasis - Entropion - Fatty cysts (likely pannus) - Conjunctival dryness - Corneal ulcer and opacities.

4-Spring Catarrh

- Chronic eye allergy that increases in summer and spring.
- The patient complains of severe eye itching - tearing and photophobia - inability to tolerate light - stringy discharge.
- Often affects children.
- Upon examination, hexagonal-shaped papillae are found on the upper eyelid and limbus of the cornea.
- The patient is treated with antihistamines and cold compresses - corticosteroid drops are used under continuous doctor supervision to avoid complications like cataracts and glaucoma.

(5) Pterygium

- A common disease involving the growth of a wing-shaped part of the conjunctiva onto the inner part of the cornea. It may reach the level of the pupil, thus affecting visual acuity.
- Exposure to sunlight for long periods is considered one of the most important causes.
- Treated by surgical removal if it affects visual acuity or for cosmetic



Common Corneal Diseases
Corneal Ulcer

- It is the loss of the surface epithelium of the cornea.
- **Its causes**:
  (1) Eye injuries.
  (2) Hypopyon Ulcer.
  (3) Viral ulcer caused by the herpes virus.
  (4) Deficiency of proteins and vitamin A.
  (5) Secondary ulcer resulting from trachoma.
- **Symptoms:**
  Severe eye pain - Inability to face light (photophobia) - Weakness of vision.
- **Diagnosis:**
  The presence of an ulcer is confirmed using fluorescein dye.
- **Complications:**
  - Corneal opacity (clouding).
  - Iritis and glaucoma.
  - Corneal perforation leading to endophthalmitis and hemorrhage.
- **Treatment:**
  - Atropine drops - Antibiotics or antiviral medications.
  - Eye patch.
  - Cauterization of the ulcer.
  - Treatment of complications.

What is Corneal Grafting (Keratoplasty)?

- It is the replacement of a patient's cornea (due to opacity or clouding) with another transparent cornea taken from a recently deceased person in general surgery
- This surgery is successful in cases of severe corneal scarring that obstructs vision.
- The patient needs careful follow-up after surgery and cortisone drops to prevent the body from rejecting the new cornea.

The Crystalline Lens

- **What is the lens?**
  It is a transparent, biconvex body located between the iris and the vitreous body, and it is connected to the ciliary body by the lens zonules.
- **What is its function?**

1. To refract (bend) light rays falling on the eye so they reach the retina.
2. The function of accommodation, which is increasing the eye's power to allow reading.

The most important disease affecting the lens is cataract.

CATARACT

- **Definition**: Partial or complete loss of the lens's transparency.
- **Types:** *

1. Congenital cataract.
2. Acquired cataract: These are divided into:

▪ Senile cataract (in the elderly).
▪ Cataract resulting from injuries.
▪ Secondary cataract.

DEVELOPMENTAL CATARACT

- A child may be born with it or it may appear shortly after birth.
  - The mother contracting German measles (rubella) during pregnancy may lead to cataract in the child.
  - It is treated by surgical removal as soon as possible.

Senile Cataract

- Occurs in the elderly – in both men and women – and usually affects both eyes.
- The patient complains of a gradual weakening of vision.
- In advanced stages, the patient only sees hand movements.

Lens Examination:

It is done with a Slit-Lamp, which provides a magnified view of the front part of the eye.

- Cataracts are treated surgically by removing the opaque lens and placing another artificial lens inside the eye to perform the function of the removed lens.
- The diseased lens is removed either through surgery or by breaking it up inside the eye using ultrasound (phacoemulsification) or laser.

How is a patient prepared for cataract surgery?

1. Thorough eye examination to ensure it is free from conjunctivitis and dacryocystitis.
2. Examination of the retina or performing an ultrasound if it cannot be seen due to the cataract.
3. Performing necessary tests such as blood sugar, liver, and kidney function.
4. Measuring the power of the intraocular lens (IOL) to be implanted.
5. Sterilizing the eye surface with povidone-iodine drops and covering the eyelashes.

Glaucoma

*** Definition:**

- It is the elevation of intraocular pressure above normal levels (10 – 21 mmHg) which leads to atrophy of the retina and optic nerve and deterioration of the visual field.
- **Types:**

(1) Congenital glaucoma.
(2) Acquired glaucoma.

- Open-angle glaucoma.
- Closed-angle glaucoma.

Buphthalmos

- It arises from a developmental defect in the angle of the eye.
- Symptoms noticed by the family: Tearing (epiphora) – Child's inability to face light (photophobia) – Increase in eye size – Change in corneal color.
- Treated with urgent surgery to open the eye's angle.

Open-angle glaucoma

- Usually affects the elderly and often has no specific complaints, hence it is called (the thief of sight).
- Diagnosis depends on the following:
  - Elevated intraocular pressure.
  - Cupping of the optic disc.
  - Changes in the visual field.
  - Open eye angle.
- Treatment of this disease involves reducing the eye pressure to normal levels via:
  - Drops such as Timolol.
  - Laser.
  - Surgery.
- Neglecting treatment leads to complete and permanent loss of vision.

Closed-angle glaucoma

- This type usually begins with a sudden (acute) rise in eye pressure, causing severe pain, blurred vision, nausea, and vomiting.
- The patient is admitted to the hospital and given Pilocarpine drops and Timolol + cidamex tablets + intravenous Mannitol solution

*** What is glaucoma surgery?**

It is creating an opening in the eye wall at the junction of the cornea and sclera in the upper part under the eyelid. Through this opening, fluid exits from the anterior chamber to the subconjunctival space where it is absorbed, thus lowering the intraocular pressure.

The Retina

The retina is the sensitive membrane upon which the image is formed.

*** How to examine the retina and fundus:**

1 - Direct ophthalmoscope.
2 - Indirect ophthalmoscope.
3 - Fundus photography with intravenous fluorescein injection.

Diabetic Retinopathy

It is the effect of diabetes on the retina and its blood vessels. The rates of infection increase in insulin-dependent diabetes and with longer duration of the disease.

The disease goes through several stages:

1. Mild Diabetic Retinopathy:
   Characterized by the presence of microaneurysms and hemorrhages in the retina. The patient usually has no symptoms at this stage.

2) The stage of growth of new blood vessels from the retina and optic nerve:

This stage may lead to complications such as vitreous hemorrhage, traction, and retinal detachment.

Treatment of Diabetic Retinopathy:

1. Controlling blood sugar levels.
2. Argon laser for the retina.
3. Surgical removal of the vitreous (Vitrectomy).

Refractive Errors

*** What does 6/6 visual acuity mean?***

This means the person can identify the direction of all the signs on the vision chart from a distance of six meters.

*** What does 18/6vision mean?**

The person reads the18/6line clearly, meaning they see at 6 meters what a normal person can see at 18 meters.

Refractive errors are divided into:

- Myopia - Short-sightedness
- Hypermetropia - Long-sightedness
- Astigmatism

*** Short-sightedness (Myopia):***

1. In a normal person, light rays focus on the center of vision on the retina, whereas in myopia, the rays focus in front of the retina.
2. In most cases, the eye is larger than its normal size.
3. The patient complains of difficulty seeing distant objects, and this difficulty increases at night.
4. Myopia is corrected using concave lenses.
5. Contact lenses can be used, or myopia can be corrected using Excimer laser.

***Long-sightedness (Hypermetropia**):

1. In this case, light rays entering the eye focus behind the retina.
2. The patient complains of eye strain, especially while reading.
3. Hypermetropia may lead to esotropia (convergent squint) in children.
4. Corrected with convex lenses that focus light on the retina.

***Astigmatism**:

- It is the difference in the refractive power of the eye in different meridians.
- Astigmatism can be regular or irregular, as in keratoconus.
- Corrected with glasses or rigid contact lenses.

Retinal Detachment

- It is the separation of the retina from the choroid which nourishes it.
- Most cases are associated with a retinal tear.
- The patient complains of severe deterioration in vision.
- Treated with prompt surgical intervention

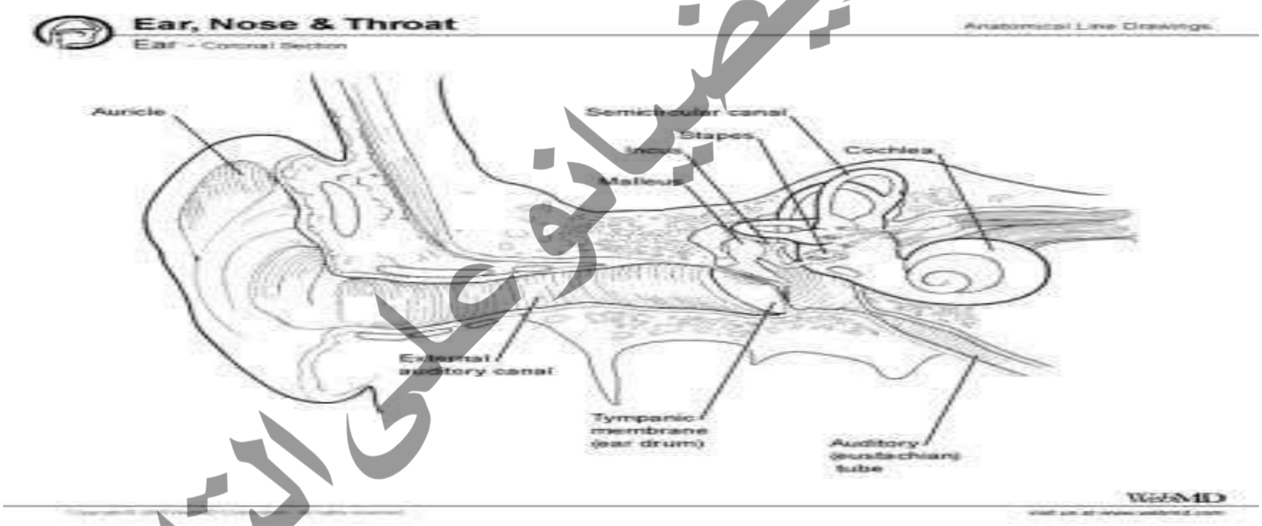
Chapter Nine

Diseases of the Nose, Ear, and Throat

The Ear

Anatomy of the Ear:
The ear is the organ of hearing in humans. Most parts of the ear are located inside the skull. Anatomically, the ear is divided into three parts:

1- External Ear
2- Middle Ear
3- Inner Ear



1- External Ear:
Consists of the Pinna (Auricle - the visible part on the side of the head), the External Auditory Canal, and the Eardrum (Tympanic Membrane), which is a delicate membrane at the end of the auditory canal.

2- Middle Ear:
Located behind the eardrum.

The middle ear is an air-filled cavity within the skull bones, situated between the eardrum and the inner ear.
This cavity contains three ossicles (small bones), in order from the outside inward: the Malleus, Incus, and Stapes.

- These ossicles transmit vibrations from the eardrum to the inner ear. The middle ear connects to the pharyngeal cavity via a tube called the Eustachian tube, which works to balance the pressure on both sides of the eardrum.

Inner Ear: Cochlea and Auditory Nerve.

The inner ear, also called the labyrinth, consists of a complex system of membranous channels inside a bony shell called the cochlea. The cochlea is filled with fluid where the hair cells of hearing are located; these are small hairs stretched like strings.

The Sense of Hearing

Sound travels through the air as waves. Sound waves reach the ear, are collected by the pinna, and travel through the auditory canal to the eardrum. Sound vibrations "hit" the eardrum, making it vibrate. The middle ear conveys these sound vibrations from the eardrum to the inner ear via the three ossicles. These ossicles act as amplifiers for the sound vibrations due to their connection as a series of levers. The end of the stapes is connected to a delicate membrane covering a narrow opening between the middle ear and the inner ear.

The vibrations transmitted by the auditory ossicles in the middle ear reach the fluid in the inner ear and agitate it. The fluid vibrations cause the hair cells to vibrate. The hair cells are connected to nerve endings. From these nerve endings, the auditory sensation is transmitted, via the auditory nerve, to the hearing center in the brain. The hearing center translates the captured vibrations into understandable sounds for us.

Ear Wax (Cerumen)

Ear wax is a natural substance secreted by the outer part of the external auditory canal where there are ceruminous glands (a modified type of sweat gland). This waxy substance works to trap dust that may enter the ear. Under normal circumstances, one should never attempt to clean the ears with anything; the ear naturally does not need cleaning. This waxy material turns into dry flakes and comes out of the ear canal naturally. However, in some cases, waxy material may accumulate inside the external auditory canal, which can lead to temporary hearing loss or a blocked ear.

If you feel blockage or hearing loss, consult a doctor without trying to clean the ear yourself. If the doctor discovers that the blockage or hearing loss is due to ear wax, they will remove the ear wax by: syringing (irrigation), suction, or removing it with a special instrument, depending on their assessment of the case. They may give drops to soften the wax a few days before syringing.

Inflammations of the External Ear: Otitis Externa

1- **Diffuse Otitis Externa:**

- Pain in the ear, especially when pressing on the tragus.
- Narrowing of the external auditory canal.
- There may be purulent discharge.
- Treatment is with antibiotics and placing a small wick in the ear containing a topical antibiotic cream and anti-inflammatory medication.

2- **Furuncle:**

- Characterized by severe ear pain that increases with movement of the pinna and chewing.
- Narrowing of the external auditory canal.
- Treatment with antibiotics: placing a small wick in the ear containing a topical antibiotic cream and anti-inflammatory medication or glycerin ichtyol.

3- **Malignant Otitis Externa:**

- Often affects patients with diabetes mellitus.
- Severe inflammation in the lower part of the external auditory canal; the inflammation spreads and causes bone necrosis.
- Treatment involves using antibiotics like Cipro.

4. **Otomycosis:**

   - Occurs due to a fungal infection.
   - Causes itching in the ear.
   - Sometimes leads to temporary hearing loss.
   - Leads to discharge from the ear.
   - Treatment of the disease involves repeated cleaning and topical antifungals.

Otitis Media:

Middle ear infection in children is one of the most common diseases, especially in preschool age. The infection must be treated quickly so that the condition does not progress and affect the child's hearing ability. Otitis media is a bacterial inflammation in the middle ear area and usually occurs after a sore throat, cold, or respiratory tract infections. The middle ear space is normally filled with air, but fluids can accumulate in the middle ear due to upper respiratory tract infections like the common cold or even due to allergies.

How does otitis media occur?

Microbes reach the ear by one of the following ways:

1- Through the Eustachian tube, usually after a cold, which causes secretions in the middle ear where bacteria find a suitable medium for multiplication, thus causing inflammation.
2- Through the bloodstream.
3- Through the external auditory canal if there is a perforation in the eardrum.

Types of Middle Ear Infections

Acute and Chronic:

Acute is often caused by the transfer of germs from the nasopharynx to the middle ear via the Eustachian tube. Patients with acute otitis media suffer from sharp pain in the ear with headache and high fever and hearing loss. After two or three days, the eardrum may perforate. As for chronic otitis media, it often results from pus remaining in the middle ear after acute inflammation and is characterized by purulent or mucous discharge with hearing loss. If middle ear infection is neglected and continues for many years, it may lead to osteomyelitis and thus destruction of the auditory ossicles.

Factors that predispose children to infection

Anyone is susceptible to middle ear infection, but children are more susceptible because their Eustachian tube is shorter and straighter than in adults, thus contributing to the easy accumulation of secretions in the middle ear area.

Symptoms and Signs

(A) In adults:
1- Earache
2- Hearing impairment
3- Discharge of pus from the ear

(B) In chidren:
1- High temperature (usually in infants)
2- The child cries a lot and cannot sleep at night
3- The child constantly presses on, pulls, or scratches their ear
4- Yellow purulent discharge from the ear

Diagnosis

In addition to the symptoms of ear infection, the doctor will use an otoscope to view the eardrum. In case of inflammation, the doctor will notice that the eardrum is bulging and very red, and it may be perforated due to fluid accumulation.

Complications of Otitis Media

Prompt administration of appropriate treatment contributes to rapid recovery. But delay in treatment or giving inappropriate treatment may lead to progression of the condition. Repeated occurrence of inflammation may lead to damage to the eardrum, ear ossicles, or auditory nerve and may cause hearing loss. Since children learn to speak by listening to others, hearing loss leads to delays in the ability to converse and learn language.

Prevention

- Avoid feeding the baby while lying on their back, as milk and other nutritious fluids can ascend into the ear through the Eustachian tube. In the presence of bacteria, these fluids become a suitable medium for multiplication.
- Do not insert any solid object or sharp tool to scratch the ear.

Treatment

1- Oral antibiotics
2- Topical antibiotics via ear drops
3- Analgesics
4- Nasal drops
5- Cleaning purulent discharge
6- If pus forms inside the middle ear to the point of bulging the eardrum, the doctor may perform

a small surgical incision in the eardrum (Myringotomy) to allow drainage of the collected fluid and reduce pressure and pain.

Foreign Bodies in the Ear

Types:

1- Hard foreign bodies: like a bead or pebble.
2- plant foreign bodies: like a wheat or bean grain.
3- Insects: like mosquitoes.

Treatment:
1- **Hard foreign bodies**:
Can often be removed by syringing (ear irrigation). Forceps should absolutely not be used for removal as it may lead to perforation of the eardrum, injury to the external auditory canal, or pushing the object further inside.

2- plant foreign bodies:
If difficult, ear irrigation can be attempted. If they are large, it is preferable to remove them using micro-instruments.

3- Insects:
They must be killed first by putting paraffin oil, then the ear is syringed.

Snoring:
Snoring is a symptom that bothers the patient and those around them. Snoring has a negative impact on human health.

Causes:
Snoring occurs due to narrowing of the airway, which results in turbulence of the incoming air stream through the nose and throat, causing the loud sound of snoring. This narrowing can occur in several areas in the nose or throat due to diseases, including:

- **In the nose:**
  - Enlargement of the nasal turbinate mucosa due to nasal allergy.
  - Deviated nasal septum.
  - Presence of nasal polyps.
- **In the throat:**
  - Enlarged tonsils.
  - Enlarged uvula.
  - Large tongue size.
  - Receding chin.

- Other causes:
  - Weight gain leading to fat accumulation around the neck and pressure on the airway.
  - Excessive relaxation of the throat wall muscles during sleep due to taking sedatives or sleeping pills, or due to extreme fatigue combined with being overweight.

Snoring can affect a person's health due to repeated obstruction and oxygen deficiency during sleep, which disturbs the patient's sleep, makes them feel unrefreshed upon waking, and causes a desire to sleep, drowsiness, and lack of concentration during daytime hours. Also, repeated obstruction during sleep may raise the level of some hormones that regulate blood circulation, which may lead to long-term problems with the heart and respiratory system. However, this only occurs in advanced cases.

Treatment:

- Treatment for snoring focuses primarily on correcting the narrowing in the airway, but the location of the narrowing during sleep must be determined. This is done through special tests that monitor the patient during sleep and perform many measurements to know the severity, cause, and impact of the narrowing on the patient, done in a sleep lab.
- One of the most common causes of snoring is weight gain, and this can often be treated by weight loss alone.
- In cases where the source of snoring is known, operations can be performed to remove the obstruction, such as:
  Correction of the nasal septum - Removal of polyps - Tonsillectomy - Uvuloplasty - Chin correction.
- For cases where the cause of snoring is primarily muscle relaxation, the problem is usually solved by using a special device that increases the pressure of air entering the throat during sleep.

Nose and Paranasal Sinuses

Secondly: The Nose and Paranasal Sinuses

Epistaxis (Nosebleed)

Nasal bleeding is a common condition, especially in summer due to high temperatures which lead to dilation of the blood vessels present in the nasal mucosa, making them prone to rupture and causing nosebleeds. Bleeding usually occurs from the anterior part of the nasal septum, which is the wall separating the two nostrils and contains a network of blood vessels.

Causes of Nosebleeds:

General Causes:

1- High blood pressure or heart and blood vessel diseases. (Main cause in the elderly)
2- Liver dysfunction, such as liver cirrhosis due to schistosomiasis or hepatitis viruses, which affects its efficiency and ability to produce clotting factors, making bleeding easier.
3- Blood clotting disorders.
4- High temperature in febrile illnesses (especially in children, leading to dilation of the blood vessels in the nasal mucosa) (Main cause in children).

Local causes in the nose:

1- Presence of dilated blood vessels in the mucosa of the anterior part of the nasal septum with weak vessel walls (Main cause in young adults).
2- Deviated nasal septum.
3- Nasal injuries.
4- Atrophy of the nasal mucosa.
5- Inflammation of the nose and sinuses.
6- Benign or malignant tumors of the nose and sinuses.

When facing a case of nasal bleeding, some questions should be asked:
1- Does the patient have high blood pressure?
2- Is there a nasal injury?
3- Is the patient taking any medications that prevent clotting like aspirin or anticoagulants?
4- Is the patient suffering from shock (by measuring vital signs)?
5- Is the bleeding severe or recurrent?

Protocol for treating nosebleeds:
1- First aid for nosebleeds:
- If the bleeding is minor and the patient's condition is stable, the affected person should sit leaning slightly forward, pinching the nose with two fingers and breathing through the mouth for at least 5 to 10 minutes.
- The patient should avoid tilting the head back as is common, because this allows the bleeding to continue internally, leading to stomach upset.
- A cotton pledget can be placed inside the nose while continuing to pinch the nose and leaning the face down until the blood clots, along with applying ice around the nose.
- If the bleeding continues, the patient should see a doctor for further blood tests such as: clotting time, hemoglobin, and blood pressure, along with an ENT examination.

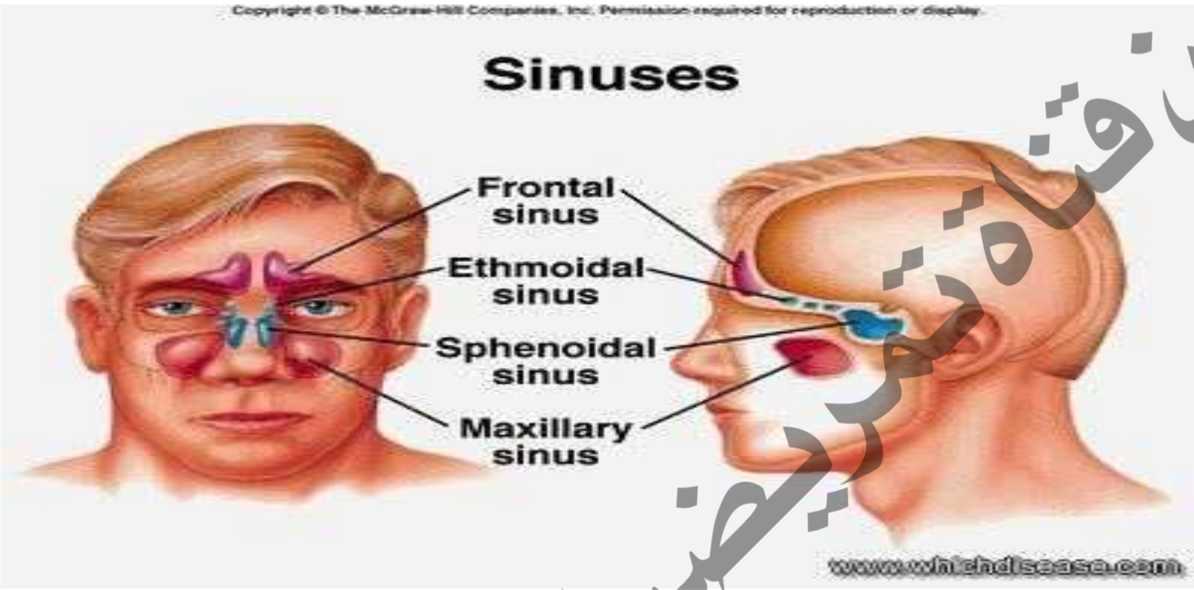
2- Stopping the nosebleed:
1-If the bleeding is minor, the doctor can cauterize the dilated vessel under local anesthesia using electrocautery or chemical cautery.
2-If the bleeding is more severe, the doctor may perform anterior or posterior nasal packing.
3-If the bleeding is severe and hemorrhagic shock occurs, the shock must be treated immediately with a blood transfusion for the patient, along with the necessary procedures to stop the bleeding by the doctor.

4- In some severe cases, surgical intervention is required to ligate the artery causing the bleeding.

Sinusitis
The paranasal sinuses are air-filled spaces distributed in the skull around the nasal area and consist of four main groups:

- Frontal sinuses (above the nose and eyes)
- Ethmoid sinuses (between the nose and eyes)
- Maxillary sinuses (under the eyes on either side of the nose).
- Sphenoid sinuses (behind the nose in the center of the head).

These sinuses connect to the airway in the nose through small openings. Many factors can cause swelling of the nasal mucosa inside the nose, such as catching a cold, allergy attacks, or sudden weather changes, leading to blockage of these openings. This helps secretions collect inside the sinuses and bacteria multiply, causing sinusitis.



Acute Sinusitis:
The main cause of acute sinusitis is the blockage of the openings of the paranasal sinuse

Symptoms:

- Headache
- The pain in sinusitis is concentrated in the sinus areas. The pain usually increases with head movement, such as bending forward and during prayer.
- Fever.
- Purulent (pus-like) discharge from the nose and the back of the throat, which causes a sore throat.

Treatment:

- Administering the appropriate antibiotic (the antibiotic must be used for at least one week after the inflammation symptoms disappear).
- Continuous nasal cleaning with saline water.
- Using nasal sprays or drops to remove the blockage.
- Analgesics (pain relievers).

Chronic Sinusitis

Chronic sinusitis is defined by its persistent symptoms, which are somewhat similar to the symptoms of acute inflammation. However, chronic sinusitis often involves the recurrence of acute inflammation episodes. Chronic inflammation usually requires a longer treatment time, and in advanced cases, it may require surgical intervention to open and clean the sinuses using an endoscopic sinus procedure.

The Pharynx

Tonsillitis:

The tonsils are two masses of lymphoid tissue located in the oropharynx, in the posterior part of the mouth, and can be seen when opening the mouth. They help us filter germs and prevent them from entering the body. The tonsils are frequently inflamed along with the pharynx in children, and often the tonsils and pharynx are affected together. The infection of the tonsils may be more prominent, which is then called acute tonsillitis.
There are two types of tonsillitis: acute inflammation and recurrent or chronic inflammation. It is important to differentiate between them because the treatment differs (medical or surgical).

Acute Tonsillitis:

This disease is an acute inflammation affecting the tonsils and may cause purulent inflammation. Tonsillitis is one of the most common diseases in children, but adults can also suffer from it. Often, tonsillitis in children is accompanied by enlargement of the adenoids (pharyngeal tonsil) behind the nose, which is also composed of lymphoid tissue similar to tonsil tissue.

Causes of Acute Tonsillitis:

Acute tonsillitis can be caused by a viral or bacterial infection. This disease often occurs due to infection with bacteria belonging to the cocci class, especially the beta-hemolytic streptococcus type (Streptococcus).

Symptoms of Acute Tonsillitis:

[1] Sore throat, which may be severe
[2] Difficulty swallowing, sometimes the child refuses even to drink fluids
[3] Fever that may reach 40°C, sometimes accompanied by chills
[4] Headache, general weakness, and loss of appetite
[5] Change in voice (hot potato voice)
[6] Nausea, vomiting, and abdominal pain may sometimes occur in children

It is very important not to confuse acute tonsillitis with pharyngitis (sore throat) because they give the same symptoms, and the doctor's examination is the deciding factor. It should be remembered that not every sore throat is tonsillitis.

Signs:

[1] Enlarged tonsils, which can sometimes be severe
[2] Redness of the tonsils with white purulent spots on them
[3] Enlarged lymph nodes in the neck with pain upon palpation

Complications:

1- Spread of the inflammation to the area around the tonsil and formation of a peritonsillar abscess.
2- Dehydration in the child due to inability to swallow or painful swallowing.
3- Delayed complications of the infection may occur, namely Rheumatic fever and acute glomerulonephritis. These complications appear 1-3 weeks after the onset of acute tonsillitis.

Treatment:

1- Administration of antibiotics, which should be continued for a period of 7-10 days to completely combat the germs.
2- Administer antipyretics (fever reducers).
3- Throat gargles or antiseptic lozenges can be given to relieve the pain resulting from the inflammation and to help clean the tonsils of purulent secretions that may have accumulated on them.
4- Administer plenty of fluids.
5- Rest.

It is important to emphasize the necessity of completing the treatment for 10 days and not stopping the medication as soon as the fever drops or symptoms improve, which happens after 2-3 days of starting treatment.

Chronic tonsillitis

When acute inflammation recurs five to six times in one year, or three times a year for two consecutive years, we can say that the person has recurrent tonsillitis. However, it must be confirmed that all the recurrent episodes the patient suffers from are cases of tonsil inflammation, not just pharyngitis, and that they are accompanied by fever, not just minor or transient throat pain. Therefore, a doctor should be consulted in each case.
It should be emphasized that it is normal for a child to get tonsillitis about twice a year, and in this case, there is no need for tonsillectomy, especially if the size of the tonsils is normal. However, tonsillectomy becomes necessary if the inflammation recurs more than five or six times in a single year, or if the size of the tonsils is so large that it causes difficulty breathing for the child during sleep and causes episodes of sleep apnea (obstructive sleep apnea). Because it is unfair to decide on tonsillectomy if they are not the cause of the inflammation, and the patient

would subject themselves to surgery without finding the expected benefit afterward since the real cause of the disease was not treated.

Symptoms of Chronic Tonsillitis:

The tonsils may suffer from chronic inflammation causing some annoying symptoms such as:

- Recurrent inflammation (five to six times during one year, with each episode accompanied by fever).
- Pain when swallowing and a feeling of dryness in the throat.
- Bad breath .

Signs:

1- Enlarged tonsils to the degree that they cause difficulty breathing for the child during sleep and cause episodes of sleep apnea or snoring.
2- Permanent redness and congestion of the tonsils.
3- Difference in size between the two tonsils.
4- Enlarged lymph nodes in the neck.

The treatment for recurrent tonsillitis is removal (tonsillectomy) when all the conditions we mentioned are met.

Tonsillectomy Operation:

Indications for the operation:

1- Chronic inflammation of the tonsils.
2- Peritonsillar abscess.
3- Tonsillar hypertrophy to a degree that causes obstruction.
4- Enlarged tonsils to the degree that they cause difficulty breathing for the child during sleep.

and cause him episodes of sleep apnea or snoring.
5- Suspicion of a tumor in the tonsil.

Contraindications for the operation:

a) Absolute contraindications: meaning the operation cannot be performed at all
1- Tendency to bleed in cases of various blood diseases like hemophilia.
2- Advanced heart, kidney, and liver diseases.

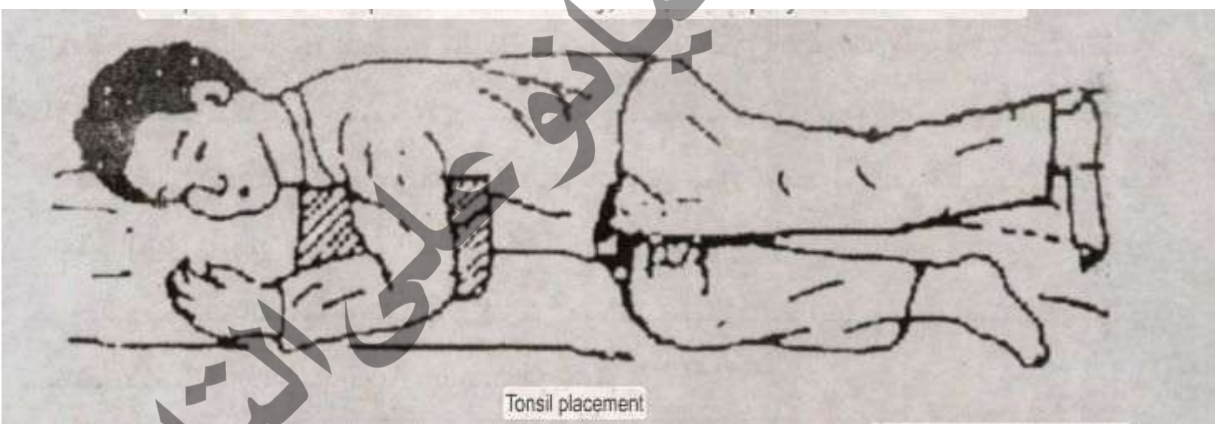
b) Relative contraindications: The operation should be postponed until the contraindication is resolved:
1- The patient suffering from an acute tonsillitis infection shortly before the operation.
2- High temperature on the day of the operation.
3- During menstruation in females.
4- During pregnancy.
5- In case of an epidemic like polio.

Preparing the patient before surgery:

1- Perform a blood test for the patient including hemoglobin level, bleeding time, clotting time, prothrombin time and percentage.
2- Urine analysis for sugar and albumin.
3- Examination of the heart and lungs by a cardiologist or internist.
4- Recording vital signs like pulse, blood pressure, temperature, and respiration on the morning of the operation.
5- Light dinner the day before the operation and fasting for six hours before the operation.

Post-Operative care

1- The patient is placed in the recovery position (the patient is placed on their side, slightly forward, with a pillow under their upper shoulder and the upper leg bent while the lower leg is extended, and the patient's lower arm is behind the body).
2- Observing for symptoms of bleeding after the operation, externally and internally (red blood with sputum,).
and gradual increase in pulse then it becomes weak, drop in temperature, and vomiting a large amount of blood.)
3-Give the patient cold fluids 6 hours after recovery from anesthesia.
4-Give the patient soft foods in the evening.
5-Emphasize to the patient to avoid dry, hot, and spicy food for a week.



Tonsil placement

Complications of the operation:
1 - Severe bleeding may sometimes occur.
2 - Complications of anesthesia.
3 - Occurrence of infections in the wound after the operation.

Adenoids in Children: Nasopharyngeal Adenoids

It is very important to differentiate between adenoids in children and nasal polyps in the nose and sinuses that affect adults in cases like allergies.

What are adenoids in children?

- They are a collection of lymphoid tissue located directly behind the nose in the nasopharynx (and not inside the nose).
- They are present naturally in all children from birth and may increase in size from after the first year until the age of 4-5 years, after which they begin to atrophy.
- They are very important in fighting microbes and contributing to the body's immunity (like the tonsils).
- The presence of adenoids itself is not the disease, but their enlargement or exposure to chronic inflammation is the real problem.

Enlarged adenoids in children cause obstruction of the natural breathing passage from the nose, forcing the child to breathe through their mouth all the time. Obstruction of the nasal airway in a child leads to intolerance to exertion like running or playing, and reduced activity and concentration. They also cause snoring due to narrowing of the airway. Enlarged adenoids and tonsils can be considered the most important cause of snoring in children. If the adenoids are infected with chronic inflammation, the symptoms are persistent discharge from the nose, nasal blockage, and may be accompanied by repeated ear infections. Obstruction of the nasal breathing passage in a child and their compulsion to breathe through the mouth all the time may lead to changes in the child's facial features (protrusion of the teeth - shortening of the upper lip - atrophy of the cheeks).

Treatment for enlarged adenoids or chronic inflammation is surgical removal (Adenoidectomy operation).

Fourth: The Larynx
Laryngeal Obstruction

Causes:
1-Infiltration of fluid under the mucosa of the larynx (Laryngeal edema) resulting from:

- Severe allergic reactions.
- Inhalation of chemical gases or prolonged inhalation of fire smoke.
- Laryngitis in children.

2-Chronic laryngitis (Laryngeal Sclerosis disease).

3-Vocal cord paralysis.

4-Laryngeal tumors.

5-Laryngeal injuries due to accidents.

6-Inhalation of foreign bodies in the larynx.

Symptoms:

1. Inability to breathe easily (suffocation).
2. A distinctive whistling sound during breathing (Stridor).
3. Hoarseness of voice.
4. The patient feels a state of exhaustion and weakness, with profuse sweating.

Signs:

1. Rapid pulse.
2. Rapid breathing.
3. The patient is in a state of tension.
4. Contraction of the chest muscles and the intercostal muscles to aid in breathing.
5. Excessive flaring of the nostrils.
6. Weak pulse, low blood pressure, and a bluish discoloration of the patient's skin indicate a delayed/advanced case.

Treatment:

The patient must receive treatment as quickly as possible to be saved:

1. In simple cases, medical treatment can be tried by giving the patient corticosteroids and decongestants, in addition to oxygen.
2. If the laryngeal obstruction is severe or after the failure of medical treatment, a tracheostomy must be performed as quickly as possible.

Tracheostomy:

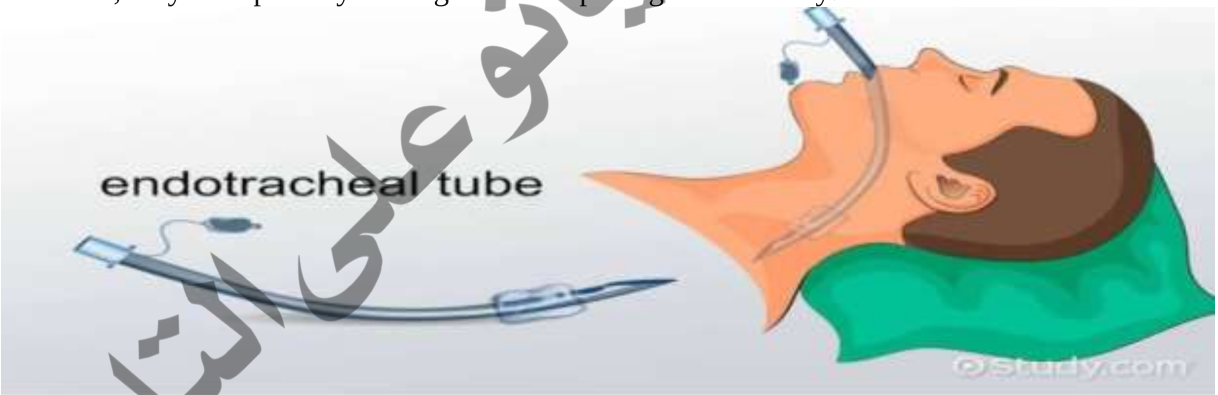
Making an opening in the trachea and inserting a plastic tube into this opening to allow air to enter the lungs.

Indications for the operation:

1. Laryngeal obstruction.
2. Profuse secretions in the lungs due to the patient's inability to cough, as in cases of coma. In this case, the procedure is performed to suction the secretions.
3. As a preliminary step in some surgeries.

Post-Tracheostomy Care Notes:

1. The patient is placed in a semi-sitting position because this position facilitates coughing and clearing accumulated mucus from the bronchial tubes.
2. Suctioning secretions using an electric suction device by inserting a thin catheter into the laryngeal tube.
3. Cleaning the inner tube as needed.
4. Monitoring for bleeding, fever, rapid breathing, or the return of suffocation.
5. Humidifying the room air by placing a pot of boiling water.
6. After the operation, the patient cannot speak for only a few days as air exits through the trachea, but after that, they can speak by closing the tube opening momentarily.



Questions

Mention what you know about the anatomical parts of the ear**.**

1. Briefly explain how humans hear.
2. Mention what you know about ear wax.
3. Mention what you know about otitis externa and its main types.

4. Explain the types of otitis media.
5. Explain the symptoms, treatment, and complications of otitis media.
6. Explain how to treat otitis media? And how to prevent this disease.
7. Mention the causes of epistaxis (nosebleed).
8. Explain the treatment and first aid required in cases of nasal epistaxis.
9. Mention what you know about acute and chronic sinusitis.
10. Mention the symptoms, treatment, complications, and how to treat acute tonsillitis.
11. Symptoms, treatment, and how to treat chronic tonsillitis.
12. Mention the indications and contraindications for a tonsillectomy.
13. Explain how to prepare a patient for a tonsillectomy.
14. Explain post-tonsillectomy nursing care.
15. Explain how to position the patient in the tonsil position
16. What is the adenoid? And what are the symptoms of adenoid hypertrophy?
17. What are the complications of adenoid hypertrophy? And what is the treatment for adenoid hypertrophy?
18. Mention the causes of laryngeal obstruction.
19. Mention the symptoms and signs of laryngeal obstruction.
20. What is the treatment for laryngeal obstruction?
21. Mention what you know about the tracheostomy procedure.

Chapter Ten

Diseases of the Mouth and Teeth

Introduction

The human mouth is the first part of the digestive tract and is responsible for receiving food and grinding it via the posterior molars, as well as mixing food with saliva to facilitate swallowing and the passage of food to the stomach and intestines preparatory to its digestion and then absorption. The human mouth consists of a cavity lined with a delicate mucous membrane, surrounded above by the maxillary bones and below by the mandibular bones and the tongue together. The mucous membrane lining the jaw bones covers part of the teeth with what is called gum tissue.

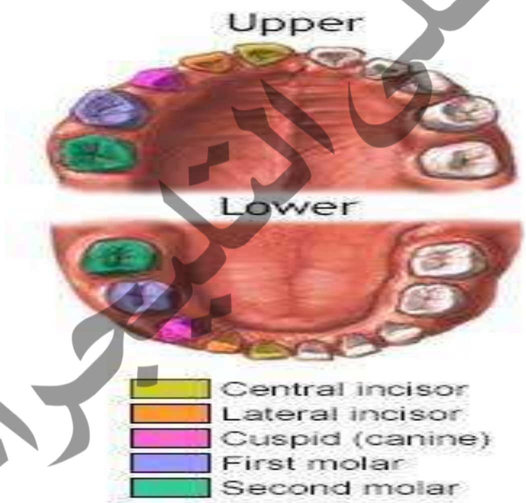
Human teeth are of two types: the first type, called primary (milk) teeth, numbering twenty, are the teeth that appear first in children starting from the sixth month of age and are later replaced by the second type, called permanent teeth, numbering thirty-two. Table (1) shows the eruption times of primary and permanent teeth in the mouth, as well as the shedding times of primary teeth in children.

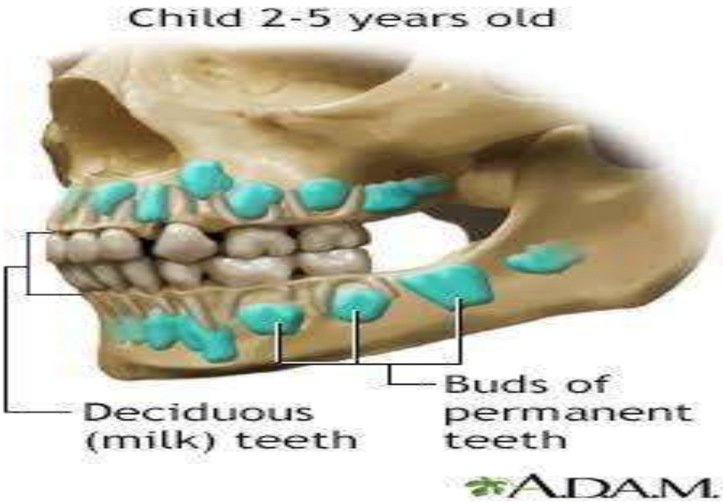
Tooth Type	**Eruption Time (Months)**	
Central Incisor	8 (6-10)	Lower primary teeth
Lateral Incisor	13 (10-16)	
Canine	20 (17-23)	
First Molar	16 (14 - 18)	
Second Molar	27 (23 - 31)	
Central Incisor	10 (8 - 12)	Upper primary teeth
Lateral Incisor	11 (9 - 13)	
Canine	19 (16 - 22)	
First Molar	16 (13 - 19)	
Second Molar	29 (25 - 33)	
Tooth Type	Eruption Time (Years)	
Central Incisor (I1)	(6 - 7)	

Lateral Incisor (I2)	(7 - 8)	Lower permanent teeth
Canine (C)	(9 - 10)	
First Premolar (P1)	(10 - 12)	
Second Premolar (P2)	(11 - 12)	
First Molar (M1)	(6 - 7)	
Second Molar (M2)	(11 - 13)	
Third Molar (M3)	(17 - 21)	
Tooth Type	**Eruption Time (Years)**	
Central Incisor (I1)	(7 - 8)	Upper permanent teeth
Lateral Incisor (I2)	(8 - 9)	
Canine (C)	(11 - 12)	
First Premolar (P1)	(10 - 11)	
Second Premolar (P2)	(10 - 12)	
First Molar (M1)	(6 - 7)	
Second Molar (M2)	(12 - 13)	
Third Molar (M3)	(17 - 21)	

- Table (1) (shows the eruption times of primary and permanent teeth in the mouth)

Note that the timing of the shedding of primary teeth is the same as the eruption timing of the permanent teeth, starting from the incisors to the premolars.





Diseases of the Mouth and Teeth

Diseases of the mouth and teeth occur in three specific regions:

- Teeth
- Mucous membrane, gums, and salivary glands
- Jaw bones

First: Dental Diseases

Teeth consist of calcified organic layers covering a soft tissue inside known as the pulp or nerve of the tooth. These layers are the outer layer known as tooth enamel, which covers a thicker layer known as dentin, which directly surrounds the tooth nerve.
Teeth are affected by many diseases, among which a group occurs more frequently and commonly, namely:

- Dental Caries
- Acute and Chronic Pulpitis
- Attrition, Erosion and Abrasion

(Dental Caries)

Definition:

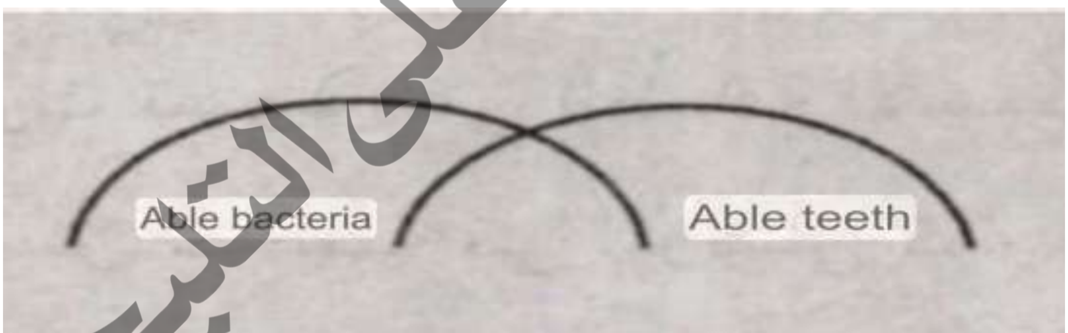
- **Dental Caries** is considered the most common disease in human teeth and is the primary cause of tooth loss in middle-aged patients (from 20 to 40 years old).
- **Dental Caries** is the process of removing calcium from the different layers of the tooth (enamel then dentin) preparatory to dissolving the organic (protein) part and creating a cavity within the tooth layers. The process of calcium removal occurs through some chemical reactions between a group of organic acids resulting from the fermentation of food residues by bacteria present in large numbers in the mouth and the tooth surface.

Basic factors for Dental Caries occurrence:

There are several factors that must be present together for decay to occur:

1. Teeth with an anatomical nature that helps initiate the decay process, such as the presence of protrusions, pits, and fissures, or roughness on the tooth surface, as well as crowding of teeth that helps retain food debris for long periods and allows fermentation to occur.
2. Bacteria causing **Dental Caries**, which are bacteria with special characteristics that secrete a group of enzymes that help them complete the fermentation process and produce acids in abundant quantities. The type of hemolytic streptococci, especially *Streptococcus mutans*, is the most important type in causing decay.
3. [*Numbering in original is inconsistent*] Fermentable materials by bacterial enzymes, which are carbohydrates, especially sugars. Sticky sugars that adhere to the tooth surface for long periods rank first among the factors causing tooth decay.
4. Accumulation of dental plaque on tooth surfaces due to lack of daily dental care, leading to the concentration of fermented sugars by bacteria and the secretion of sufficient quantities of organic acids causing decay.

The following figure shows the factors required for **Dental Caries** to occur:



[*A diagram is described here showing the interaction of: Teeth (Host) + Bacteria leading to Caries*]

Types of Dental Caries

There are many types of tooth decay, the most important of which are:

1. Pit and Fissure Caries - The most common type.
2. Smooth Surface Caries
3. Acute Caries –The most common type in adolescents and young adults
4. Chronic Caries – The most common type in patient and the elderly
5. Rampant Caries - This occurs in specific conditions such as diseases accompanied by reduced salivary gland secretion and in cancer patients due to exposure to therapeutic radiation.

Clinical Manifestation of Dental Caries

1. **Dental Caries** usually begins as small, white, opaque spots affecting the tooth enamel on any surface of the tooth, and it is difficult to diagnose at this stage.
2. **Dental Caries** can reach the dentin layer without a visible cavity in the enamel.
3. **Dental Caries** takes a conical shape in the enamel or dentin layer; the base of the cone is in the enamel towards the outer surface of the affected tooth and its apex towards the dentin layer. Its base in the dentin is towards the enamel and its apex is directed towards the pulp (nerve) of the tooth.
4. When a cavity occurs within the enamel layer, it gradually expands inside the dentin layer, and its color begins to change from whitish-yellow to dark brown and then black due to bacteria and coloring materials found in some types of food and drink.
5. In some cases, decay can occur in the cementum layer covering the tooth root. This type usually starts at the tooth neck, especially in elderly patients when gum tissue recedes from the tooth neck, and anaerobic bacteria are the main cause of this type of decay. This type usually takes the form of a smooth cavity parallel to the tooth neck and is accompanied by extreme sensitivity to many types of drinks.

Areas of Dental Caries

Dental Caries can affect any surface of the tooth, but some surfaces are more susceptible to decay:

- Pits and fissures on the occlusal surface of the posterior molars and premolars.
- The buccal surface of upper and lower molars, as well as the lingual surface of upper incisors and canines
- The buccal or lingual surfaces of all teeth at the gingival margin .
- Rough surfaces that help retain food debris and accumulate plaque for long periods.

Treatment of Dental Caries

- Complete removal of tissues hardened by decay.
- Replacing the lost part of the tooth with an appropriate filling material, such as amalgam fillings in posterior molars and composite (light-cured) fillings for front teeth.

Prevention of Dental Caries

Prevention of **Dental Caries** is achieved by following:

1. Continuous dental care using a suitable toothbrush and toothpaste containing fluoride, especially for children.
2. Not overconsuming fermentable sugary substances, especially those that stick to the tooth surface and are difficult to clean.
3. Increasing the intake of foods containing natural fiber due to their positive effect on dental cleanliness.
4. Increasing health awareness and educating about the importance of teeth and their care.
5. Periodic visits to the dentist for early detection of tooth decay and thus treatment that preserves teeth from premature loss.
6. Removing ill-fitting prosthetic restorations that help plaque accumulation and making new, more precise restorations to preserve the teeth.
7. Consuming fresh fruits containing vitamins due to their effective impact on increasing the secretion of saliva, which resists **Dental Caries**

(Pulpitis)

Causes of Inflammation

1. Inflammation of the tooth nerve by bacteria causing tooth decay.
2. Microbes reaching the tooth nerve through:

- A tooth fracture resulting from severe trauma or an accident.
- Fracture of a dental filling and recurrence of decay.
- The presence of a deep, unsealed crack in the tooth enamel.
- The microbe reaching via the bloodstream in some rare cases like diabetic patients.
- Exposure of the tooth nerve to some chemical materials sometimes used in dental clinics.
- Exposing the tooth nerve to high temperatures, such as during decay removal using a high-speed drill without water coolants, and also during tooth preparation for fixed prostheses.

Types of Tooth Nerve Inflammation

- Acute Pulpitis
- Chronic Pulpitis

First: Acute Pulpitis

This type usually occurs as a result of the tooth nerve being exposed to strong irritants, such as dental caries bacteria that attack the pulp tissue in large numbers, causing acute inflammation.

Characteristics and Symptoms of Acute Inflammation:

This type of inflammation occurs in a short period and leads to symptoms appearing over a short time, summarized as follows:

- A sensation of sharp pain that usually comes at the beginning of the inflammation in intermittent periods, especially after the patient drinks cold beverages.
- The pain then becomes continuous throughout the day and is at its peak during the night and when sleeping.
- A slight increase in body temperature may occur.

Treatment of Acute Tooth Nerve Inflammation

Treatment is directed towards relieving the patient's pain by:

- Removing the inflamed nerve under local anesthesia and performing root canal treatment and filling of the nerve canals.
- Sometimes, tooth extraction is the effective treatment for this condition.

Second: (Chronic Pulpitis)

This type usually occurs as a result of the tooth nerve being exposed to weak irritants, such as caries-causing bacteria that attack the pulp tissue in small numbers, causing chronic inflammation, or the inflammation occurring in a patient with a strong immune system.

Characteristics and Symptoms of Chronic Inflammation:

This type of inflammation occurs over a long period and leads to symptoms appearing over a long time, summarized as follows:

- A sensation of weak, tolerable pain that usually does not cause disturbance to the patient.
- Pain with hot or cold drinks is weak.

Treatment of Chronic Tooth Nerve Inflammation:

- Removing the inflamed nerve under local anesthesia and performing root canal treatment and filling of the nerve canals.
- Tooth extraction in some cases.

Third: Chronic Hyperplastic Pulpitis in Children

This type occurs in children up to the eighth year of age. It is a distinct type that affects specific teeth due to the abundance of blood supply to the nerve, resulting in hypertrophy of the nerve tissue and its protrusion from inside the tooth through a deep decay cavity.

Characteristics and Symptoms:

This type occurs in primary molars and also in the first permanent molar. Its symptoms are summarized as follows:

- The presence of a large decay cavity with the protrusion of part of the hypertrophied nerve tissue that can be seen with the naked eye.
- A sensation of weak, tolerable pain in the child, and possibly no pain at all.
- Pain with hot or cold drinks is weak.

Treatment:

- Removing the inflamed nerve under local anesthesia and performing root canal treatment and filling of the nerve canals.
- Extracting the affected tooth in some cases.

Enamel erosion

Causes:

1. Physiological wearing of tooth enamel occurs with aging (Attrition), especially in men.
2. Mechanical wearing of tooth enamel (Abrasion) occurs due to direct friction between the tooth enamel and a hard object such as:

   - Using a hard toothbrush to clean teeth.
   - Incorrect use of the toothbrush in a horizontal direction instead of vertical.
   - Some bad habits like placing nails and needles between the front teeth, habits common among some craftsmen like carpenters and tailors.

3. Chemical wearing of tooth enamel (Erosion) occurs due to exposure of the tooth enamel to a chemical substance from one of the following sources:

   - An external source, such as exposure to acid vapors in dry battery factories or consuming large amounts of acid-containing juices (erosion occurs on the buccal surface of front teeth in these cases).
   - An internal source when gastric juice reflux occurs in some individuals (erosion occurs on the lingual surface of front teeth in these cases).

Treatment of Enamel erosion Cases

- Treatment is usually done by making crowns or dental bridges to cover the affected teeth.

Second: Gum Diseases

The gums are soft tissue covering the bony area surrounding the teeth and also extend to cover the tooth neck from all surrounding surfaces. The natural gum color is pale red, and its texture is granular like an orange peel, ending above the tooth neck a short distance with a fine edge like a knife blade.

The gums are affected by many diseases, among which a group occurs more commonly:

1. Chronic Periodontitis
2. Acute Necrotizing Ulcerative Gingivitis
3. Gingival Hyperplasia and Hypertrophy

(Chronic Periodontitis)

The most common type among gum diseases and affects a large percentage of patients visiting dental clinics.

Causes:

1. Neglect of daily dental hygiene and not brushing with toothpaste regularly during some periods like puberty and also during pregnancy.
2. Failure to reach all different tooth surfaces with a toothbrush, leading to the accumulation of plaque on the gum edge and beneath it, containing bacteria and food residues causing inflammation over time.
3. Incorrect use of the toothbrush.
4. Neglecting to chew equally on both sides, leading to calculus accumulation and inflammation due to bacterial proliferation.
5. Mouth breathing due to nasal obstructions, leading to drying of gum tissue and activity of inflammation-causing bacteria.
6. Some diseases accompanied by bacterial activity like diabetes.

7. Cases of chronic indigestion and vitamin deficiency, especially vitamin C.

Symptoms of Chronic Gingivitis:

1. Change in gum color to dark red.
2. Swelling of the gum tissue, which becomes smooth with enlargement of the gum edge on the teeth.
3. Bleeding from the gums, its degree varies according to the severity of inflammation, occurring frequently, especially during toothbrush use.
4. Presence of calculus deposits, especially on tooth surfaces facing salivary gland openings (lingual surface of lower front teeth and buccal surface of upper molars).
5. In extreme cases, the following is observed:

- Discharge of a pus-like fluid when pressing on the gums.
- Formation of pockets between the gums and teeth where diseased tissue, bacterial waste, and their secretions collect.
- Recession of the gums from the tooth neck and the beginning of exposure of part of the tooth root.
- When the inflammation progresses, the jaw bones around the teeth begin to erode gradually, eventually causing tooth loss.

Treatment of Chronic Gingivitis:

1. Removing the main cause.
2. Performing gum care sessions and emptying pockets of diseased tissue and removing calculus deposits.
3. Paying attention to dental hygiene with a suitable brush and paste.
4. Using a mouthwash containing antibacterial substances.

Acute Necrotizing Ulcerative Gingivitis (ANUG)

It is a special type of gum inflammation accompanied by ulcers in the gum tissue and erosion in some areas.

Causes:

1. Decreased tissue resistance to infection due to plaque accumulation and calculus deposits, and also cases of nervous stress accompanying some individuals.
2. The dental bacteria for this type of inflammation are spirochete bacteria that coexist peacefully with the mouth's flora and begin to multiply and cause inflammation when the person's resistance is low.

Symptoms of Necrotizing Gingivitis:

1. Slight increase in body temperature.
2. Erosion and ulcers in the gum tissue, especially in areas covering the spaces between teeth.
3. Severe bleeding from ulcer areas may occur, accompanied by a foul odor.
4. Inflammation of the lymph glands located under the lower jaw.
5. In extreme cases, the infection spreads towards the throat and jaw bones, causing severe inflammation and erosion of the jaw bones with the possibility of gangrene Cancrum Oris occurring.

Treatment:

1. Removing the main cause.
2. Performing gum care sessions and getting rid of diseased tissue and removing calculus deposits.
3. Administering necessary antibiotics like Metronidazole (Flagyl).
4. Maintaining dental hygiene with a suitable brush and paste.
5. Using a mouthwash containing substances resistant to anaerobic bacteria, such as hydrogen peroxide.

Other Causes of Ulceration of the Oral Mucosa, Lips, and Gums:

1. Infection with some types of viruses like measles and herpes virus.

- Infection with some types of bacteria like those causing syphilis and tuberculosis. Acute deficiency in white blood cell functions like leukemia.
- Acute decrease in white blood cell count.
- Nutritional disorders.
- Deficiency of some vitamins like Vitamin A and Vitamin C.
- Advanced stages of some types of oral mucous membrane cancer.
- Some diseases of immune system dysfunction.
- Wrong habits practiced by some patients, like placing aspirin tablets on the gums and teeth causing ulcers due to their acidic content.

Gingival Hyperplasia and Hypertrophy

Causes of Gum Hypertrophy:

The causes of gum hypertrophy are many, the most important of which are:

- Hereditary causes like hereditary gingival fibromatosis (Fibromatosis Gingivae).
- Taking some medications for long periods, like the drug (Dilantin)used in treating epilepsy patients, and also some blood pressure medications like the drug (epilat).
- Occurrence of a disorder in the testosterone hormone in men, as it is a stimulating factor for the proliferation of cells forming the fibrous tissue of the gums.
- Chronic inflammations of the gum tissue.

Symptoms of Gum Hypertrophy:

- Increase in the amount of gum tissue covering the teeth.
- Swelling of the gum tissue, which may be accompanied by redness in its color due to inflammation.
- In some cases, severe fibrous swelling may occur, completely covering the tooth crown.

Treatment of Gum Hypertrophy:

1. Searching for the main cause and trying to treat it or avoid its occurrence in the future.
2. Cosmetic surgery for the gum tissue.
3. Treating the inflammation accompanying gum hypertrophy in some cases.

Third: Diseases of the Jaw Bones

The jaw bones (maxilla or mandible) may be affected by many diseases; some occur commonly and others rarely. The causes of jaw diseases can be summarized in several points as follows:

1. Bacterial inflammations of the jaw bones.
2. Diseases resulting from congenital defects in the jaw bones, both hereditary and non-hereditary.
3. Tumors of the jaw bones, both benign and malignant.
4. Diseases resulting from a hormonal imbalance affecting the jaw bones.

Bacterial Inflammations of the Jaw Bones

Inflammation may occur inside the jaw bones due to the spread of bacteria from an inflamed tooth nerve, causing an acute purulent inflammation around the tooth root called a dental abscess (Dento-alveolar Abscess).

Symptoms of Purulent Dental Abscess:

1. Severe pain upon vertical pressure on the tooth causing the abscess.
2. Swelling and redness in the gum tissue adjacent to the point of the inflamed bone.
3. A feeling of looseness in the tooth, varying in degree according to the severity of the inflammation.
4. Elevated body temperature.
5. Swelling and inflammation in the lymph glands surrounding the jaw.
6. In chronic cases, pus discharge is observed at varying intervals from a small opening in the gum opposite the root tip of the inflamed tooth.
7. In some cases, swelling of the tissues and muscles of the cheek opposite the abscess may be observed.

Treatment of Purulent Dental Abscess:

1. Giving the patient appropriate doses of antibiotics to control the inflammation, as well as suitable analgesics.
2. Incising the gum overlying the abscess to clean the jaw bone of pus.
3. Performing root canal treatment for the tooth causing the abscess to preserve it, or in some cases resorting to its extraction.

Some Hereditary Diseases and Congenital Defects Accompanied by Dental Symptoms

1. **Down Syndrome:**

   This disease results from a defect in the number of chromosomes of the fetus and is a non-hereditary disease. The incidence rate increases with pregnancy in women of advanced age (35-40 years).

- **Symptoms:**

General symptoms:

- Congenital heart, nervous system, and reproductive system defects.
- Varying degrees of mental retardation and memory loss.
- Appearance of the eye opening in a manner similar to the Mongolian race with a short distance between the eyes.
- Congenital defects in the skull bones.
- Immune system dysfunction.

Symptoms specific to the mouth and teeth:

- Enlarged tongue size with fissures.
- Nasal polyps and enlarged tonsils leading to mouth breathing.
- Increased saliva secretion, giving these patients natural resistance against tooth decay.
- Increased rates of gum inflammation due to immune system dysfunction.
- Delayed eruption of primary and permanent teeth with a reduction in the number of each.
- Small size of existing teeth, which often resemble thin pegs.

Ectodermal Dysplasia:

It is a group of hereditary diseases where there is a defect in all or some of the tissues formed from the embryonic ectoderm layer, including teeth.

Symptoms:

- **General symptoms:**
  - Deficiency in the formation of sweat gland tissue.
  - Incomplete formation of the skin layer.
  - Incomplete formation of nails.
  - Appearance of a small number of weak hair follicles in the head area.
  - Varying degrees of mental retardation.
- **Symptoms specific to the mouth and teeth:**
  - Incomplete formation of the mucous membrane lining the mouth.
  - Incomplete formation of salivary glands.
  - Severe reduction in the number of primary and permanent teeth or their complete absence, especially in male children.

Delayed Eruption

The natural eruption time of teeth is affected by several factors that may cause delayed appearance in the mouth in some cases or failure of teeth to erupt completely in other cases. Among these factors are the following:

1. Crowding of teeth and lack of sufficient space for eruption.
2. Formation of supernumerary teeth which impede the eruption of natural teeth.
3. Formation of cystic tumors around teeth inside the jaw bones.
4. Some hereditary diseases causing delayed tooth eruption like Down's disease.
5. Some diseases accompanied by delayed tooth eruption like osteomalacia .
6. Therapeutic radiation harmful to teeth in the formation stage.
7. Underactive thyroid gland (Hypothyroidism).

Diseases Requiring Special Precautions in Dental Clinic

There are many patients suffering from dental pain who simultaneously suffer from more serious diseases, requiring special handling to avoid exacerbating the patient's underlying problem. Among these diseases are the following:

First: Cardiovascular Diseases

(Angina Pectoris)

- **Precautions to be taken:**
  - Avoid long dental appointments.
  - Do not change the patient's position suddenly from lying down to sitting to avoid dizziness resulting from a sudden drop in blood pressure. These patients must be regular in taking Nitroglycerin medication to prevent heart attacks.
  - It is preferable to start dental treatment in the morning to avoid increasing the load on the heart muscle after digesting daily meals.
  - Prefer giving the patient a sedative before starting treatment, half an hour prior, like the drug (Valium 5-10 mg).
  - Give the patient a sublingual tablet of Nitroglycerin as a preventive measure.
  - It is possible to give the patient a local anesthetic containing a vasoconstrictor (Adrenaline), provided the dose of this substance does not exceed 1:100,000 and ensuring the patient receives only one dose of anesthetic per treatment session.
  - If the patient shows symptoms of an impending heart attack, dental treatment must be stopped immediately, and the patient given a sublingual pill of Nitroglycerin.

(Myocardial Infarction)

- **Precautions to be taken:**
  - Avoid long dental appointments, preferably starting in the morning.

- Prefer giving the patient a sedative before starting treatment, half an hour prior, like the drug (Valium 5-10mg)
- It is possible to give the patient a local anesthetic containing a vasoconstrictor (Adrenaline), provided the dose of this substance does not exceed 1:100,000 and ensuring the patient receives only one dose of anesthetic per treatment session.
- Avoid treatment methods that cause pain.
- If the patient is still in the stage of taking blood-thinning medication (like Heparin or Marevan), it is necessary to check the prothrombin time with a blood test before performing any type of dental treatment that may be accompanied by bleeding, such as tooth extraction or oral surgeries.

(Rheumatic Fever).

These are patients with heart valve defects or patients with artificial valves, who are at risk of rheumatic symptoms in the heart muscle due to bacterial activity in the blood.

- **Precautions to be taken:**

  - The patient must be given a prophylactic dose of penicillin before performing any type of dental treatment leading to bacterial activity, such as gum treatment sessions, tooth extraction, or surgeries.
  - The penicillin dose is one million units of penicillin (G) intramuscularly half an hour before treatment, followed by eight doses after treatment of penicillin (V), the dose being half a million units orally every 6 hours.
  - For patients allergic to penicillin, it is possible to give them one gram of the drug (Erythromycin) half an hour before treatment, then 8 doses of the same drug after treatment, the dose being half a gram orally every 6 hours.

Second: Liver Diseases

Liver Cirrhosis

- **Precautions to be taken:**
  - A complete blood count (CBC) must be performed to monitor clotting factors before extraction or surgeries.
  - Avoid giving the patient medications that lead to increased fibrosis in liver cells.

Viral Hepatitis

- **Precautions to be taken:**
  - The patient's condition must be followed up with their treating internist.
  - Perform dental treatment sessions only in extreme cases.
  - Follow high safety standards for sterilization procedures, such as wearing sterile gloves, a mask, and a gown, with proper and safe sterilization of tools used for the patient to prevent infection of others.
  - Avoid giving the patient medications that lead to increased damage to liver cells.

Third: Endocrine Diseases

(Diabetes Mellitus)

- **Precautions to be taken:**
  - Prefer avoiding long dental appointments.
  - The local anesthetic used must contain a vasoconstrictor, and preferably contain korbasel and not Adrenaline.
  - Give the patient antibiotics before and after gum treatment sessions or tooth extractions.
  - For patients taking insulin, the treatment session must start after they have eaten a meal to avoid a sudden drop in blood sugar and the possibility of coma.

(Hyperthyroidism)

- The occurrence of a crisis (Thyroid storm) in these patients during dental treatment must be avoided by following:
  - Avoid long treatment sessions.
  - Keep the patient calm.
  - Avoid the patient feeling pain as much as possible.
- If a crisis occurs inside the dental clinic, the following must be done quickly:

- Give the patient high doses of thyroid secretion inhibitors (Antithyroid drugs).
- Inject cortisone.
- Give the patient an intravenous solution of glucose.
- Doses of oxygen if needed.
- Monitor the patient's temperature and reduce it to normal levels.

Sterilization Methods and Infection Control in the Dental Clinic.

Dental clinics are distinguished from other clinics by the frequent use of various tools that contact each patient's condition and the treatment to be performed on the mouth and teeth, making sterilization methods an utmost necessity to prevent the transmission of any type of infection between patients or handlers, whether they are doctors or nursing staff. Among these methods are those commonly used in dental clinics due to their suitability for the materials and tools used, as follows:

(Steam Autoclave)

This device is suitable for all tools used in the clinic made of stainless steel, such as examination tools, instruments used in tooth drilling, as well as tooth extraction forceps and surgical instruments.

Device Features:

The device operates at high temperatures (about 121°C) that eliminate most infection-transmitting microbes.

Questions on this chapter

1. **Choose one correct answer for each of the following questions:**

1. Decomposition of one of the following substances is the primary risk factor for causing tooth decay:
   (1) fatty substance
   (2) Proteins of all kinds
   (3) Sugary substances
2. The accumulation of plaque is considered a fundamental factor for the occurrence of decay in:
   (1) The pits and fissures of the tooth
   (2) The smooth surfaces of the tooth .
   (3) The enamel layer covering the root
3. Decay in the dentin layer takes one of the following shapes:
   (1) A cone with its base towards the tooth enamel.
   (2) A cone with its base towards the tooth nerve.
   (3) An irregularly shaped pit.
4. The common type of tooth decay in patients treated with radiation is:
   (1) Smooth surface caries
   (2) Rampant caries
   (3) Acute caries

5. One of the most important characteristics of acute pulpitis is:
   (1) The occurrence of sharp pain when pressure is applied to the tooth
   (2) The occurrence of mild, tolerable pain with hot or cold drinks
   (3) The occurrence of sharp, continuous pain that usually comes during sleep

6. One of the following characteristics does **not** accompany **chronic** dental pulpitis in children:
   (1) The presence of a large cavity with protrusion of part of the pulp due to swelling
   (2) The inflammation occurring in the pulp of the second molar
   (3) The child experiencing weak and tolerable pain
7. Incorrect use of the toothbrush causes:
   (1) Physiological tooth erosion
   (2)Mechanical tooth erosion
   (3) Chemical tooth erosion

8. Chemical erosion of teeth occurs on the lingual surface of the anterior teeth in the case of:
   (1) The occurrence of gastric acid reflux in some people
   (2) Exposure to acid vapors in dry battery factories
   (3) Consuming large quantities of acidic juices
9. One of the following symptoms is important in differentiating between acute ulcerative gingivitis and acute gingivitis:
   (1) The presence of calculus (tartar) deposits on tooth surfaces
   (2) Swelling of the gum tissue
   (3) Severe bleeding accompanied by a foul odor and erosion of the gums

10. One of the following is **not** a symptom of suppurative dental abscess:
(1) Severe pain upon vertical pressure on the tooth related to the abscess
(2) Severe pain when consuming cold drinks
(3) Intermittent, prolonged pus discharge from a small opening in the gum facing the root tip

1) **Complete the following:**

1.Among the most important methods for preventing Dental Caries are the following:

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2. **Causes of chronic gingivitis include:**

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3. **Causes of delayed tooth eruption include:**

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4. **Symptoms of a suppurative dental abscess include:**

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5. **Precautions that must be taken with patients suffering from myocardial infarction include:**

C. Answer the following questions:

1. Mention the similarities and differences between chronic gingivitis and acute ulcerative gingivitis?
2. Mention the differences between acute and chronic pulpitis?
3. What are the methods used to sterilize dental instruments? Mention one of them in detail, explaining the advantages and disadvantages of this method.
4. What are the instructions that must be followed to prevent the occurrence of infection within the dental clinic?
5. What are the precautions that must be taken with a patient suffering from chest asthma?
6. What are the precautions that must be taken with a patient suffering from diabetes?
7. What are the causes and symptoms of gingival (gum) enlargement? Mention the treatment for this condition.
8. Mention the specific oral and dental symptoms for children afflicted with Down syndrome?
9. What are the causes of ulcers in the oral mucosa and gums?

Mention the types of dental caries, then **explain Clinical Manifestation of Dental Caries** ?

Image1: Dental Caries

The image illustrates the bacteria's need for sugary substances to form plaque and cause decay, away from the influence of saliva and fluoride.

Image2: Tooth Decay
The dark color of the decay and a gap in the grinding surface of the lower molars are noticeable

Image3: Gingivitis

Notice the recession of the gums from the teeth and the appearance of a large portion of the tooth roots, indicating the severity of the inflammation.

Image4: Physiological Tooth Wear

The image shows the wearing down of all the tooth enamel with a reduction in volume.

Image 5: Acute Ulcerative Gingivitis

The image shows the locations of ulcers occurring in the gums between the teeth.

Image6: Mechanical Tooth Wear

The arrow indicates the presence of wear on the tooth enamel at the biting edge of the teeth.

Image 7: Chronic Pulpitis in Children
Notice the protrusion of inflamed tissue from inside the first molar through a wide cavity.

Image8: Chemical tooth erosion.

The image shows erosion of all tooth enamel on the lingual surface.

Image 9: Purulent dental abscess.

The arrows indicate a tumor opposite the root tip.

Image 10: Periapical Dental Abscess
Notice the occurrence of external swelling related to the cheek, indicating the spread of inflammation and its complications.

Image 11: Down Syndrome
The picture shows a group of children suffering from Down syndrome.

Chapter Eleven
Bones

1- The Locomotor System

- **General Principles about Fractures and Joint Dislocation:**
  Fractures:
  A fracture is the separation of a bone into two or more parts.
- **Types of Fractures:**

1. **Simple or Closed Fracture:** A fracture not accompanied by a wound in the skin.
2. **Compound or Open Fracture:** A fracture accompanied by a wound. It is more dangerous due to the exposure of the wound to contamination and bleeding.
3. **Pathological Fracture:** When a fracture occurs in an area affected by bone disease.
4. **Comminuted Fracture:** When the fracture fragments are multiple.

- **Shapes of Fractures:**

1. **Transverse Fracture:** The fracture line is across the bone.
2. **Oblique Fracture:** The fracture has an oblique line.
3. **Spiral Fracture:** Where the fracture line moves in a spiral pattern around the bone.
4. **Impacted Fracture:** Where the ends of the bones interlock with each other after breaking.
5. **Greenstick Fracture:** Occurs in children and the fracture is incomplete.
6. **Depressed Fracture:** Occurs in skull fractures.

- **Symptoms of Fractures:**

1. Pain at the fracture site or tenderness to pain at the point of fracture.

- Swelling at the fracture site.
- Change in the shape of the fractured limb.
- Inability to move the fractured limb.
- Crepitus (grating sound) in the broken bones when moved.
- There may be protrusion of the broken bones through the skin.
- **Diagnosis of Fractures:**
  - Clinical examination.
  - Plain X-rays.
  - CT scan in cases of spinal fractures.
- **Complications of Fractures:**
  - **General Complications.**
    1. Neurogenic shock from pain or hemorrhagic shock due to severe bleeding.
    2. Occurrence of fat emboli in the case of multiple fractures.
    3. Complications from the patient lying in bed for long periods.
       - Blood clots in the veins (DVT).
       - Pneumonia.
       - Bedsores (pressure ulcers).
       - Chronic constipation.
  - **Local Complications.**
    - Wound infection in compound fractures.

- Nerve damage which may lead to paralysis.
- Injury to blood vessels which may lead to gangrene of the affected limb.
- Non-union and malunion of fractures.
- Joint stiffness due to long periods of immobilization.

First Aid for Fractures:

1. Administer painkillers and ensure the airways are clear and patent.
2. Stop bleeding and cover wounds in open fractures.
3. Treat shock if present.
4. Apply a splint to the fractured limb.
5. Transport the injured person correctly to the hospital.

Treatment of fractures usually follows these steps:

1. First aid.
2. Specific treatment at the hospital.
3. Discharge from the hospital followed by follow-up visits until healing is complete.

*** Splinting:**
Care must be taken with a fracture or dislocation injured by applying the appropriate splint to relieve pain and prevent complications.

*** Purpose of applying a splint:**

1. Prevent movement of the broken bones or dislocated joint during transport to the hospital, thereby reducing the pain.
2. Prevent damage to soft tissues, muscles, nerves, and blood vessels by the broken fragments.
3. Prevent piercing of the skin by the broken bones in a compound fracture.
4. Prevent pressure from the broken bones on blood vessels or causing bleeding in the fracture area.
5. Prevent limb paralysis due to pressure or damage to nerves.

*** General rules for applying a splint:**

- As a general rule, all fractures must be splinted before moving the injured.
- Fractures that result in a large angle must be straightened before applying the splint to align the broken bones and prevent damage to the nerves and blood vessels around the fracture.
- The splint must immobilize the joint above and below the fracture to prevent movement of the broken bones.
- When splinting a dislocated joint, immobilize the joint above and below the dislocation using the splint, taking care not to straighten the angle of the dislocated joint as this poses a danger to the nerves and blood vessels.
- Place the splint so that its ends do not impede blood circulation in the limb.
- Stop bleeding and dress wounds in compound fractures before applying the splint.

- Pad the splint at pressure points or for the comfort of the injured limb by placing cotton or padding under the splint.

- **Types of Splints:**

1. **Rigid Splints:** Like wooden pieces padded with foam, wire splints, cardboard splints, and long and short wooden boards. These splints are used for limb fractures, and wooden boards are used for pelvic and spinal fractures.
2. **Air Splints:** These are double-walled nylon bags that can be inflated by mouth after being placed on the fractured limb. They prevent movement of the broken bones and are used for upper limb fractures. They come in different sizes and can also be used to stop bleeding in these limbs.
3. **Traction Splints:** Used for splinting the lower limb. Among them is the Thomas splint, which is useful for fractures of the femur and fractures below the knee.

General method for using splints:
Preparing the splint: The splint should first be padded with cotton, then wrapped with a gauze bandage and secured.

firmly until it is covered. After that, the splint is applied as follows:

1. Cut the clothing covering the fracture site.
2. Hold the limb gently with one hand and the other hand under the fracture.
3. Place the splint below and above the fracture so that it does not impede blood circulation.
4. Cover all wounds with a sterile dressing before applying the splint.

How to use rigid splints:

1. Cut the clothing present on the fracture gently using scissors.
2. Measure the splint according to the length required for the fracture. It should be long enough to reach the joint above and below the fracture.
3. Gently support the limb at the fracture area and place the splint under the injured limb.
4. Secure the splint with bandages and do not tie too tightly to avoid stopping blood circulation.

How to use air splints:
Place the injured limb on the air splint with the valve open, then close the valve, and inflate by mouth until the required pressure is reached.

How to use traction splints:

1. There is a common type consisting of a fixed half-circle connected to two metal rods, connected at the end to a hook for traction, and it has straps to secure the injured limb.
2. Two people secure the splint. The first person gently pulls the lower limb at the foot area, applying light traction with caution in case of a thigh or leg fracture.
3. The other person places the splint so that it slides under the limb, raising the belt to the top of the thigh to reach the hip joint, and uses the traction part to the same degree that was applied by hand.
4. Secure the limb and the splint with the straps on the splint and tie with a circular bandage.
5. Position the injured part correctly without moving it.

Dislocation

It is the movement of the ends of the bones that form the joints, which may be accompanied by tearing of the joint capsule. The joints most exposed to dislocation are the shoulder, elbow, fingers, hip, and ankle joints, and the least exposed are the wrist and knee joints.

Symptoms of dislocation:

1. Change in the shape of the joint.
2. Pain at the joint site.
3. Inability to move the joint. The dislocated bone may press on nerves or nearby blood vessels, so the following symptoms must be observed in the injured:
   (i) Occurrence of numbness or paralysis below the site of the dislocation or fracture due to pressure on the nerves.
   (ii) Loss of pulse below the site of the dislocation or fracture due to pressure or injury to the blood vessels. When a blood vessel is injured, the affected limb becomes cold and requires rapid treatment.

Sprain

It is a partial tear or strain in the muscle fibers resulting from a sudden twisting of the joint beyond its movement limits. The sprain ranges from a minor injury to an injury leading to significant damage to the tissues surrounding the joint. It may be difficult to diagnose between a sprain, dislocation, and fracture, as they all have the same symptoms like swelling, pain, tenderness, and inability to move, although dislocation results in a change in the shape of the joint.

Fractures of the Upper Limbs:

1. **Fractures of the shoulder bones and clavicle:**
   In fractures of the shoulder bones, the clavicle or the upper part of the arm may be injured, and rarely the scapula is injured. In a clavicle fracture, the injured complains of pain and swelling at the fracture site. First aid for a clavicle fracture and the upper part of the arm involves immobilizing it with a sling or tying the arm to the victim's chest. Often, treatment involves immobilizing the clavicle bones by applying a figure-of-8 bandage after placing two pieces of padding in the patient's armpits, or the condition can be treated by placing a sandbag between the shoulder blades, and the patient sleeps on their back for a week.

   - Fracture of the arm bone :
   The injured feels pain when moving the arm and when touching the arm at the fracture site. It can be treated by making a sling and tying the arm to the chest. A short splint can be placed on the arm made of wood, cardboard, or wire. If there is a change in the shape of the arm, traction must be applied and the splint fixed on the outer part of the arm.

   - Elbow joint fracture:
   It is considered one of the serious fractures, causing severe pain, swelling, and internal bleeding due to the possibility of injury to the main artery of the arm and the nerves near the joint. There is a clear change in the shape of the joint and an inability to move it. A splint must be applied to

the joint in the same position it was found in, taking care not to attempt to return the joint to its original position. A wooden splint or a wire ladder splint can be used. A sling and bandage can be used to secure the limb to the chest.

- Wrist and forearm fracture:
A fracture of the two forearm bones may lead to a change in the shape of the bones and pain. It can be treated by applying a rigid splint or an air splint, taking care to immobilize the elbow and wrist joints. If there is a severe change in the shape of the broken bones, the fracture requires traction of the broken limb, then the splint is applied and the injered is transported to the hospital.

- Shoulder joint dislocation:
It occurs due to falling on the elbow or hand. The dislocation can be anterior or posterior. In the case of an anterior dislocation, the victim holds their arm away from the body and cannot place the arm at their side. There is a protrusion in front of the shoulder. First aid involves splinting the arm in the same position it is found in, without attempting to reduce the dislocation. In a posterior dislocation, the arm is in front of the chest with an inability

to move the arm and severe shoulder pain. It can be treated by applying a bandage around the chest to secure the arm, and the victim is transported sitting on a stretcher, not lying down.

1. **Elbow joint dislocation:**
   It occurs due to falling on the hand. The dislocation can be anterior or posterior. The victim cannot move the elbow due to severe pain. The elbow should not be pulled; instead, it should be immobilized with a posterior splint in the position it is found in. The victim is transported sitting. A sling can be made to limit movement of the joint.
2. **Wrist joint dislocation:**
   The wrist joint is rarely dislocated. The joint should not be reduced. A splint with padding is applied, and the victim is transported to the hospital.
3. **Pelvic bone fracture:**
   It occurs due to a severe injury leading to serious complications in the urinary system, such as the bladder and urethra. Severe bleeding may occur inside the pelvis or a rupture of the urethra. Severe bleeding inside the pelvis can occur due to the large blood vessels inside the pelvis. Treatment involves using the long wooden splints used for spinal fractures.

Fractures of the Lower Limbs

1. **Femur (thigh bone) fracture:**
   The fracture may be in the head and neck of the bone or in the shaft of the bone. These fractures occur in accidents or in the elderly from falling on the ground. The injured suffers severe pain with an inability to move and a change in the shape of the thigh. One leg appears shorter than the other, and the foot on the injured leg is turned outward.

First aid:

1. Use a hare traction splint or.
2. Use a long wooden board or.
3. Tie the legs together , where the healthy leg acts as a splint for the broken leg.

- **Knee joint fracture:**
  It occurs when the knee hits a solid object, such as falling on the ground or in car accidents due to the knee hitting the car dashboard. Obvious deformity occurs with swelling in the front part of the knee. The leg should be gently aligned, a splint applied, and circulation in the foot and leg monitored.
- **Tibia and fibula (leg bones) fracture:**
  It is a common fracture resulting in swelling, pain, and a change in the shape of the leg with an inability to walk. The fracture can be treated by applying a wooden splint, air splint, or traction splint after aligning the fracture by applying traction to the leg from below while immobilizing the knee joint.
- **Hip joint dislocation:**
  A rigid splint can be applied whether in cases of fracture or dislocation. It is preferable to transport the victim on a rigid stretcher in the position found.
- **Knee joint dislocation:**
  It is a rare injury but very serious. The joint must be immobilized without reduction. A rigid splint is applied, preferably wooden types, and the victim is transported immediately to the hospital.
- **Ankle joint dislocation:**
  It is often difficult to differentiate from a fracture. A rigid splint or a pillow splint should be applied.

Spinal fracture:

Fracture and dislocation of the spine are of great importance, as they may be accompanied by damage to the spinal cord resulting in paralysis. Because the spinal cord is protected by the spine, as long as the spine is intact, the spinal cord is safe. The first aider, when transporting an injured with a spinal fracture or even just suspected fracture, must do so correctly and with extreme caution to prevent complications. Spinal injuries occur in car accidents, falls from height, or diving injuries.

Signs and symptoms of spinal fracture:

1. **Pain:** If the injured is conscious, they feel pain and can indicate the location of the injury. If unconscious, this important sign is absent .
2. **Tenderness to pain:** If the first aider palpates the injured area, the patient complains of pain.
3. **Painful movement:** If the injured attempts to move, they feel severe pain.
   The displaced bones may press on the cord tissue, causing bleeding in the cord or damage to the veins or venous plexuses located above the cord.
4. **Deformity of the spine:** Occurs secondarily in severe injuries, and there may be an unusual curvature or bony prominence in the spine.
5. **Abrasions or wounds over the spine.**
6. **Occurrence of paralysis** due to damage to the spinal cord with loss of sensation and movement.

First aid for spinal fracture patient :

1. Maintain a clear airway and ensure breathing is adequate.
2. Apply a long wooden splint for the spine or use a scoop stretcher.

3. Stop bleeding and treat shock if present.
4. The patient should be lifted by their head and legs when transported to the hospital.

Skull Fracture

A skull fracture is life-threatening if it results in internal complications in the brain.

Types of skull fractures:

- **Types of skull fractures:**

1. Linear skull fracture.
2. Penetrating injuries to the skull like gunshot wounds.
3. Depressed skull fracture. The depressed bones may press on brain tissue, causing bleeding in the brain or damage to the veins or venous sinuses located above the brain.

Signs of skull fractures:

1. Deformity in the shape of the skull.
2. Flow of blood or cerebrospinal fluid from the nose and ears.
3. Presence of a blood bruise (raccoon eyes) under the conjunctiva of the eye.
   Generally, a skull X-ray must be done for all head injuries to confirm the presence or absence of a fracture.

Care for skull fractures:

1. Maintain a clear airway.
2. Stop bleeding.
3. Cover open wounds.
4. Observe and record diagnostic signs carefully, especially the size and reaction of the pupils, blood pressure, pulse, temperature, level of consciousness, and occurrence of convulsions.

Rib Fracture

A rib fracture often occurs from compression between two solid objects, like a wall and a car, or falling on a solid object like a curb or stairs. Often the fracture is in the outer region of the breastbone (sternum) on both sides, which is the most curved part of the rib. It can be treated by placing an adhesive plaster on the fractured area in the position of deep inspiration for a month, or placing a chest bandage around the chest for a month, or bandaging the patient with an elastic bandage around the chest. The lung is injured with rib fractures, and hemorrhage or pneumothorax may occur due to air escaping from the injured lung into the space surrounding it. Its treatment involves inserting a needle connected to a underwater seal drainage bottle.

Lumber disc herniation

Between the vertebrae of the spine are cartilages (discs) responsible for the flexibility of movement between the vertebrae. The disc consists of strong fibers in the form of a circular fibrous disc, in the center of which is a rubbery, round body called the nucleus, which absorbs shocks to the vertebrae. The disc may slip backward and press on the nerves passing through this area, suddenly causing pain in the lower back if the disc herniation is in the lumbar region. It may cause nerve shock and pain in the thigh, leg, and foot, weakness in the muscles, and joint instability.

Treatment:

- Complete rest on a hard bed.
- Administering painkillers.
- Applying skin traction to the limb over the edge of the bed to relax muscles contracted from pain.
- Using supportive belts and corsets for the lumbar vertebrae.
- Performing exercises to strengthen back muscles so the patient recovers faster and to reduce the chance of pain returning.
- Surgical intervention in case of severe pain or if muscle atrophy occurs.

Cervical Disc Herniation

It is a prolapse affecting the discs between the cervical vertebrae, causing pain in the neck, shoulder, and arm. It is treated with rest, applying traction, or using a collar that supports the cervical vertebrae, then physical therapy. If necessary, surgery is performed.

Amputation Surgery

This operation is performed in the following cases:

- Severe injury, especially those accompanied by compound fractures and damage to arteries and nerves.
- Gas gangrene.
- Diabetic gangrene.
- Gangrene resulting from narrowing or blockage of blood vessels.
- Malignant tumors in the limbs that have spread to the tissues or bones of the limb.

Joint Inflammation (Arthritis)

Joint inflammation occurs for multiple reasons, the most important being rheumatoid arthritis, pseudo-rheumatoid inflammation, and osteoarthritis.

1. **Rheumatoid Arthritis:**
   It is a chronic disease that generally affects joints, particularly the small ones in the fingers. This disease occurs in the synovial membrane lining the joint. This membrane swells and creeps over the joint cartilage, which erodes as a result, eventually leading to adhesion of the joint surfaces. The joint becomes loose, and along with what happens inside the joint, swelling occurs around the joint which takes on a spindle shape.

Symptoms:

- The small joints of the fingers are affected first and are the most affected.
- Spindle-shaped swelling around these fingers.
- Deviation of the fingers towards the ulnar side, and some tendons may rupture.
- The disease spreads gradually to other joints.

- Stiffness in the joints, especially in the morning.
- Flare-ups occur where pains increase, and temperature may rise with a general deterioration in the patient's health.

Treatment:

- Administering cortisone , salicylates, and gold compound.
- Physiotherapy to preserve joint movement and strengthen muscles.
- Surgical correction to stop the severity of the disease and protect the joint, such as synovectomy (removal of the synovial membrane) or joint fixation if the joint cartilage is damaged, or correction of deformities.
- Performing joint replacements (arthroplasty), such as in the hip joint, knee joint, or finger joints.

Osteoarthritis

A disease that occurs in the joint cartilage and causes pain, especially when moving the affected joint or loss of the normal range of motion. It occurs gradually due to joint roughness and increased fibrosis in its flexibility. If an X-ray is taken, it shows changes in the joint surface.

Treatment:

1. **Analgesic** like aspirin and butazolidin.
2. Physical therapy like short-wave diathermy with muscle exercises and joint movement.
3. Surgical treatment: Surgical fixation of the joint (arthrodesis) or performing a joint replacement (arthroplasty).

Osteomyelitis

(a) Acute Osteomyelitis:
It mostly occurs in children aged 5-15 years, especially children suffering from general weakness and malnutrition, where resistance is low. This inflammation occurs at the ends of long bones, where growth is rapid, such as the lower end of the femur, the upper end of the tibia, the upper end of the humerus, and the lower end of the radius. However, it can also occur anywhere in the bones and can occur in people other than children. The causative organism is a staphylococcal bacterium.

Symptoms:

1. High fever with increased pulse.
2. Severe pain at the inflammation site and appearance of swelling around the inflammation site.
3. Symptoms of toxicity like pallor, coated tongue, and delirium.
4. Loss of movement in the affected limb.

- Severe tenderness upon touching to the extent that the patient screams when the affected bone is touched.

Complications:

1. septicemia and pyogenic poisoning(pyaemia), which appears as abscesses in other parts of the body.
2. Septic inflammation in the adjacent joint.
3. Pathological fracture in the affected bone.
4. Appearance of a subperiosteal abscess.
5. The acute inflammation turns into chronic, necrosis or caries.

Treatment:

1. Hospitalize the patient.
2. Immobilize the affected limb with an appropriate splint.
3. Administer broad-spectrum antibiotics after performing a culture for the microbe.
4. If a subperiosteal abscess is found, open the abscess immediately and place the leg in a cast with a window opposite the wound for dressing.
5. Follow up on the condition until complete recovery.

(b) Chronic Osteomyelitis:

It occurs after acute inflammation if it is not treated quickly and decisively.

Symptoms:

1. The patient gives a long history of the disease since the acute phase.
2. There are symptoms of chronic pyaemia like weakness, emaciation, and pallor.
3. There may be a fistula over the inflammation site from which pus exits.
4. Swelling of the affected bone.
5. X-ray examination shows signs of inflammation.

It may show the presence of an abscess inside the bone or the presence of a dead piece of bone (sequestrum) or general thickening of the bone.

Treatment:

- Surgical treatment to remove the tumor if present or to create drainage tube for the abscess.
- Treatment with antibiotics according to culture and sensitivity testing if pus is present.
- Amputation may be resorted to sometimes in cases that cannot be controlled surgically or with antibiotics.

Tuberculosis of Bones and Joints

This disease often affects the spine, hip joint, knee,humerus, shoulder, and phalanges of the fingers.

Symptoms:

- Slow onset Chronic pain in the affected area.

- Pain in the affected joint with deformity of the joint due to muscle contraction.
- Atrophy of the muscles surrounding the affected bone.
- Slow appearance of an abscess around the infection site, known as a cold abscess.
- Appearance of general symptoms of tuberculosis: which are emactations - slight rise in temperature in the evening, profuse sweating at night, loss of appetite, rapid pulse, and sudden pains that wake the patient at night.

Diagnosis:

- Perform an X-ray.
- CBC and erythrocyte sedimentation rate (ESR).
- Pathological examination of a sample from the infection site or a lymph node.

Treatment:

- Physical, mental, and psychological rest, and exposure to fresh air and sunlight.
- Good nutrition with proteins and vitamins.
- Administering specific antibiotics like Rifampicin capsules, Streptomycin injections, Para-aminosalicylic acid, or Isonicotinic acid hydrazide (Isoniazid).
- Local treatment through rest and traction in appropriate splints gradually until the pain decreases and the deformity disappears.
- Surgical treatment: Performing fixation of the affected joint (arthrodesis) or excision of disease foci from the bones to prevent reactivation.

Spinal Tuberculosis (Pott's Disease)

affects children in their early years and causes deformity in the spine. Paralysis of the limbs may occur.

Symptoms:

1. General symptoms of tuberculosis.
2. Pain in the spine.
3. Formation of a cold abscess in the spine and deformity of the back .
4. Muscle contraction leads to shock in back movement.
5. Paralysis of the lower limbs (paraplegia).

Treatment:

1. The general treatment mentioned previously.
2. Using a plaster cast (body jacket) or special beds for that purpose.
3. Aspiration of the cold abscess.
4. Surgical intervention in case of paralysis.

Bone Diseases
Bone Tumors:

1. **Benign Tumors:**
   a. Osteomas: occur in the skull.
   b. Osteochondromas: occur at the ends of long bones like the femur.
   c. Chondromas: usually occur in the metarsal bones and fingers.
   d. Bone cysts.

   Treatment:
   Excision of the tumor with bone grafting.

- **Malignant Tumors:**
  Usually occur at a young age from 10 – 30. The most important are:

  - Malignant bone tumor: Osteosarcoma.
  - Malignant cartilage tumor: Chondrosarcoma.
  - Ewing's tumor (Ewing's sarcoma).
  - Secondary tumors: These are malignant tumors in the bones resulting from the spread of cancer from other places like the breast or prostate.

  Treatment:

1. Chemotherapy and radiotherapy.
2. Complete removal of the tumor and placement of alternative artificial joints.
3. A few cases require amputation surgery.

Pediatric Diseases

1. **Rickets:**
   Occurs during bone growth due to calcium deficiency.

   Causes:
   a. Vitamin D deficiency in diet.
   b. Lack of exposure to sunlight.
   c. Kidney function disorder.

   Symptoms:

- Delayed walking and teething.
- Bone deformities, the most famous being bowlegs (inward) or (outward).

Treatment:
Administering the child appropriate amounts of calcium and vitamin D. Some cases that do not improve with treatment need corrective surgeries (osteotomy).

Congenital Hip Dislocation:

It is a congenital defect that affects females more than males.

Symptoms:

- Limited movement of the hip joint.
- The affected limb is shorter than the healthy one.
- Hearing sounds when reduction of the dislocation occurs .

Diagnosis:

- Clinical examination.
- Ultrasound examination.
- X-rays only after 3 months of age.

Treatment:

- Reduction of the dislocation and placing the child in a hip spica cast.

Clubfoot (Talipes Equinovarus):

A congenital defect of the foot that occurs in males more than females.

Symptoms:
The child is born with a deformed and crooked foot.

Treatment:

- Correcting the deformity and placing the foot in a cast multiple times.
- In some cases that do not respond to casting, surgery is performed to lengthen the tendons.

Osteoporosis

Osteoporosis is occurs as a result of deficiency in the bone cell formation.

Causes:

- Old age.
- Post-menopause in females.

Common Sites of Injury:
A - Spine.
B - Neck of the femur.
C - Wrist bones (Carpals).

Symptoms:

- Pain in the affected part.
- Occurrence of pathological fractures.

Treatment:

- Suitable meals with supply of calcium, vitamin D, and protein.
- Physical exercises.
- Exposure to sunlight.
- Drug therapy:
  - Calcium and vitamin D.
- Hormones to activate bone cells formation.

Arthroscopy

Arthroscopy is a procedure used to diagnose and treat some problems inside any joint.

Indications for Arthroscopy:

1 - Knee joint:

- Cartilage tear.
- anterior cruciate ligament tear.
- Loose body inside the joint.
- Inflammation of the synovial membrane of the knee joint.
- **Shoulder joint:**
  - Tear of the shoulder tendons.
  - Recurrent dislocation of the shoulder joint.

Advantages of Using Arthroscopy:
1- Accurate diagnosis of the disease.
2- Treatment.
3- Quick return of the patient to normal life.
4- Very small wounds.
5- Reduced rate of complications.

Artificial joint

These are operations aimed at changing the damaged joint by removing the ends of the joint bones and replacing them with artificial joints.

Aim:

1- Relieve pain.
2- Obtain a movable joint.

Indications:

1- Osteochondritis of the joint .
2- Patients with rheumatoid arthritis .
3- Post-fracture deformities around the joint.

Contraindications:

1- Septic (purulent)inflammation of the joint .

2- Cases of paralysis of the muscles around the joints.

Most commonly replaced joints:

- Total hip joint.
- Total knee joint.
- Shoulder joint.

Questions on the Locomotor System

1- What are the types of fractures, their symptoms, and complications?
2- Talk about the types of splints?
3- What is the general method for using splints?
4- What is the difference between dislocation and sprain?
5- What are the types of upper limb fractures?
6- What are the signs and symptoms of spinal fracture?
7- What are the types of skull fractures?
8- What is the difference between lumbar and cervical disc herniation?
9- In which cases is amputation performed?
10- Talk about bone tumors, their types and treatment methods?
11- Mention what you know about:
- osteomalacia
- Osteoporosis .
12- What is the importance of arthroscopy in treating bone diseases?
13- Talk about artificial joints?
14- Mention what you know about:
- Clubfoot in children .
- hip dislocation (corrected from).

Chapter Twelve

Components of the Urinary System

The urinary system consists of:

1- The upper urinary tract, which includes the kidneys and ureters.
2- The lower urinary tract, which includes the urinary bladder and urethra.

Symptoms of Urinary System Diseases

A symptom refers to the complaint the patient is suffering from. The most important symptoms indicating a malfunction in the urinary system are:

1- Pain, such as:

- Renal colic: Usually caused by a ureteric stone and presents as severe colicky pain in the flank that may "radiate" towards the external genitalia or thigh, and may be accompanied by other urinary symptoms.
- Kidney pain: Usually caused by kidney infections and comes as a continuous ache or pain in the flank.
- Bladder pain: It is in the form of "burning" during urination, which may be accompanied by pain in the suprapubic area.

2- Fever (may be accompanied by chills and a feeling of coldness), indicating the presence of infection in the kidney (or prostate or testicle in males).

- Hematuria (red urine) is the most important urinary symptom and has many causes, the most important of which are tumors, stones, inflammations, and kidney injuries.

3- Lower Urinary Tract Symptoms (LUTS) (specific to the bladder and urethra) and are divided into:

- Obstructive symptoms such as: weak urine stream, interrupted urination, and urinary retention.
- Irritative symptoms (storage dysfunction) such: as increased frequency of urination and inability to delay urination.

4- Urinary Incontinence: Inability to control urination.

- **types:**
  - Stress incontinence: Leakage of urine upon coughing, laughing, or lifting heavy objects.
  - Urge incontinence: Accompanied by a strong desire to urinate that cannot be delayed.
  - Total incontinence: Continuous leakage of urine day and night.

5- General symptoms: The most important are:

- Fever
- High blood pressure
- Loss of appetite, nausea, and vomiting due to renal failure.

Investigations

These are the basic tests and investigations for the urinary system:

- Complete urine analysis: The most important findings are the presence of "pus cells" in the urine, blood in the urine, or albumin "proteinuria" in the urine.
- Kidney function tests: the percentage of urea and creatinine in the blood.
- Ultrasound examination on the abdomen and pelvis (USI): The basic investigation.
- Contrast X-ray (IVP) on the urinary system to evaluate the urinary tract stones.
- Renal scan: To assess the function of both kidneys.
- CT scan of the abdomen and pelvis: For cases of kidney and urinary bladder tumors.
- Urodynamic examination: To evaluate urinary incontinence and neurogenic bladder.
- Cystoscopy to evaluate the presence of blood in the urine.

The most important congenital defects of the urinary system

1- Obstruction of the upper ureter(UPJ obstruction) (ureteropelvic junction)
2- Vesicoureteral reflux (Urine reflux from bladder to ureter)
3- Defects of the urethral meatus
4- Urethral meatal stenosis
5- Hypospadias (Urethral opening located on the underside of the penis)
6- Epispadias (Urethral opening located on the upper side of the penis)
7- Congenital urethral valves

Urinary Tract Tumors

A tumor, in simple definition, is: the proliferation of cells in a random manner that do not perform a useful function, resulting in a "mass" of tissue we call a "tumor". The most common urinary tract tumors are:

1- Bladder tumors: Generally malignant (cancerous) tumors.
2- Kidney tumors: Generally malignant (cancerous) tumors.
3- Prostate tumors: Mostly benign (Benign Prostatic Hyperplasia - BPH) and some are malignant (cancerous).
4- Testicular tumors

We will learn about each in detail.

Bladder Tumors

It is the most common tumor in men in Egypt due to increased smoking, environmental and industrial pollutants, along with endemic schistosomiasis infection.

Symptoms:
1- Urinary bleeding (blood in the urine) (red-colored urine) is the most important symptom.
2- A steady increase in the frequency of urination that does not respond to treatment.

Investigations:
1- Urinalysis: Presence of blood.
2- Ultrasound: Presence of a bladder tumor.
3- Cystoscopy: The cornerstone of diagnosis.

Treatment:
1- Superficial bladder tumor:
- Removal of the tumor via endoscope (TUR - TransUrethral Resection).
- Intravesical BCG instillation.
- Intravesical chemotherapy instillation.

2- Deep bladder tumor (invasive):
1- Radical removal of the urinary bladder with urinary diversion via:
- An intestinal conduit to the skin with an external bag to collect urine.
- Creating an artificial bladder from intestine connected to the urethra, allowing the patient to urinate through the natural passage.
- Ureterocolic anastomosis, where urine exits with stool.
5- Radiation therapy and chemotherapy for some advanced cases.

Kidney Tumors

The kidneys can be affected by different types of tumors, and most kidney tumors are malignant, i.e., cancerous.

Malignant Kidney Tumor (Cancer)
Causes: Smoking is considered one of the most important causes of kidney cancer.

Symptoms:
1- Blood in the urine.
2- Appearance of a swelling in the kidney area.
3- Steady weight loss.

Investigations:
1- Presence of blood in urinalysis.
2- Appearance of a tumor on ultrasound and CT scan of the kidney.

Treatment
1- Radical nephrectomy is the optimal treatment.
2- Immunotherapy with Interferon for advanced cases.
3- Conservative treatment with radiofrequency ablation in special cases.

Note:
It is necessary to differentiate between kidney tumors and kidney "cysts", as the presence of one or more cysts in the kidney (usually appears on ultrasound) generally has no pathological significance and does not pose any danger to the patient.

Prostate Tumors

Divided into benign tumor / malignant (cancerous) tumor.

Benign Prostatic Hyperplasia (BPH) (Senile)
It affects men over the age of fifty, causing disturbance in urination function.

Symptoms:
1- Obstructive symptoms (impaired urine expulsion).
2- Irritative symptoms (impaired urine storage).
3- Urinary retention.

Investigations:
1- Examination of the prostate by a urologist via the rectum.
2- Measuring prostate size by ultrasound.
3- Measuring urine flow rate (urodynamics).
4- Performing a PSA test to rule out cancerous tumors.

Treatment
Includes medical treatment and surgical treatment.

Medical Treatment
1- Use of alpha-blockers like Doxazosin.
2- Use of male hormone inhibitors.
3- Herbal medicines, which have no proven objective benefit.

Surgical Treatment (most effective method)
1- Transurethral resection of the prostate (TUR-P).
2- Prostatectomy via open surgery.

Temporary or emergency treatment:
1- Insertion of a Foley catheter into the urethra in cases of urinary retention.

Experimental Treatment
1- Laser prostatectomy.
2- Thermal therapy for the prostate.

Malignant Prostate Tumor (Cancer)

Affects men over the age of sixty.
Symptoms: Similar to benign prostatic hyperplasia.

Investigations:
1- High PSA level above 10.
2- Microscopic examination of prostate samples taken via ultrasound through the rectum.

Treatment

1- Radical prostatectomy or radiation therapy.
2- Orchidectomy (removal of testicles) in advanced cases or hormonal therapy.

Urinary Tract Stones

Urinary tract stones are among the most common urinary diseases and affect the kidney, ureter, and urinary bladder.

Types of Stones:

- Calcium oxalate: Represents most cases.
- Uric acid (Radiolucent stones on X-ray) 10%.
- Stones associated with urinary tract infection (Struvite) 5% .
- Rare stones: Cystine .

Symptoms

- Acute renal colic or persistent kidney pain.
- Urinary bleeding.
- Recurrent urinary tract infections.

Investigations

- Complete urinalysis: Shows the type of salts and presence of blood in urine.
- Contrast X-ray (IVP) on the urinary system.

Treatment of Urinary Tract Stones

The appropriate method is chosen by the urologist based on many factors, so there is no single method for treating all stones.

- Dissolving stones: Only suitable for radiolucent stones made of uric acid (about 10% of stones).
- Stone fragmentation using shock wave lithotripsy device(ESWL). Fragmentation is done without surgery, but may require placing an internal "stent" in the ureter called a "DJ stent".
- Stone extraction via endoscopy, such as nephroscopy, cystoscopy, and ureteroscopy.

Stone extraction via surgery, especially if the stone is large or accompanied by a stricture below the stone that requires repair

Preventive treatment for stones depends on:

1. Drinking plenty of fluids (at least 2-3 liters per day).
2. Reducing intake of foods high in uric acid, such as red meat, liver, brain, kidneys, tuna ,shrimp, dried legumes, cocoa, chocolate, and nuts.
3. Treating urinary tract infections.
4. Chemical analysis of the stone to determine the type of salts it contains.

Urinary Tract Infection (U.T.I.)

Definition: It is the inflammatory reaction of the urinary system due to the invasion of bacteria (or other pathogens) and is defined by an increased number of white blood cells (pus cells) in urine analysis.

Routes of Infection:

1. **Ascending route** (via the urethra) from the perineal area: The most common route.
2. **Via the blood**, such as in cases of renal abscess.
3. Via the lymphatic vessels(?)

Causes of Urinary Tract Infection:

1. **Gram-negative bacteria:** The most common cause, coming from the anus via the urethra (ascending route).
2. **Gram-positive bacteria:** Rare, except in renal abscess.
3. Tuberculosis bacillus.
4. Fungi.
5. Viruses, including the AIDS virus.

Investigations:

1. Complete urine analysis with **"urine culture and sensitivity"**.

Methods of Urine Sample Collection:

- Urinating directly into a container: An incorrect method that gives inaccurate results

MSU (Mid-Stream Urine) Sample

1- Clean the external genitalia.
2- Start urinating and discard the first part of the urine (10cm).
3- Collect the midstream portion of urine in a sterile container.
4- Collect the sample via a urinary catheter (especially in children).
5- Collect the sample via suprapubic aspiration (especially in children).
6- Imaging test to rule out causes of recurrent infection.

Signs:
1- Increased number of white blood cells (pus cells in urine analysis over 5 per high power field (5/HPF)).
2- Growth of pathogenic bacteria in urine culture.
3- Symptoms of bladder infection (burning urination - increased frequency of urination).
4- Symptoms of kidney infection (flank pain - fever and chills).

Treatment:
1- Appropriate antibiotic according to "urine culture and sensitivity" analysis and for a sufficient duration.
2- Treating causes of recurrence and persistence of infection, such as:

- Resistance to treatment like TB cases or fungi.
- Presence of stones in the urinary system.
- Presence of a foreign body in the urinary system like permanent catheter.
- Presence of a vesicoenteric fistula.
- Vesicoureteral reflux (especially in children).
- Bladder tumor.
- Neurogenic bladder cases.

3- Surgical treatment such as cases of renal and perinephric abscess and cases of urinary tract obstruction.

4- Persistence of UTI with fever for more than three days requires hospitalization.

Renal Failure

Definition: Inability of the kidneys to perform their functions regarding regulating body fluid levels, blood acidity, and cleansing the body of harmful substances like creatinine and urea.

Types of Renal Failure
1- Acute renal failure
2- Chronic renal failure
3- End-stage renal failure

Chronic Renal Failure

Definition: Decrease in glomerular filtration rate below 60 ml/min, corresponding to an increase in creatinine level above 2.5 mg/dl.

Causes:
1- Chronic kidney inflammation due to UTI or other immune conditions.
2- Diabetes, high blood pressure, and high uric acid.
3- Hereditary polycystic kidney disease.
4- Chronic urinary tract obstruction.
5- Misuse of some antibiotics and anti-inflammatory drugs (analgesics).

Signs
1- Loss of mental concentration and easy fatigue.
2- Loss of appetite, nausea, and persistent vomiting.
3- Inflammation of the pericardium in the heart and lungs .
4- hemoglobin deficiency (chronic anemia) and general wasting.

Treatment

- Conservative medical treatment:
  - Regulating diet and reducing protein intake.
  - Calcium compounds and active vitamin D hormone.
  - Iron compounds and erythropoietin hormone to stimulate red blood cell production.
  - Monitoring kidney functions: Creatinine.
  - Treating causes like regulating sugar and blood pressure.
- **Surgical treatment (End-stage renal failure):**
  - Kidney transplant.
  - Renal replacement therapy: either peritoneal dialysis or hemodialysis.

Kidney transplant: The kidney is taken from a relative of the patient (living donor) or from a recently deceased person (cadaveric donor). Tissue matching tests between the patient's (recipient) tissues and the donor's tissues must be performed to prevent immune rejection of the kidney so it can function.

Acute Renal Failure

Definition: Acute decrease (within days) in kidney function resulting in increased blood creatinine and usually decreased urine output (less than 400 ml in 24 hours).

Normal blood creatinine level: 0.5mg - 1.5 mg/100mg

Normal urea level: 20-40 mg/100mg.

Causes:
1- Decreased renal blood perfusion like cases of shock due to bleeding and fluid loss (vomiting, diarrhea, burns) and cases of heart failure.
2- Obstruction of urinary passages due to ureteric stones, prostate enlargement, and urinary bladder tumors.
3- Kidney-specific causes such as:

a- Glomerulonephritis.
b- Misuse of some medications like: Aminoglycosides (antibiotics), anti-rheumatism drugs.
4- Acute rejection of a transplanted kidney.

Main Signs:
1- Acute decrease in urine output (less than 400 ml / day).
2- Deterioration of the patient's consciousness, may progress to coma.

3- Hiccups, nausea, and vomiting.
4- Shortness of breath and increased respiratory rate.

Investigations:

- Ultrasound examination of the abdomen.
- Plain X-ray of the abdomen (KUB).

Treatment:
1- Treat the cause like blood and fluid loss, and ureteric stones.
2- Charting fluid intake and output, giving the patient fluids equal to the output volume + 500cm / kg/day .
3- Daily weighing of the patient.
4- Regulating patient's diet (low protein) and reducing potassium.
5- Treating emergency complications like blood tests, high blood potassium level, and high fluid overload .
6- Dialysis.
7- Relieving kidney obstruction via DJ stent or nephrostomy tube (PCN - Percutaneous Nephrostomy).

Acute Urinary Retention

Definition:

Inability to urinate accompanied by a strong desire to urinate and suprapubic pain, with a quantity of urine in the bladder (usually more than 300 ml).

Causes:

1- Benign prostatic hyperplasia (males over fifty).
2- Urethral stricture.
3- Bladder stone.
4- Following some surgeries like hemorrhoids and lumbar disc surgery.
5- Associated with some neurological diseases like spinal fractures.

Treatment:

1- Insertion of a urethral catheter to drain the bladder. If urethral catheterization fails, a suprapubic catheter is inserted.
2- Treating the cause later, such as prostate enlargement.

Chronic Urinary Retention

Definition:

Loss of the bladder's ability to empty completely, so a amount of urine remains in the bladder after urination (more than 100 cm3) permanently, which may lead to a swelling in the lower abdomen.

Complications:

1- Overflow incontinence (False incontinence): Continuous dripping of urine without control despite a full bladder.
2- Obstructive renal failure.

Causes:

1- Long-term bladder outlet obstruction and neglect of treatment.
2- Some types of neurogenic bladder, like complications of diabetes.

Treatment:
1- Insertion of a urethral catheter.
2- Treating the cause.
3- Ultrasound examination of the pelvis "measuring the amount of urine in the bladder after urination .

Complications of Urinary System Surgeries (corrected from)

1- Bleeding
a-Bleeding from the wound: This appears in the change in the site of the catheter or the presence of a large amount of blood in the red vac, such as in kidney surgeries..
b- Urinary bleeding: Appears in urinary catheters, e.g., after prostatectomy surgery.
c- General signs of bleeding.
2- Urinary leakage: Appears as an increase in the amount of urine coming from the wound or the catheters than expected and for a long period, due to obstruction below the site where the surgery was performed.
3- Wound infection or urinary tract infection, which may result in fever and pus in the urine.
4- Pain from the wound or due to the presence of the urinary catheter, requiring appropriate analgesics.
5- Urinary retention.
6- Imbalance of sodium, potassium, and body fluids.
7- Blockage of urinary catheters due to blood clotting inside them or incorrect placement.

Urinary System Injuries

Kidney Injuries:
Causes:
1- Road accidents.
2- Falling from high places.
3- Direct injury to the kidney area with a sharp object or gunshot.

Signs and Complications

1- Urinary bleeding.
2- Hematoma or urinoma around the kidney area, which may lead to a perinephric abscess.
3- Rib fractures and bruises in the kidney area.
4- Signs of other associated injuries like intraperitoneal bleeding.
5 – hydatid or purulent cyst in the kidney later on.

Note:

Signs of bleeding: Rapid pulse, low blood pressure, and pallor.

Treatment:

1- Emergency treatment of the general condition and treatment of bleeding until the patient's condition stabilizes.
2- Complete bed rest, catheter insertion, and monitoring urine color and quantity.
3- Surgical treatment in severe cases to save the patient's life by exploring the kidney and suturing the laceration or performing a nephrectomy.

Ureteral Injury

Causes

1- Improper use of a ureteroscope.
2- Some surgeries, especially gynecological surgeries and pelvic tumors surgeries.
3- Direct injury with a sharp object or gunshot.

Signs and Complications

1- Urinary bleeding.
2- Hematoma or urinoma.
3- Signs of other associated injuries like intraperitoneal bleeding.
4-. hydatid or purulent cyst in the kidney later on.

Treatment:
1- Placing a DJ stent in the ureter via endoscopy.
2- Surgical repair with reimplantation of the ureter into the bladder if necessary.

Bladder Injury

Causes
1- Road accidents, especially those associated with pelvic fractures.
2- Direct injury with a sharp object or gunshot.

Signs and Complications
1- Urinary bleeding.
2- Extraperitoneal or intraperitoneal hematoma or urinoma.
3- Signs of other associated injuries.

Investigations:
1- Placement of a catheter in the urethra with ascending x-rays of the urinary bladder

Treatment:
1- Surgical exploration of the bladder and suturing the laceration, placing a solar catheter in the bladder.

Questions about the Urinary System

1- Discuss the main causes of urinary bleeding?
2- Put atrue mark (√) in front of the correct statement or false mark (X) in front of the incorrect statement:
1-Most bladder tumors are malignant (cancerous) ( )
2-all urinary tract stones appear on plain X-ray.( )
3- Every patient with prostate enlargement needs prostatectomy.( )
4- To diagnose malignant prostate tumor use PSA ( )

5- The main symptom of acute renal failure is decreased urine output with increased blood creatinine level ( )

3-What are the main signs of chronic renal failure?
4- Discuss methods of collecting a urine sample for analysis and methods of treating urinary tract infections.

Chapter Thirteen
Anesthesia

Introduction:

Definition of Anesthesia:

It is the temporary loss of sensation in a part or the whole body through the use of anesthetic drugs to perform surgery without pain.

Types of Anesthesia:

- General Anesthesia
- Regional Anesthesia

General Anesthesia: involves

- Analgesia
  - Loss of pain sensation.
  - Loss of consciousness.
- Muscle relaxation

Local Anesthesia: involves

Loss of sensation in a specific part of the body through:

- Local Infiltration or Spray
  - : Injection or spray of local anesthetic at the surgery site.
- Nerve Block
  - : Injection of the nerve supplying the surgery site.
- Plexus Block
  - : Injection of the nerve plexus supplying the surgery site.
- Central Neuronal Blockade
  - : Injection of central nerves.
  - Includes:
  - Spinal Anesthesia
  - Epidural Anesthesia

Pre-anesthetic Assessment

Objectives:

1- Knowing the patient's medical history in detail.
2- Calming the patient.
3- Prescribing pre-anesthetic medication (premedication).
4- Writing the consent form for performing surgery and anesthesia.
5- Determining the patient's fasting period.
6- Preparing the operating site.

This is to ensure patient safety and avoid problems that may occur due to exposure to anesthesia and surgery.

Knowing the Patient's Medical History:

The patient is assessed comprehensively before anesthesia to know the patient's previous and current medical history, in addition to the family history of various diseases.

Knowing the medications the patient is taking for the following reasons:
1- Interaction may occur between some medications the patient is taking and anesthetic drugs, e.g., blood pressure medications.
2- Some medications must be adjusted before surgery, like anticoagulants and diabetes medications.
3- Some medications must be continued during surgery, like corticosteroids.

Knowing details of previous surgeries performed, previous anesthesia methods, and any history of complications due to anesthesia or surgery.

Existence of a medical history of allergy to any medications.

Family history of complications during anesthesia, as a different response to some anesthetic drugs may occur due to genetic influence.

3- Examination: Examining the Patient

- The patient is given a comprehensive examination covering all body systems.
- Examination of the airway.

4- Investigation: Performing Pre-operative Tests

- General: Coagulation profile, clotting rate, complete blood count (CBC).
- Liver function tests, kidney function tests, electrocardiogram (ECG), echocardiography (ultrasound on the heart). Pulmonary function tests, chest X-ray, sodium and potassium levels, according to the patient's health condition.
- Preparing blood for some patients according to the type of surgery.
- Requesting necessary medical consultations according to the patient's condition, such as cardiologist, pulmonologist.
- All this is written in the special form for pre-operative patient examination, stating the patient's name, age, hospital number, and specifying the operation site.

5- Calming the Patient:

- It is normal for any patient to feel anxious before any surgery, so the patient must be calmed by:

  - Explaining the procedures that will happen to the patient since hospital admission.
  - Clarifying the steps of anesthesia and surgery.
  - Clarifying the method of pain relief after the operation.
  - Prescribing sedative medications for the patient by the anesthesiologist.

6- Premedication: Pre-anesthetic Medications
7- Medications for calming the patient:

- **Diazepam (Valium):**

  - Given orally the night before surgery.
  - Can be given on the morning of surgery, one hour before, intramuscularly.

disadvantages:

- May cause respiratory depression in patients with respiratory failure.
- Patient's breathing must be monitored.
- Causes pain upon intramuscular injection.
- **Midazolam (Dormicum):**

  - Does not cause pain upon intramuscular injection.
  - More of a hypnotic than a sedative.
- **Flumazenil:**

  - It is a diazepim antagonist.
  - Given when there is difficulty breathing due to overdose of diazepim.

b- Analgesics:
Like Morphine, given to relieve pain before surgery, e.g., in fracture cases to avoid pain during transfer, and for cardiac patients.

c. Medications of the patient:
Like blood pressure medications - heart medications - corticosteroids.

d. Medications to reduce gastric acidity, increase gastric motility, and antiemetics:
Like Zantac and Primperan for patients undergoing emergency surgery without fasting.

4- Writing the pre-operative consent:

- A consent form must be signed agreeing to perform the surgery and anesthesia, stating that the medical team has explained the steps of anesthesia and surgery to the patient and clarified the percentage of possible complications.
- A risk consent must be signed if the patient's general condition is poor or for dangerous surgeries that may affect the patient's life.
- Signed by the patient himself, or by the nearest relative, or by the parents if the patient is a child.

Determining the Patient Fasting Period:

1. No Food or drink should be given orally for at least 6 hours before administering anesthesia.
2. To avoid vomiting or regurgitation of stomach contents into the respiratory tract, which leads to airway obstruction and pneumonia.
3. This period is reduced in infants to 4 hours to avoid low blood sugar.
4. This is not adhered to in cases of emergency surgery, such as internal bleeding cases, to save the patient's life.

Surgical Site Preparation:

1. According to the surgeon's instructions, such as shaving hair at the operation site, bowel preparation, using eye drops.
2. The correct surgical site must be confirmed (right or left).

Role of the Nurse before Anesthesia:

1. Writing the nurse notes form.
2. Informing the anesthesiologist examining the patient.
3. Implement the anesthesiologist's instructions, send the required labs, and ensure that x-rays are performed.
4. Confirm patient fasting.
5. Administer sedatives according to the anesthesiologist's instructions, confirm the drug name, dose, and administration method, and record this in the patient's file.
6. Prepare the operating room.
7. Calm the patient.
8. Accompany the patient to the operating room.

Questions

- What is the definition of anesthesia? And what are the types of anesthesia?
- What is the objectives of pre-anesthetic assessment of patient?
- What is the importance of knowing the patient's medical history?
- How can the patient be calmed before surgery?
- What is the fasting period before anesthesia? What is its importance?
- What is the pre-operative consent?
- What is the importance of knowing the medications the patient is taking before anesthesia?
- What medications can be given to the patient before surgery?
- How is surgical site prepared?
- What is the role of the nurse before surgery and anesthesia?

General Anesthesia

Stages of General Anesthesia

- (Induction) Loss of consciousness
- (Maintenance) Continuation of anesthesia
- (Monitoring) Monitoring the patient's vital functions
- (Recovery)

Loss of Consciousness Stage (Induction): Done via:

1. Intravenous injection of anesthetic drugs.
2. Inhalation of anesthetic gases.

General Anesthesia via Intravenous Injection (Intravenous Induction):

1. One of the most common methods for administering general anesthesia as it leads to rapid loss of consciousness.
2. The anesthetic is injected directly into the vein at the appropriate dose for the patient according to weight and medical condition, and by slow injection to avoid complications.
3. The anesthetic drug reaches the brain, causing loss of consciousness.
4. The anesthetic drug also reaches other body organs like the heart, liver, and kidneys, then distributes to muscles and fat, after which it is slowly eliminated.

Barbiturate Derivatives:

1. **Thiopental (Intraval)**
   - Causes rapid loss of consciousness.
2. (Ultra-short Acting).
3. Lasts 5-10 minutes, so anesthesia must be continued with another agent.
4. Dose: 5 mg/kg of patient weight or until loss of consciousness (Sleep Dose).

5. Must be diluted with 20 ml of sterile water or saline solution to avoid causing phlebitis.

- Must be injected into intravenous cannula to avoid accidental intra-arterial injection, which causes arterial spasm and gangrene in the fingers.
- Injected very slowly to avoid circulatory complications.

Side Effects:

1. Nervous System:

- Depression of the respiratory center leads to respiratory arrest.
- Depression of the blood pressure control center leads to low blood pressure.

2. Circulatory System:

- Decreased myocardial contractility: depends on dose and injection speed.
- Decrease in blood pressure.

3. Respiratory System:

- Airway obstruction due to loss of consciousness.

4. Phlebitis:

- Arterial blockage upon wrong injection.

(Diprivan, Propofol)

- Characterized by rapid recovery .
  - Causes more decrease in blood pressure than intraval .
  - Dilates bronchi, so preferred for asthmatic patients.
  - Preferred for short surgeries, day surgeries, and endoscopy.
  - Causes pain upon injection, so 1 ml xylocaine can be added.
  - Can be used as a sedative in small doses via continuous
  - Dose: 2 mg/kg of patient weight.

Ketamine

- Characterized by preserving circulation.

Characterized by pain relief (analgesia).

Disadvantages:

Increased heart rate.

Increased blood pressure.

Increased intracranial pressure.

Hallucinations and frightening dreams upon recovery.

Etomidate

Characterized by circulatory stability.

Can be used in patients with low blood pressure due to bleeding and cardiac patients.

- **General Anesthesia via Inhalation:** (**Inhalational Induction**)
  - For anesthetizing children.
  - When there is difficulty with the airway.
  - Also used during surgery in addition to other drugs.
  - Administered using a vaporizer which converts liquid to vapor inhaled by the patient via an anesthesia mask or through an endotracheal tube during anesthesia.
- **Halothane:**
  - Does not have a pungent odor, so can be used for children's anesthesia.
  - Causes decreased heart rate.
  - Causes decreased myocardial contractility, so used cautiously in cardiac patients.
  - Causes deterioration of liver function in some patients, so not used in liver patients.
  - May cause high fever (**Malignant Hyperthermia**) in some patients (**genetic predisposition**).

Isoflurane

Has a pungent odor, so not preferred for induction in children.

Advantages:

Its effect on the liver is less than Halothane.

Circulatory stability and can be used for cardiac patients.

Sevoflurane

- Known for rapid loss of consciousness and rapid recovery .
- Circulatory stability.
- Used when there is difficulty with the airway.
- Expensive, and requires a closed anesthesia circuit to reduce consumption.

Nitrous Oxide Gas

- Gas that causes pain relief (analgesia).
- Not used alone, but added to oxygen to avoid cyanosis and added to other agents due to its analgesic effect.
- **Entonox** (used for pain relief during childbirth.)
  Not used in cases of intestinal obstruction or middle ear surgeries.

Maintenance of Anesthesia Stage (Maintenance of Anesthesia)

Comes after the loss of consciousness stage.

Muscle relaxants are given, then the laryngeal tube is inserted.

Connected to the anesthesia machine.

Continuation of anesthetic drugs (oxygen and nitrous oxide gas in addition to one of the inhalational agents: Halothane or Isoflurane or Sevoflurane) (with continuation of muscle relaxants and analgesics).

Balanced Anesthesia:

Is the use of a combination of anesthetic drugs during surgery to reduce the side effects of each one.

Muscle Relaxants: (Muscle Relaxants)

- Drugs used to facilitate insertion of the laryngeal tube, through which anesthetic gases are delivered to the patient's lungs.
- Relax abdominal muscles to facilitate surgery.
- Using muscle relaxants necessitates the use of artificial ventilation because they cause respiratory muscle paralysis.

Succinylcholine (Succinyl Choline)

- Given to facilitate insertion of the laryngeal tube.
- Broken down by an enzyme in the blood (Pseudocholinesterase).
- Onset of action is after 30 seconds, so useful in cases where the patient hasn't fasted, for rapid intubation and reducing the chance of regurgitation of stomach contents into the trachea.

Causes muscle contractions first (Fasciculations) then relaxation.

- Duration of action is 10 minutes, may be prolonged if there is a deficiency of the enzyme that breaks it down, during which the patient needs artificial ventilation until its effect ends, and may need a blood transfusion due to containing the enzyme (?).

Other muscle relaxants used during anesthesia:

(**Esmerone**) (**Arduan**) (**Tracrium**) (**Pavulon**)

Morphine Derivatives (Narcotics)

Used during anesthesia for pain relief and to reduce the dose of other drugs.

Used after surgery for pain relief.

Repeated use causes dependence.

Administered intravenously, intramuscularly, subcutaneously, or by continuous IV infusion.

1. **Morphine**

- Causes bradycardia .
- Causes depression of the respiratory center, so the patient's breathing and oxygen saturation must be monitored.
- Causes bronchoconstriction and is not used in asthmatic patients.
- Causes pupillary constriction (miosis).
- Given cautiously in liver patients.

2. **Fentanyl**

A morphine derivative.

- Lasts for half an hour.
- Does not affect circulation, so preferred for cardiac patients.
- Causes less bronchoconstriction and can be used in asthmatic patients.

3. **Pethidine**

   A morphine derivative.

- Causes increased heart rate.
- Causes less respiratory depression than morphine.
- Causes less bronchoconstriction and can be used in asthmatic patients.

Other analgesic Tramal- Stadol- nopain

Naloxone

Is a morphine antagonist and is used to treat the side effects of morphine.

Monitoring the patient's vital functions

- The patient's vital functions must be monitored during anesthesia to diagnose side effects resulting from anesthesia, surgery, and the patient's pre-existing diseases.
- The following must be monitored:

1. Electrocardiogram (ECG)
2. Heart Rate (HR)
3. Blood Pressure (BP)
4. Temperature (Temp)
5. Oxygen Saturation (SpO_2)
6. End-tidal Carbon Dioxide ($EtCO_2$)
7. Concentration of anesthetic gases in inspired and expired air.

Recovery

- Anesthesia is terminated after surgery ends.
- Antidote for muscle relaxants is given: Prostigmin with Atropine.
- Anesthetic drugs are eliminated via exhalation from the lungs.
- Suctioning from the mouth.
- The endotracheal tube is removed when the patient's breathing is regular.
- The patient is transferred to the recovery room when the patient can control the airway.

Role of the Nurse in Preparing for Anesthesia in the Operating Room:

1. **Prepare the anesthesia machine:**

- Connect oxygen from the oxygen source (wall central oxygen or oxygen cylinder).
- Check the reading of the oxygen meter on the anesthesia machine to confirm oxygen pressure.
- Connect nitrous gas and check the gauge reading to confirm pressure.
- Fill the vaporizer with the appropriate gas.
- Check the appropriate anesthesia circuit and change the carbon dioxide filter when its color changes due to use.
- Connect the vital signs monitor.

1. **Prepare anesthesia tools:**
   - Prepare anesthesia drugs : necessity of writing the drug name and concentration on each syringe.
   - Prepare the laryngoscope and check the light.
   - Prepare endotracheal tubes for the patient's size, with a smaller and larger tube available as backup.
   - Prepare the airway for the patient's size.
   - Prepare Laryngeal Mask Airway.
   - Prepare the suction device connected to a suction catheter.
   - Prepare Ringer's solution and an intravenous device.
2. **Prepare emergency drugs:**

Dealing with any emergency: Adrenaline - Atropine - Calcium - Sodium Bicarbonate - Lidocaine - xylocaine -(Inderal) - Isoptin - Lanoxin - Aminophylline - Dopamine - Dobutrex- Tridil - Levophed.

- Prepare the defibrillator and ensure it works efficiently.
- Replace consumed drugs.
- Report any malfunction in anesthesia devices to the maintenance unit.
- Fill out the anesthesia machine checklist form (Anesthesia Checklist).

Spinal Anesthesia

- Is anesthesia of the lower half of the body by injecting a local anesthetic into the cerebrospinal fluid or into the epidural space .
- Used for surgeries of the lower part like inguinal hernia repair, hemorrhoids, anal fissure, and leg fractures.
- Anesthetizes sensory, motor, and sympathetic nerves.

Complications of Spinal Anesthesia:

- Low blood pressure: Resulting from anesthesia of the sympathetic nerve, treated by giving solutions and injecting vasoconstrictor drugs like ephedrine.
- Bradycardia : Resulting from anesthesia of the sympathetic nerve, treated by injecting atropine.
- Difficulty breathing: Paralysis of respiratory nerves. Treated using oxygen and artificial ventilation if necessary.
- Headache, treated with analgesics and solutions.

Contraindications for Spinal Anesthesia:

- (anticoagulant drugs).
- Patient refusal.
- Spinal diseases.
- Circulatory collapse (shock).

Method of Administering Spinal Anesthesia:

1. Explain the anesthesia to the patient and obtain consent.
2. Sterility must be observed.
3. Preparation:

- Sterile gauze.
- Sterile bowl.

- betadine solution.
- Insulin syringe for local anesthetic.
- 2 syringes of 5 ml.
- Ampoule of heavy Marcaine .
- Spinal anesthesia needle size 22 or 24 or 25.

Drugs used:

- Heavy Marcaine.
- Heavy xylocaine
- Fentanyl or Morphine: to prolong the effect of the local anesthetic and reduce complications.

Review Questions:

- What are the stages of general anesthesia?
- What drugs are used for general anesthesia via intravenous injection?
- What drugs are used for general anesthesia via inhalation?
- What are the side effects of interval?
- What are the disadvantages of Ketamine?
- What is the best intravenous anesthetic for patients with circulatory collapse?
- What is the best intravenous anesthetic for asthmatic patients?
- What are the disadvantages of Halothane?
- What are the contraindications for using Halothane?
- What are the advantages of nitrous gas?
- Why are muscle relaxants used?
- What are the advantages and disadvantages of Morphine?
- What vital functions are monitored during anesthesia?
- What is the recovery stage?
- What is the role of the nurse in the operating room?
- What resuscitation drugs must be available inside the operating room?
- What is spinal anesthesia? What are its disadvantages? And contraindications for its use?
- How is general anesthesia prepared for?

Recovery Room

- The patient goes to the recovery room after the end of surgery and anesthesia before going to the regular room to complete full recovery from anesthesia.
- **Criteria for patient discharge from the recovery room.**

1. Neurological Recovery:

- Where the level of consciousness returns to normal (awareness of place, time, and people).

2. Respiratory Recovery:

- Regular breathing, normal oxygen saturation.

3. Circulatory Recovery:

- Normal blood pressure
- Regular heart rate.

4. **Absence of Pain.**
5. **Absence of bleeding from the surgical site**.

Importance of the Recovery Room:

- Many serious complications can occur in the recovery room.

2. **Nervous System:**

   – **Delayed Recovery**

- Resulting from the effect of anesthetic and analgesic drugs.
- Resulting from chronic diseases: decreased kidney and liver function.
- Low patient body temperature.

Convulsions (Convulsions)

- Brain and nerve surgeries.

Treatment (Valium - Epanutin).

1. Respiratory System:

- Patient airway obstruction and oxygen deficiency.
- Treatment: Use airway, oxygen, and suction.
- Bronchospasm:
  - Treatment: Oxygen and bronchodilators.

2. Circulatory System:

- Low blood pressure:
  - Treatment: solutions , blood, and circulatory stimulants.
- High blood pressure:
  - Treatment: Analgesics, blood pressure medications, and diuretics.
- Heart failure (myocardial depression):
  - Treatment: Oxygen, diuretics, and Dobutrex .

6. **Digestive System:**
   - Vomiting
   - Treatment: Lower the patient's head, suction.
   - Oxygen.
   - Antiemetics (Zofran - Primperan).

Role of the Nurse in the Recovery Room:

1. Before receiving the patient: Preparing the recovery room:
   - Ensure oxygen source.

- Ensure suction is available.
- Ensure ECG machine is available.
- Ensure pulse oximeter is available.
- Ensure availability of resuscitation drugs and solutions .
- Ensure the defibrillator is available.
- Ensure availability of endotracheal tubes, laryngoscope, and airway.

After receiving the patient:

- Connect oxygen to the patient.
- Connect ECG.
- Connect the pulse oximeter.
- Measure blood pressure every 10 minutes.
- Patient positioning:
  - Semi-sitting
  - Prone position: Surgeries of the mouth and nose (Note: "" might be typo, prone is)
- Observing bleeding from the wound. Bed rails must be raised for all patients to prevent falls during recovery.
- Fill out the recovery form.
- Notify the anesthesiologist of any change in vital signs.
- Hand over the patient to the ward nurse, including the patient file, the intraoperative anesthesia form, the surgery details form, the recovery form, and the patient's imaging, and sign for handover and receipt.

Recovery in the Intensive Care Unit:

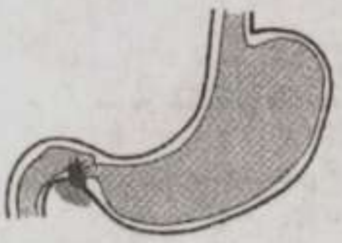
- Some critically patients or those undergoing certain surgeries like open heart, brain, and neurosurgery need to complete recovery in the intensive care unit.
- An ICU bed must be reserved before surgery.
- The ICU must be notified when surgery ends to prepare to receive the patient.
- Safe transfer of the patient must be ensured, including:
  1. Accompanying the patient by a nurse and doctor.

2. Continuation of vital sign monitoring devices during transfer like ECG and pulse oximeter.
3. Continuation of circulatory stimulants and inotropic drugs.
4. Continuation of oxygen use.
5. Maintaining patient position to prevent falls during transfer.

Review Questions:

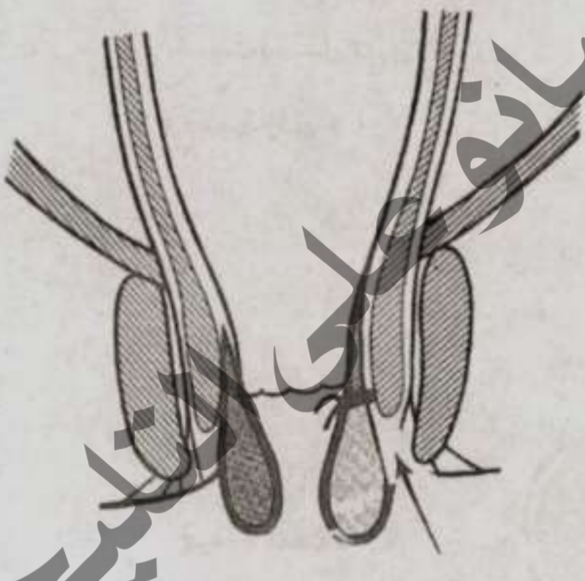
- What are the signs of full recovery?
- What are some complications that may occur in the recovery room?
- What is the role of the nurse in the recovery room?
- What are the precautions for transferring a patient to the intensive care unit?

Picture Appendix



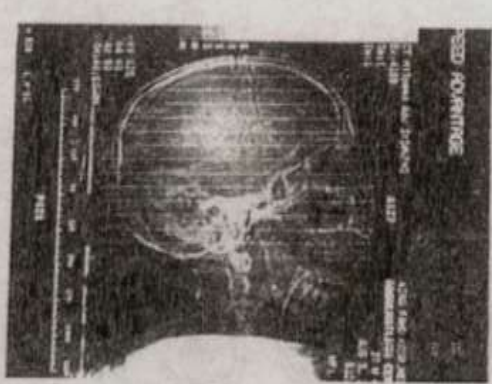
Duodenal Ulcer

(Image No. 1)



Anal Hemorrhoids

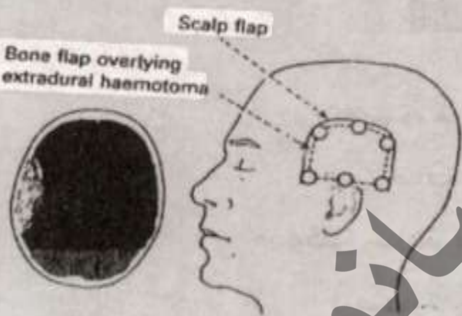
(Image No. 7)



CT Scan of Hemorrhage

Extradural

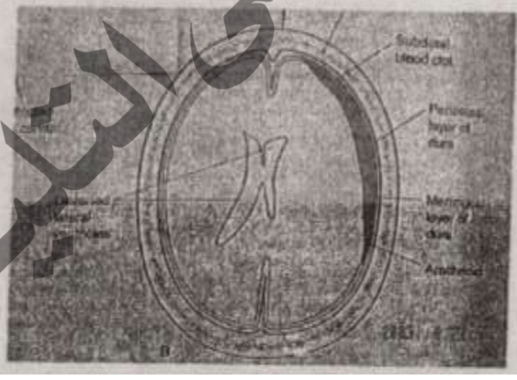
(Fig. 8)



Surgical Intervention for Hemorrhage

Extradural

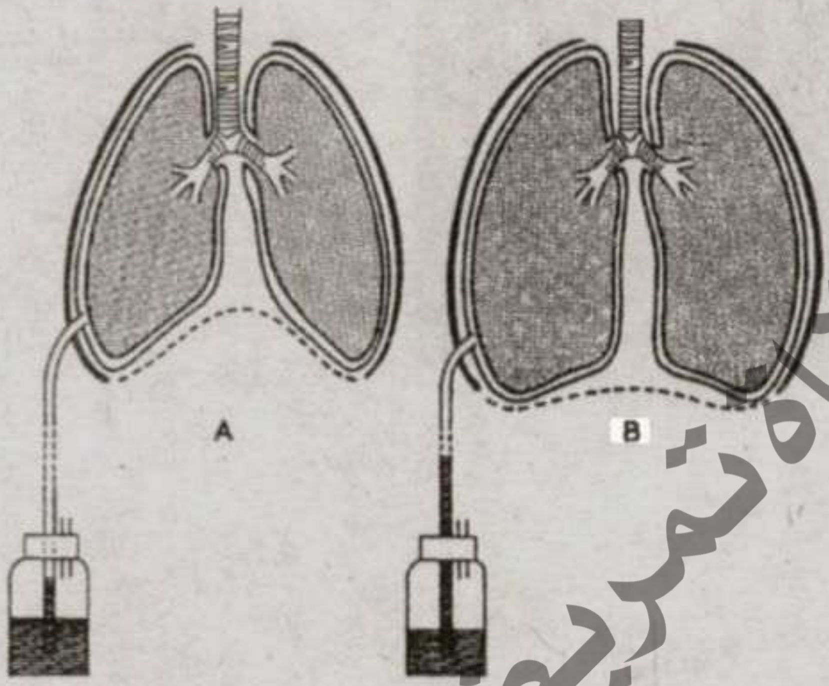
(Fig. 9)



Intradural

(Dura)

(Fig. 10)



Chest tube

8 in exhalation

B in inhalation

(Fig. 12)